

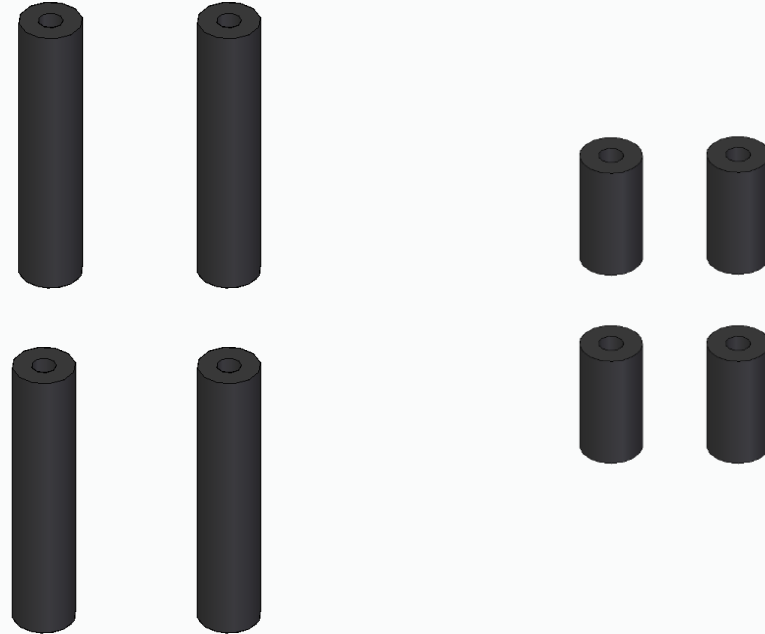
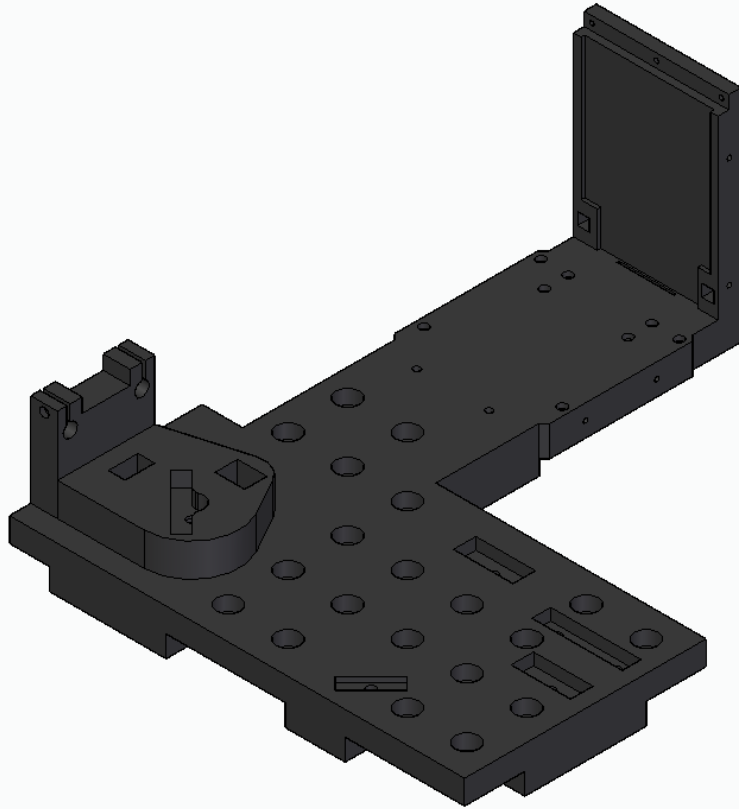
Fixation-Stimulus channel (FiSt)

Assembly instructions

v1.1

Main board

Installing PCB mounting brackets



Needs to be 3D-printed:

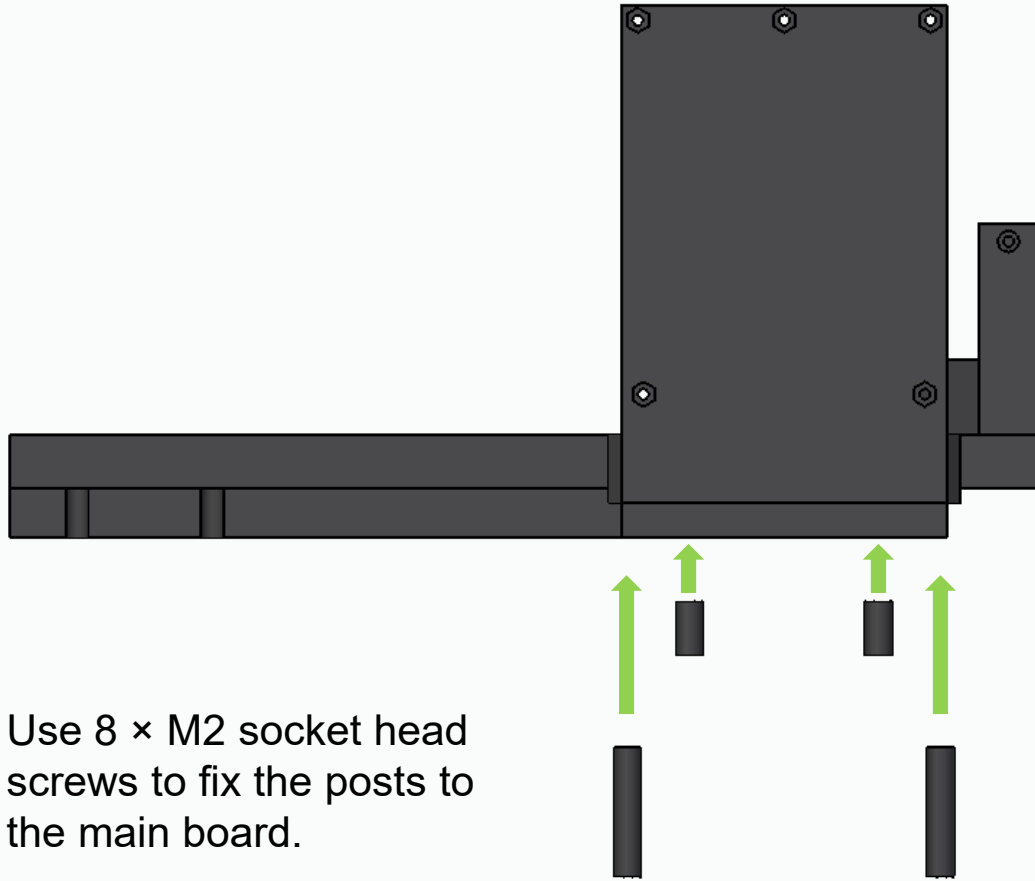
1 × 01_mainboard

4 × 02_electronics_post_24mm

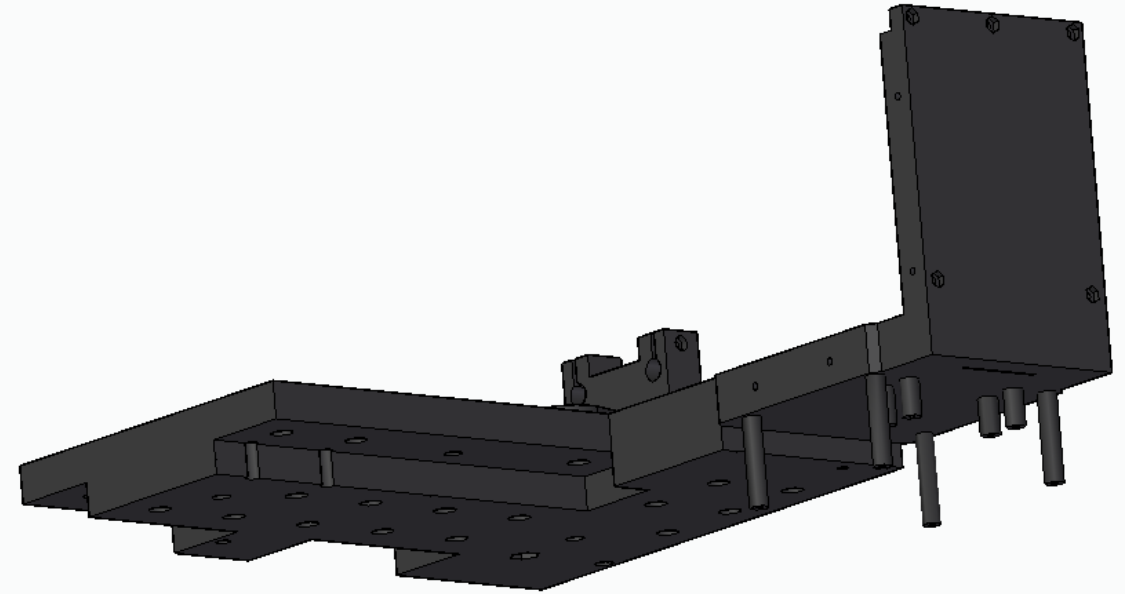
4 × 03_electronics_post_10mm

Main board

Installing PCB mounting brackets



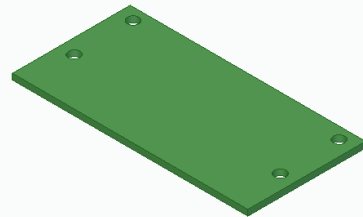
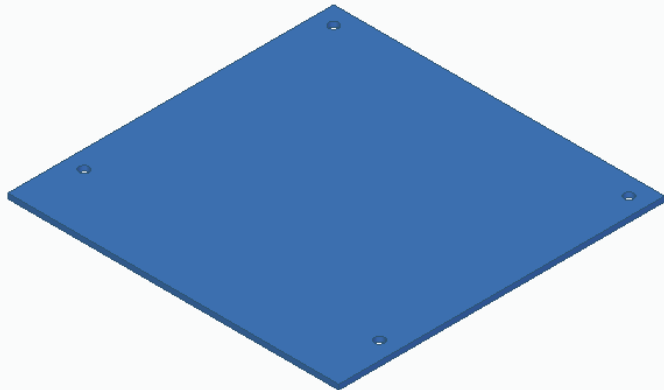
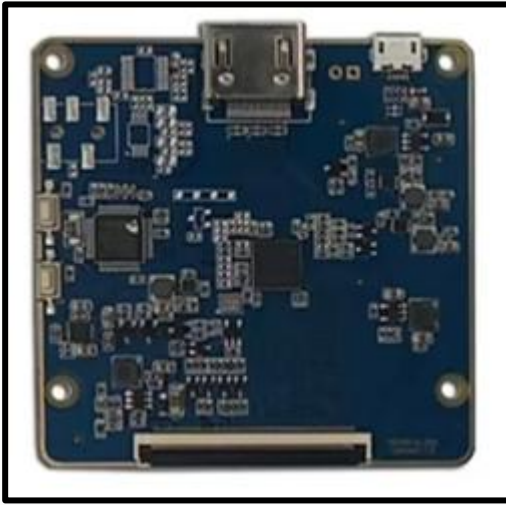
Use 8 × M2 socket head screws to fix the posts to the main board.



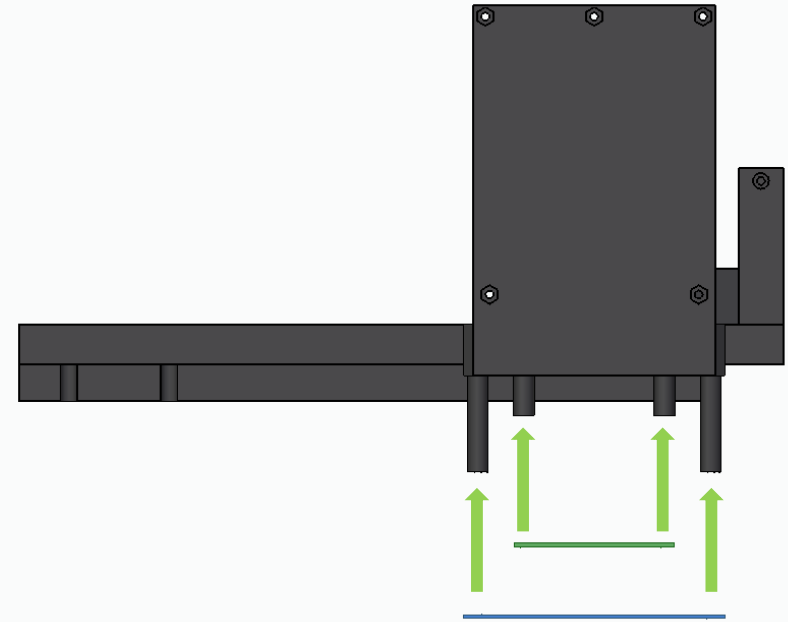
Main board

Installing PCB boards

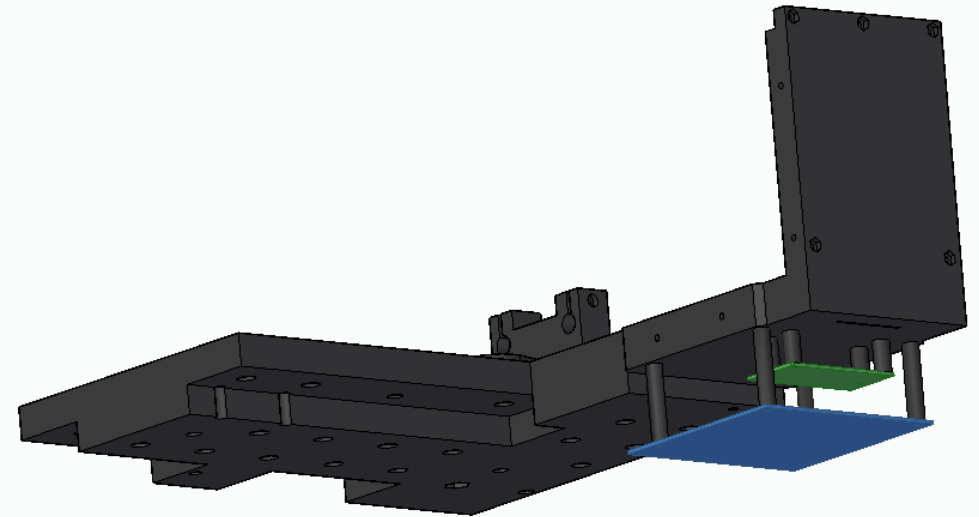
Attach the PCB boards that come with the LCD screen to the previously fixed bottom posts.



(a)



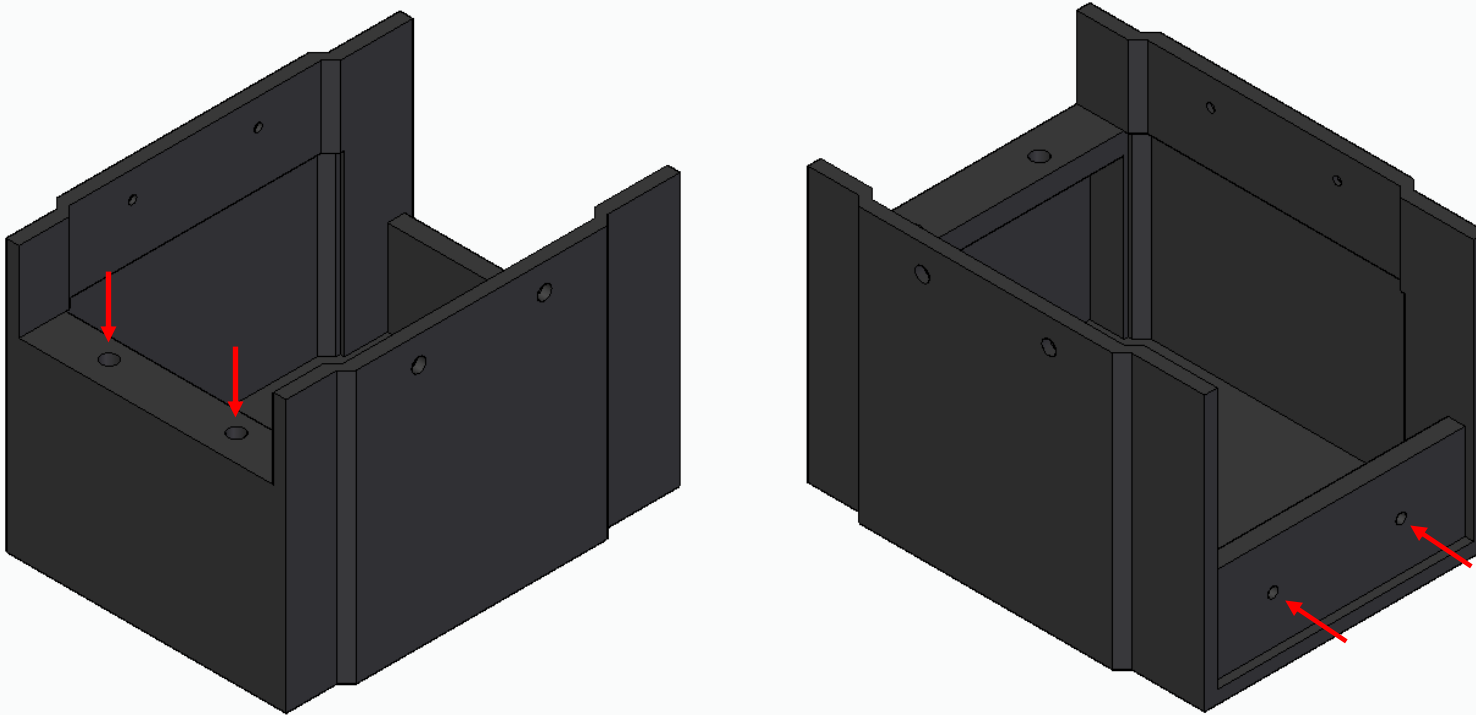
(b)



Main board

PCB enclosure

Mold the M3 threaded inserts into the designated holes (red arrows) on both sides using a soldering tool.

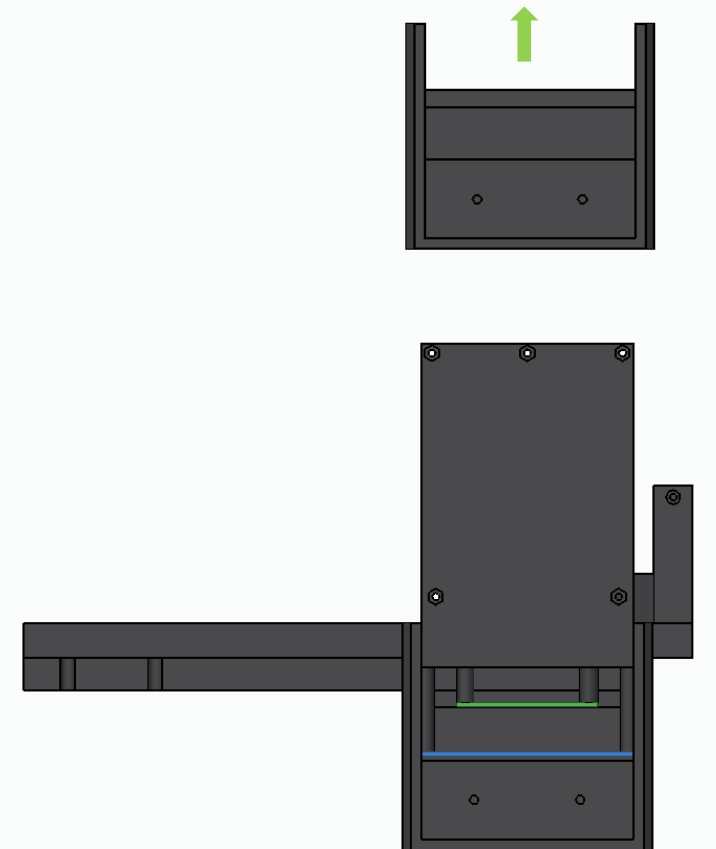
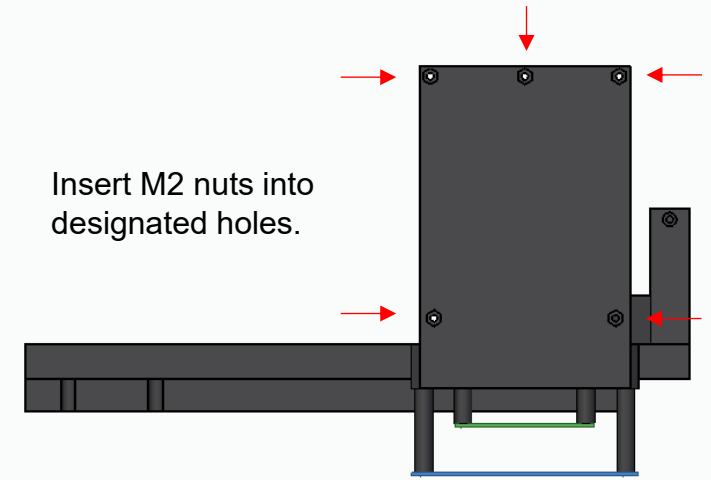
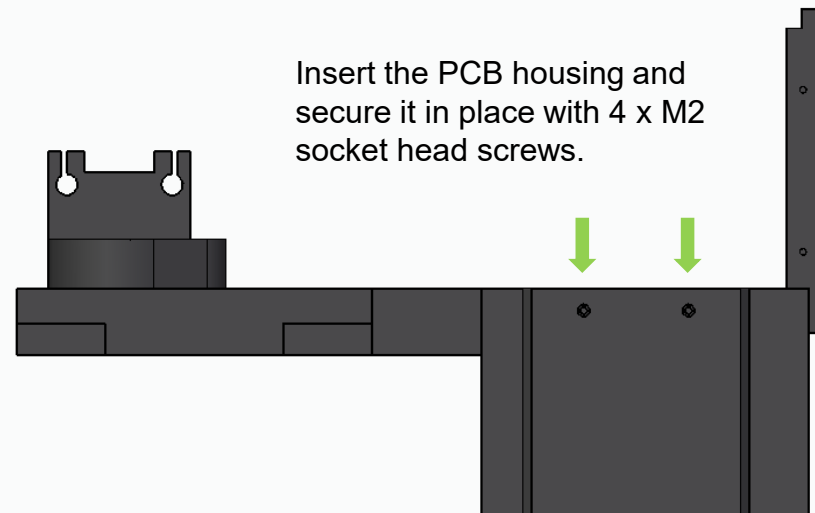
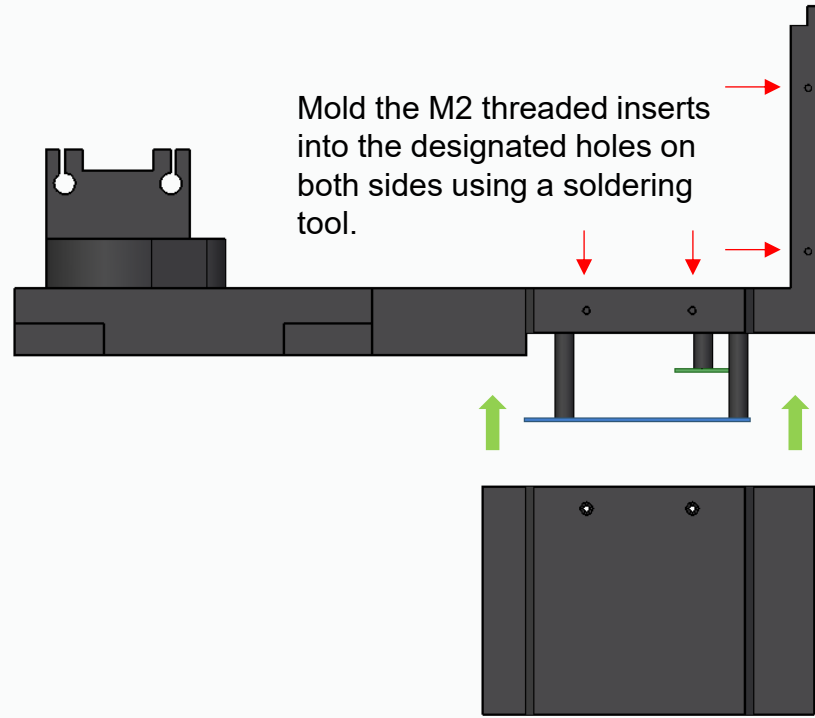


Needs to be 3D-printed:

1 × 04_PCB_boards_housing

Main board

PCB enclosure

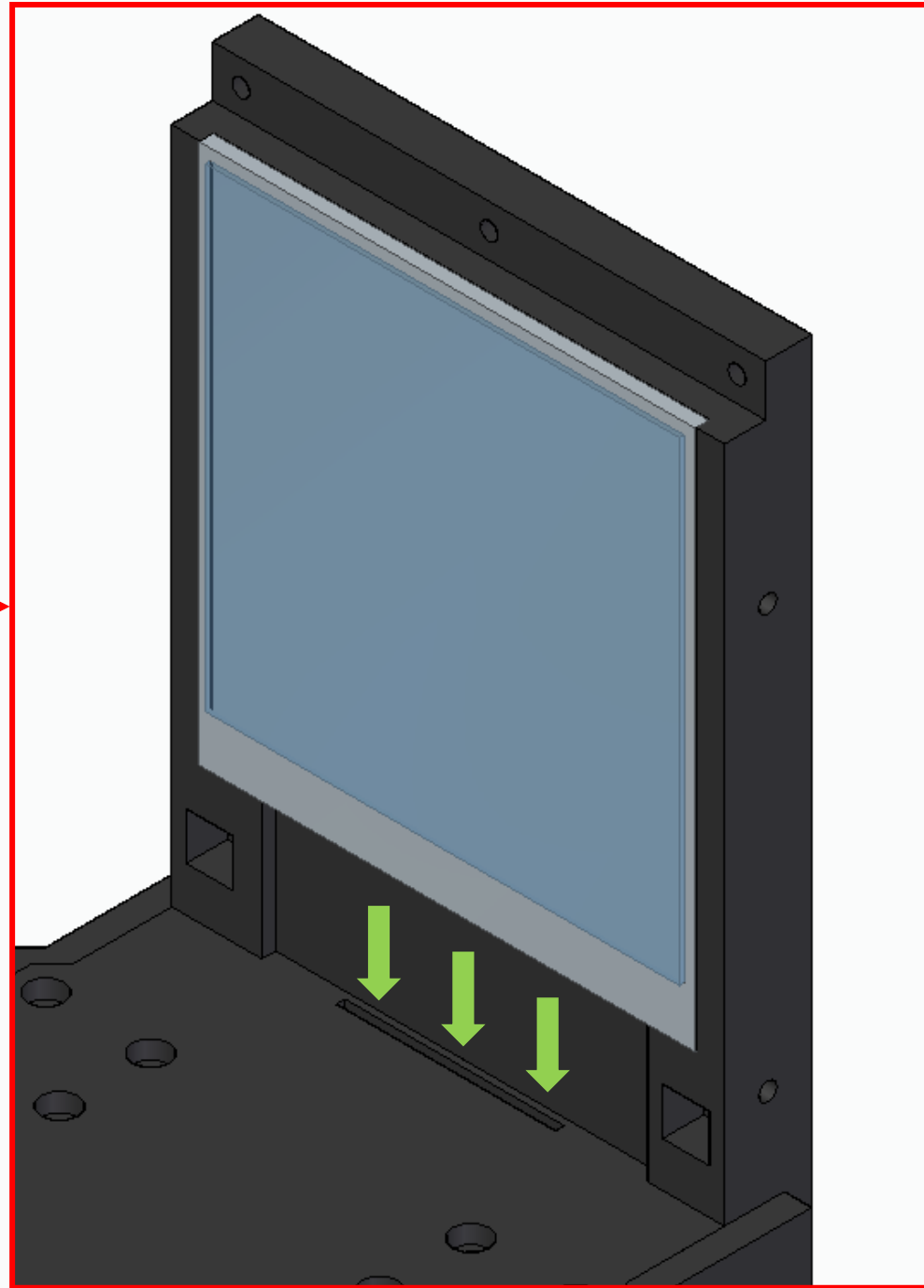
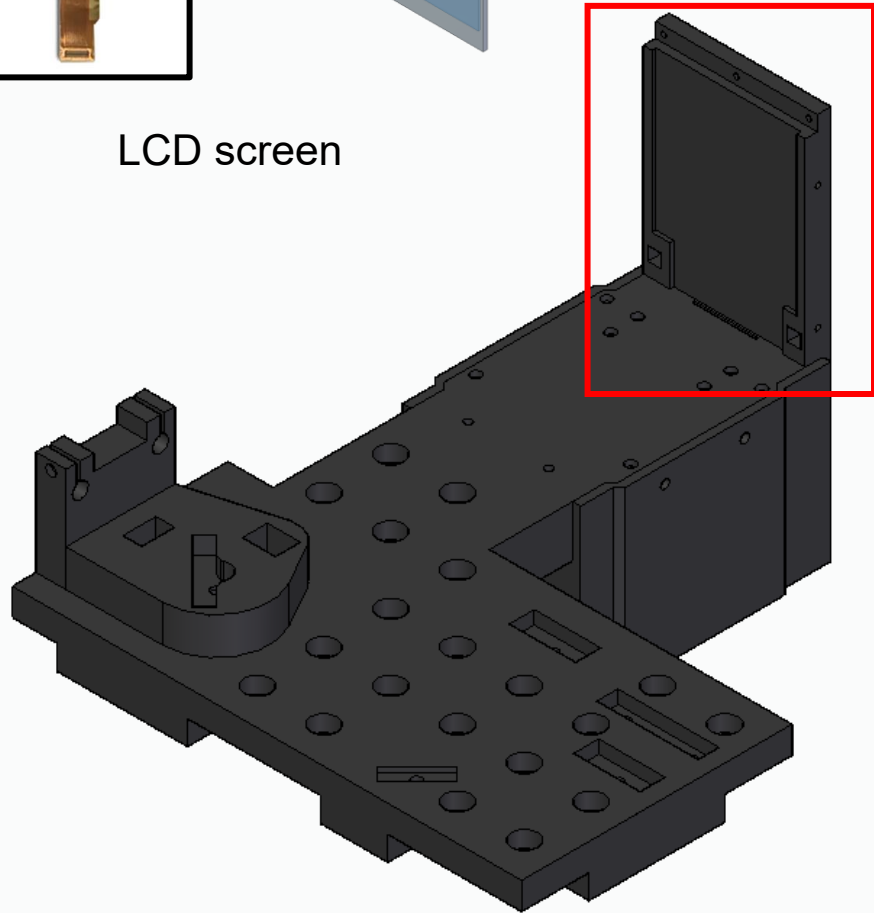


Main board

Inserting LCD screen



LCD screen



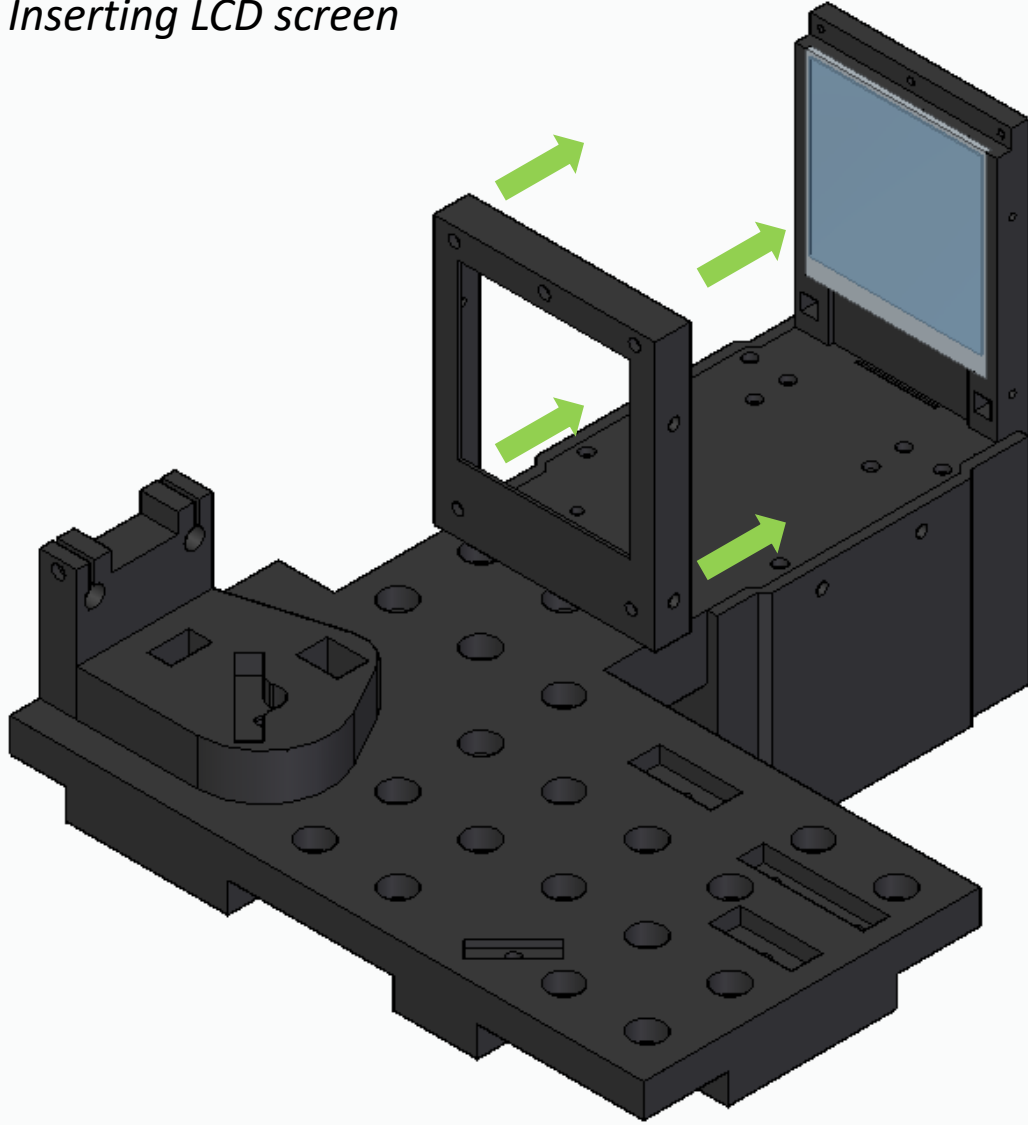
Insert the screen into designated slot.

Pass the flat flex cable (not visualized in the magnified image) through the designated slot as indicated by green arrows.

Connect the cable to the green PCB board.

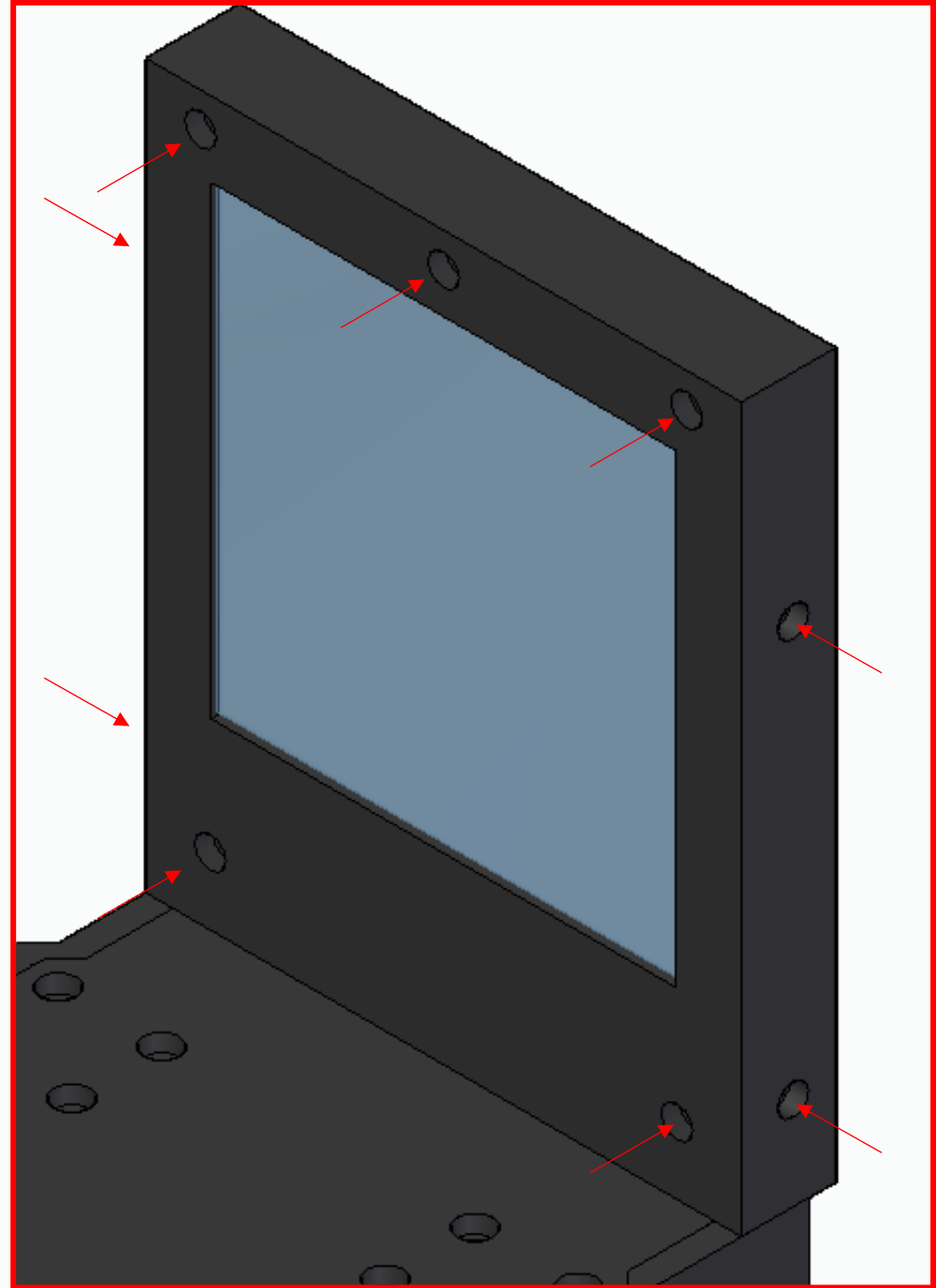
Main board

Inserting LCD screen



Needs to be 3D-printed:

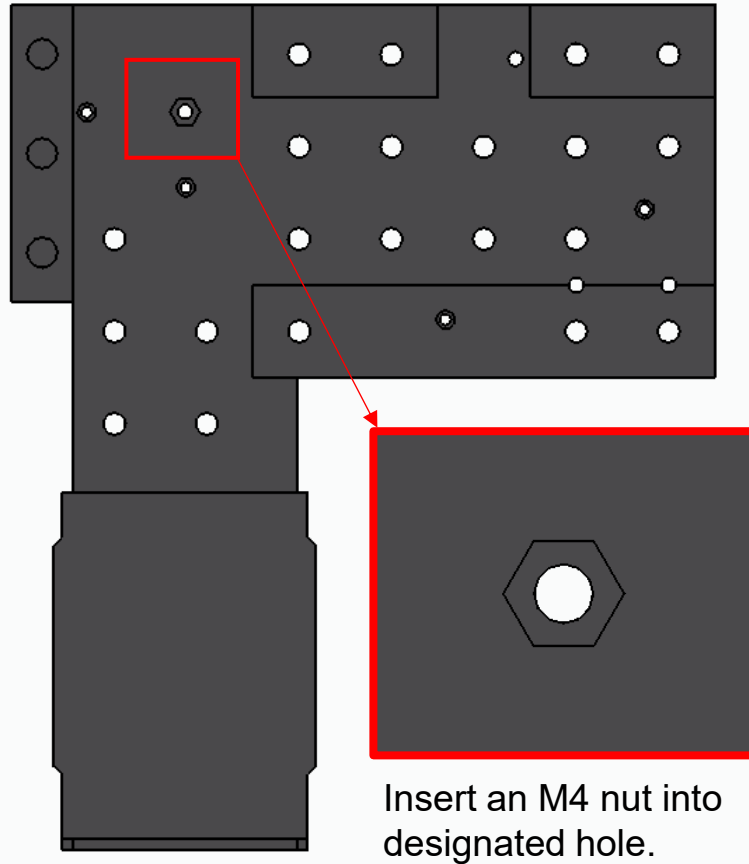
1 × 05_LCD_screen_front_panel



Secure the front panel to the main board with 12 × M2 cap screws.

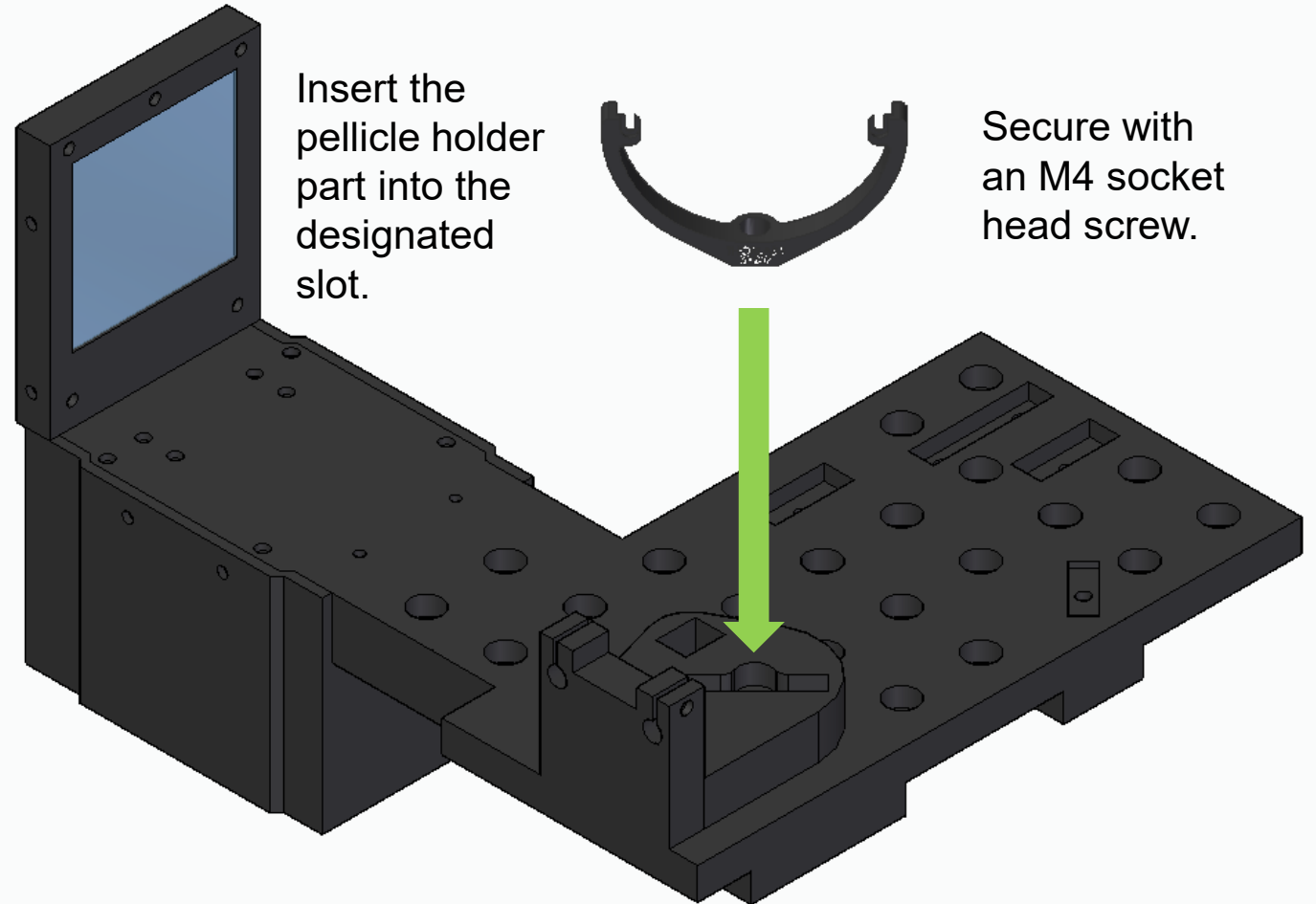
Main board

(a)



Bottom view of the main board

(b)



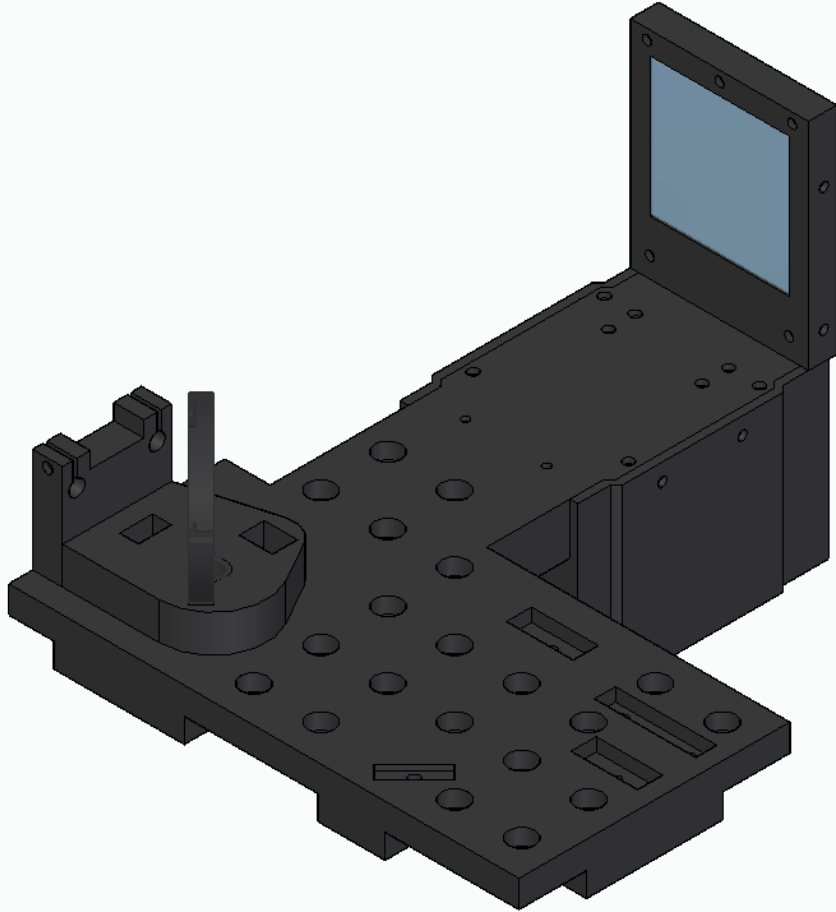
Off-the-shelf components:

1 x Thorlabs Mounting Fork for Ø2" Pellicle Beamsplitters ([BP207](#))

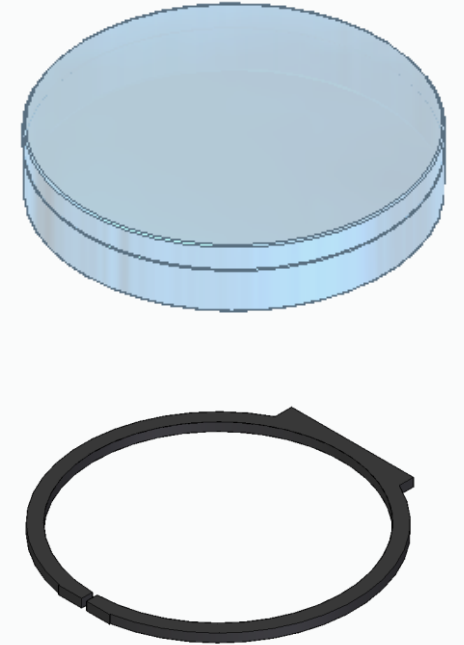
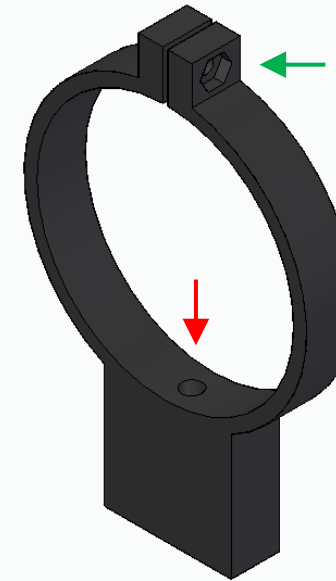
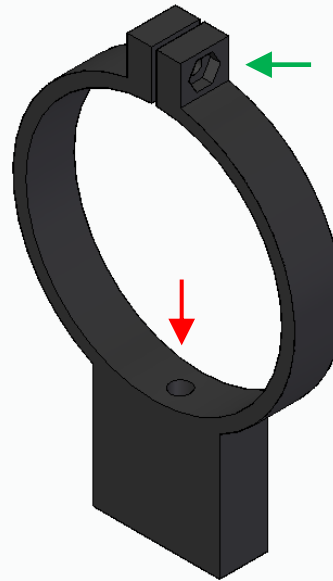
Main board

Off-the-shelf components:

2 x Achromatic Doublet ([AC508-150-A](#))



Insert M3 nuts into the designated slots indicated by the green arrows.



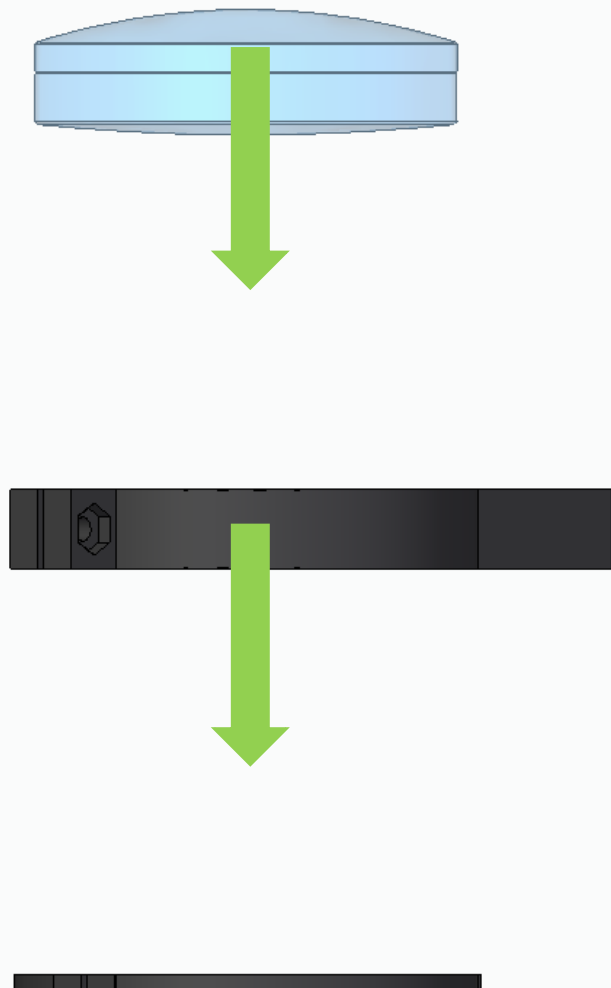
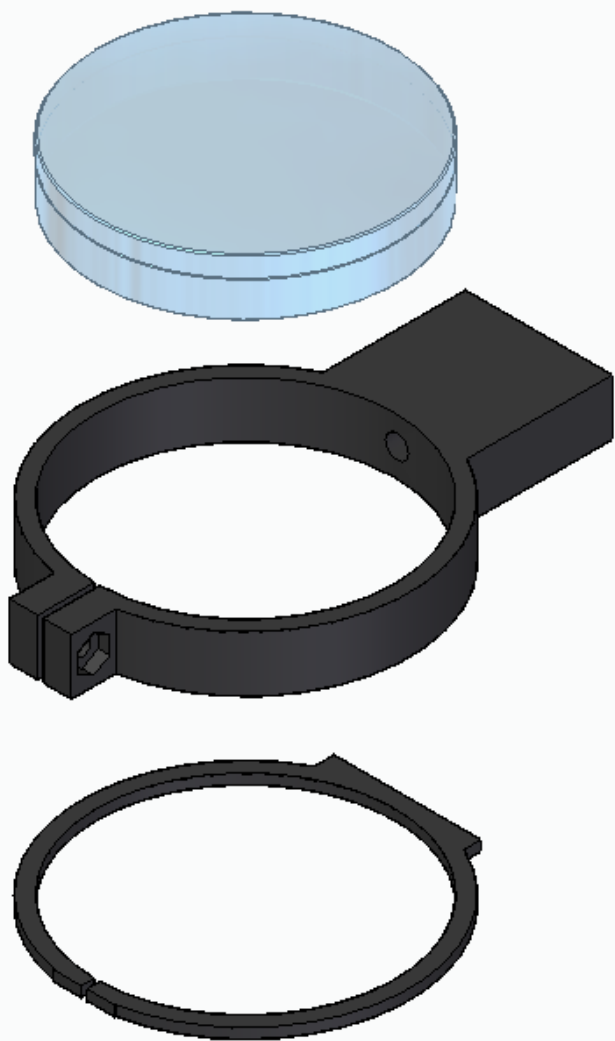
Mold M3 threaded inserts into the designated holes indicated by the red arrows.

Needs to be 3D-printed:

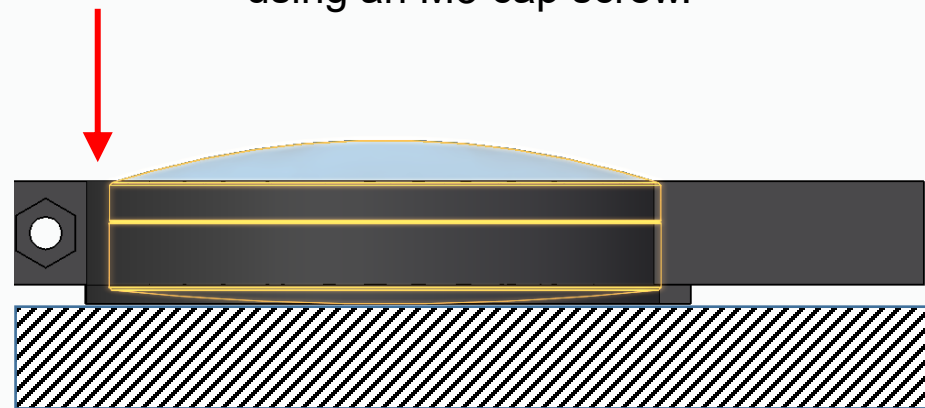
2 × 06_two_inch_lens_clamp_holder

1 × 2in_lens_clamp_holder_mounting_tool (optional)

Main board

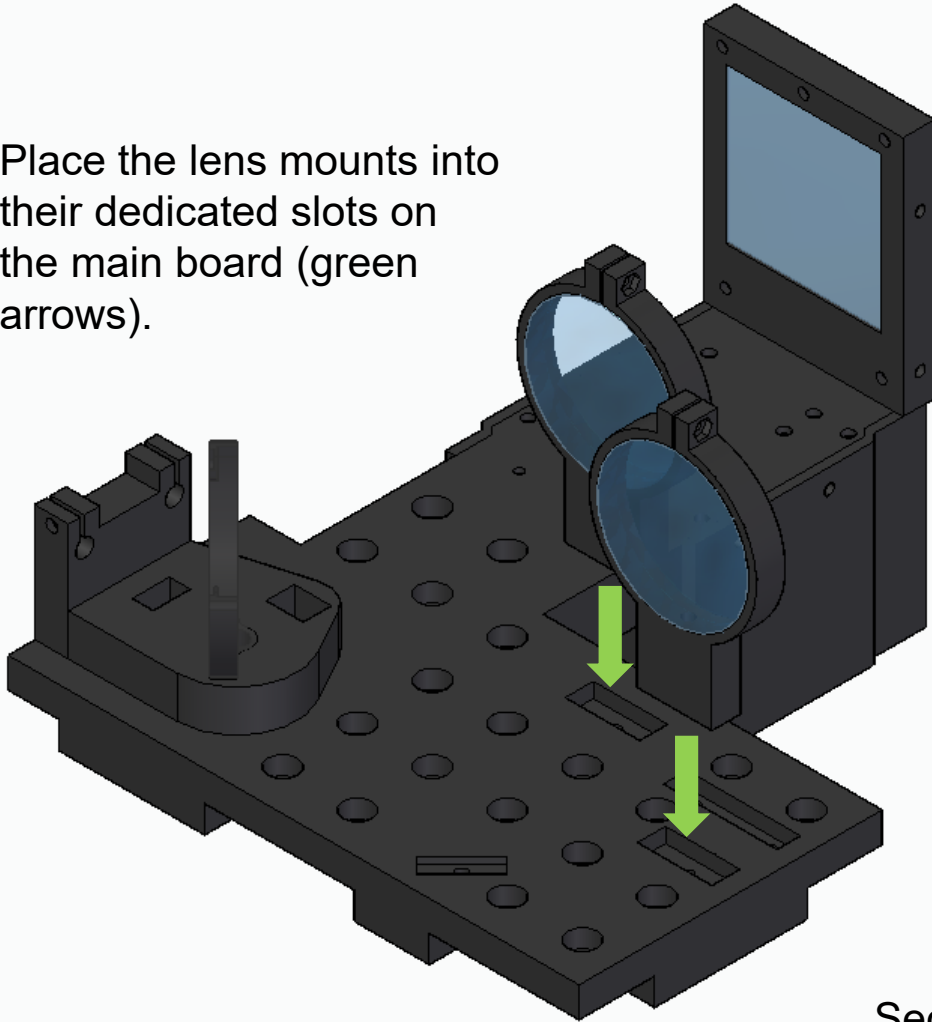


Once in the proper position, secure the lens using an M3 cap screw.

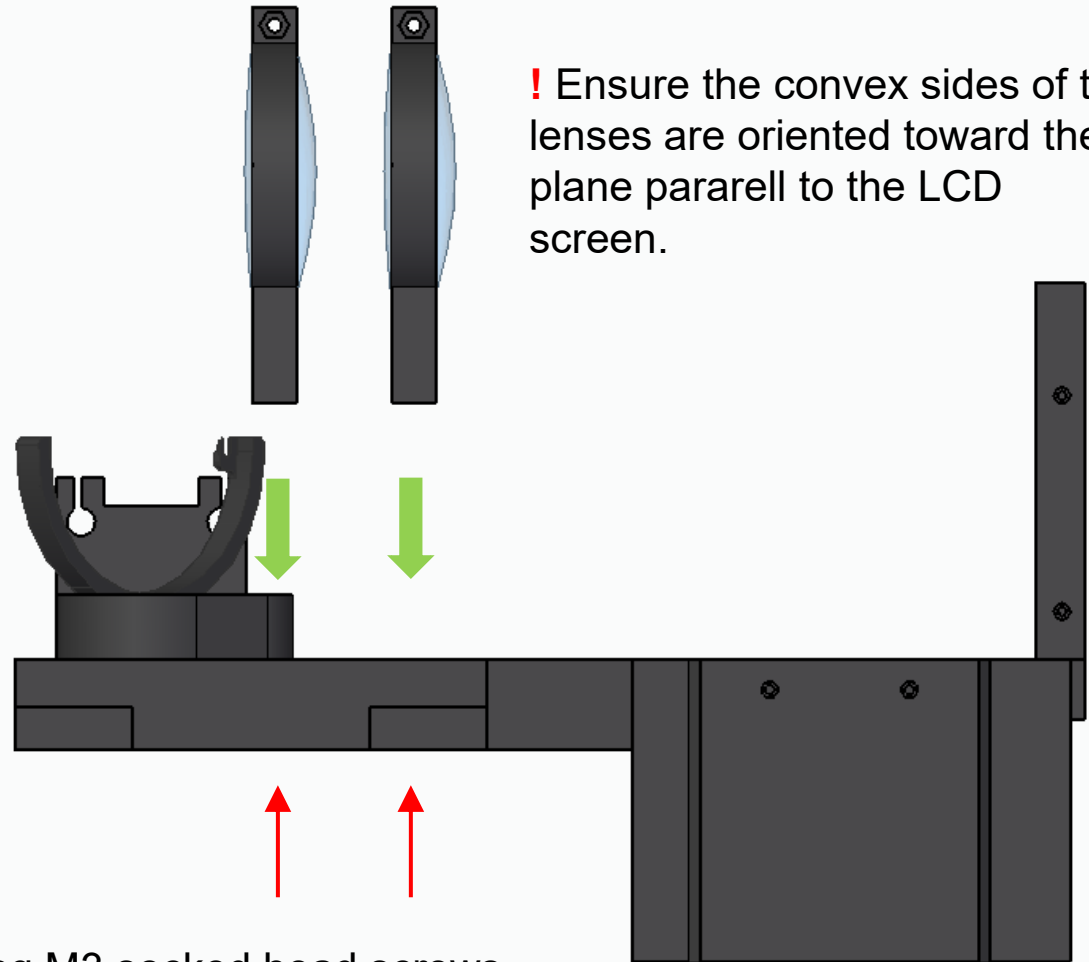


Main board

Place the lens mounts into their dedicated slots on the main board (green arrows).



! Ensure the convex sides of the lenses are oriented toward the plane parallel to the LCD screen.

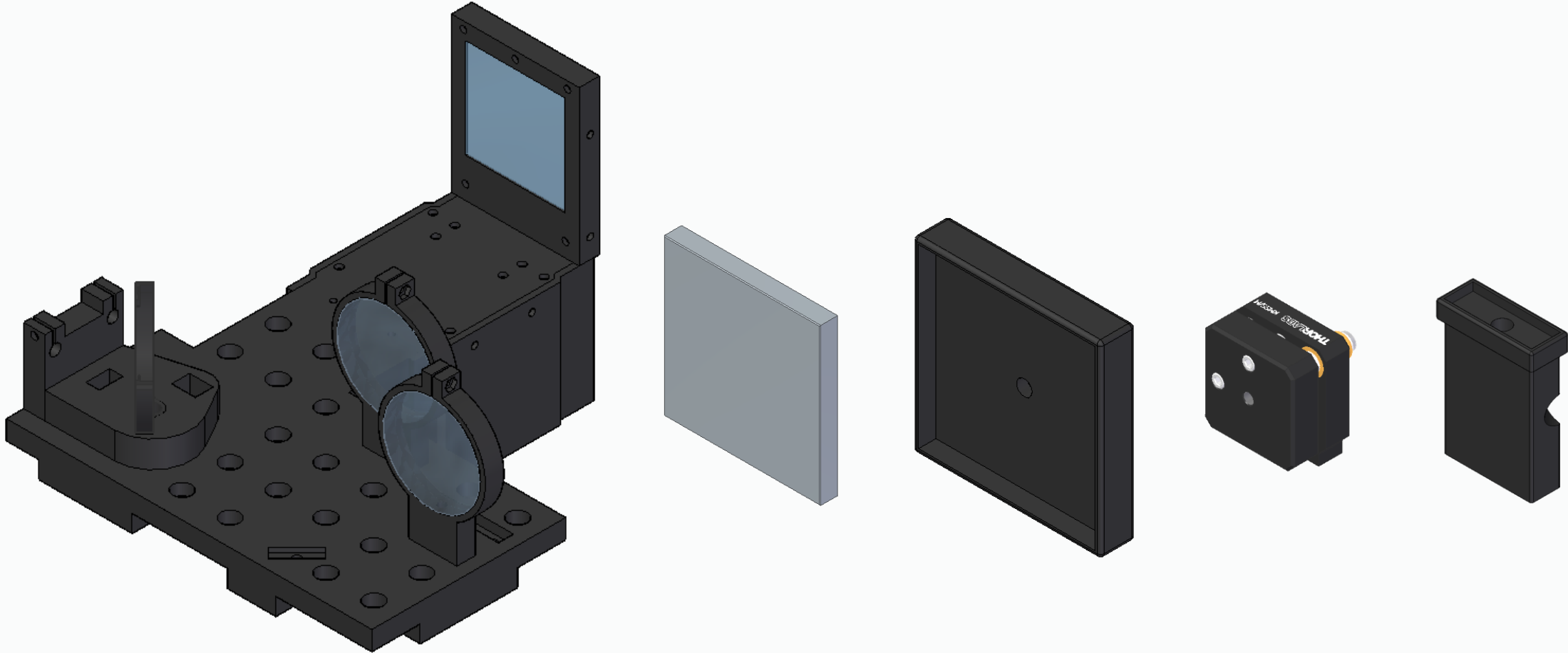


Secure using M3 socket head screws inserted from the bottom side. Ensure the screws are **not too long** to avoid drilling into the lens itself.

Main board

Off-the-shelf components needed

- 1 × Thorlabs 2" x 2" Protected Silver Mirror ([PFSQ20-03-P01](#))
- 1 × Thorlabs Compact Kinematic Mirror Mount ([KMSS/M](#))

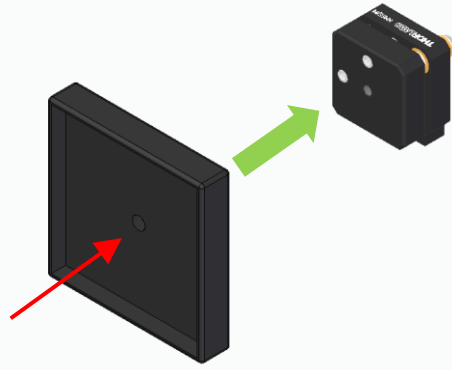


Needs to be 3D-printed:

- 1 × 07_square_mirror_holder_small
- 1 × 08_square_mirror_support_with_6mm_rod_cutout

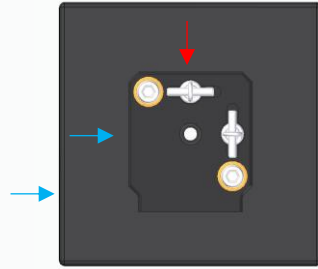
Main board

1.

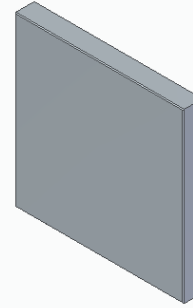


Secure using 4-40 socket head screw.

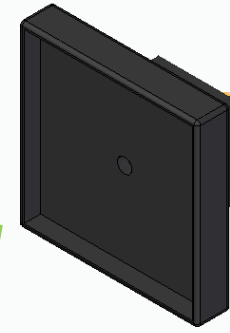
Ensure the corresponding surfaces are as parallel as possible.



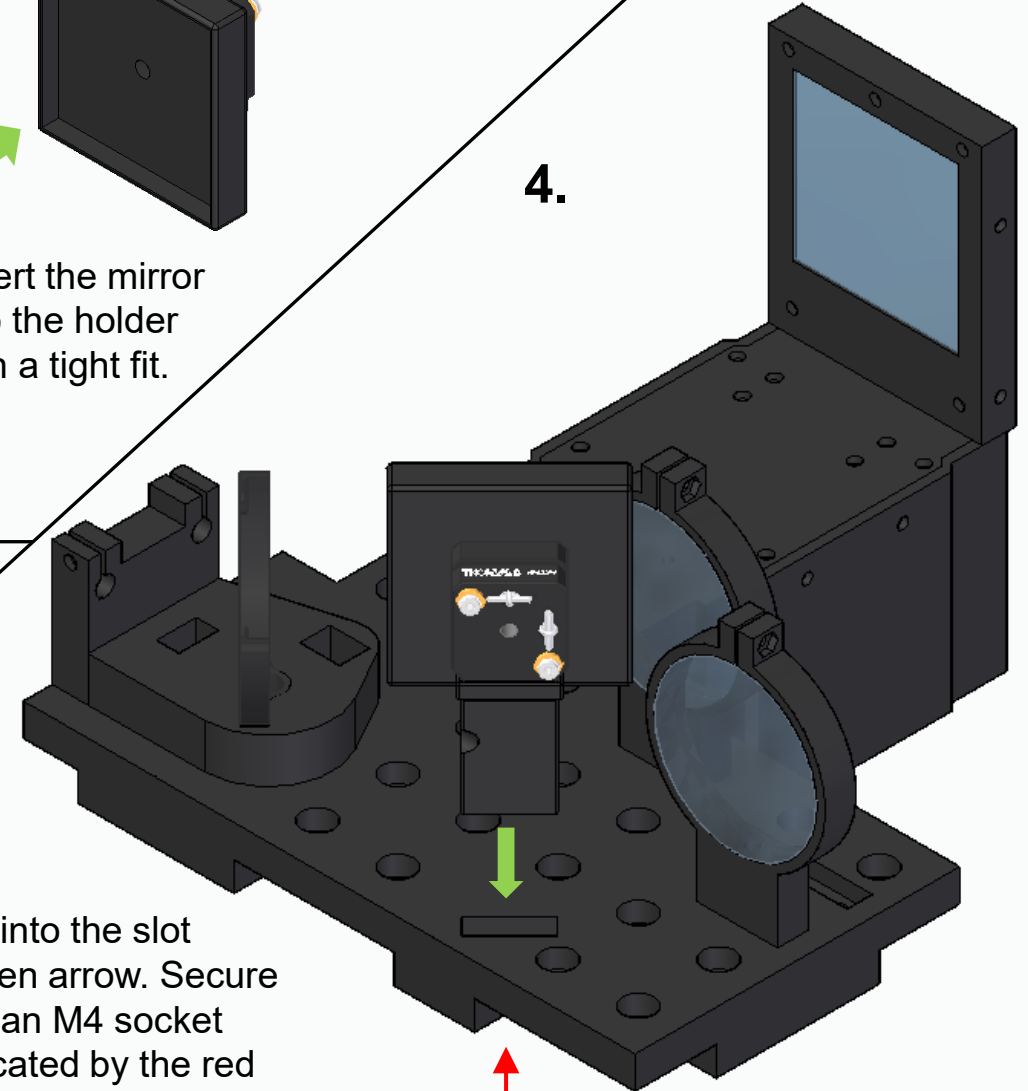
2.



Insert the mirror into the holder with a tight fit.

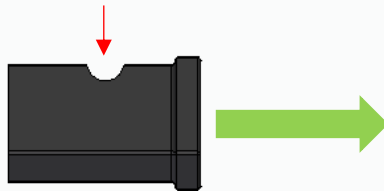


4.

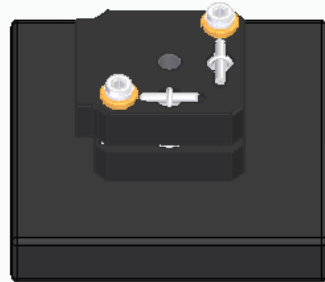


3.

Ensure the position of the cutout is as indicated.

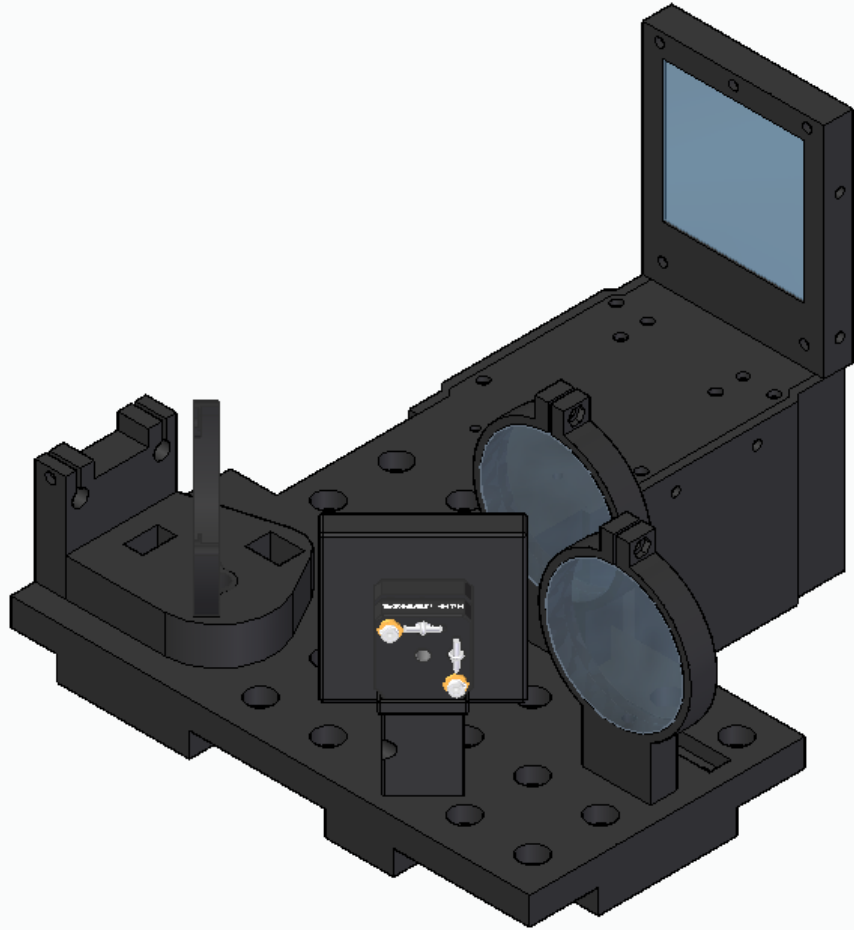


Slide the kinematic mount into the post holder as tight as possible. Be careful as it may fall off – final fixing is with the M4 cap screw in the next step.



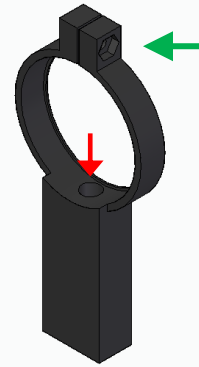
Fit the support post into the slot indicated by the green arrow. Secure the assembly using an M4 socket head screw, as indicated by the red arrow. The screw should be long enough to go through the main board and the post holder and reach the threading of the kinematic mount.

Main board

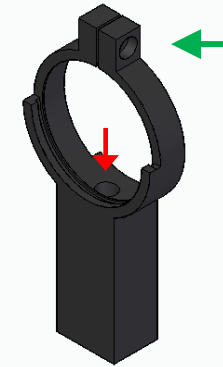


Off-the-shelf components needed

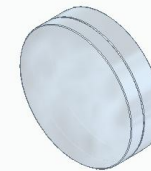
- 1 × Pellicle Beamsplitter ([BP208](#))
- 1 × Ø1" Achromatic Doublet ([AC254-075-A](#))
- 1 × Iris diaphragm ([CP20D](#))
- 1 × Ø1" Bi-Convex Lens ([LB1294-A](#))



Mold M3 threaded inserts into the designated holes indicated by the red arrows.



Insert M2 nuts into the designated slots indicated by the green arrows.

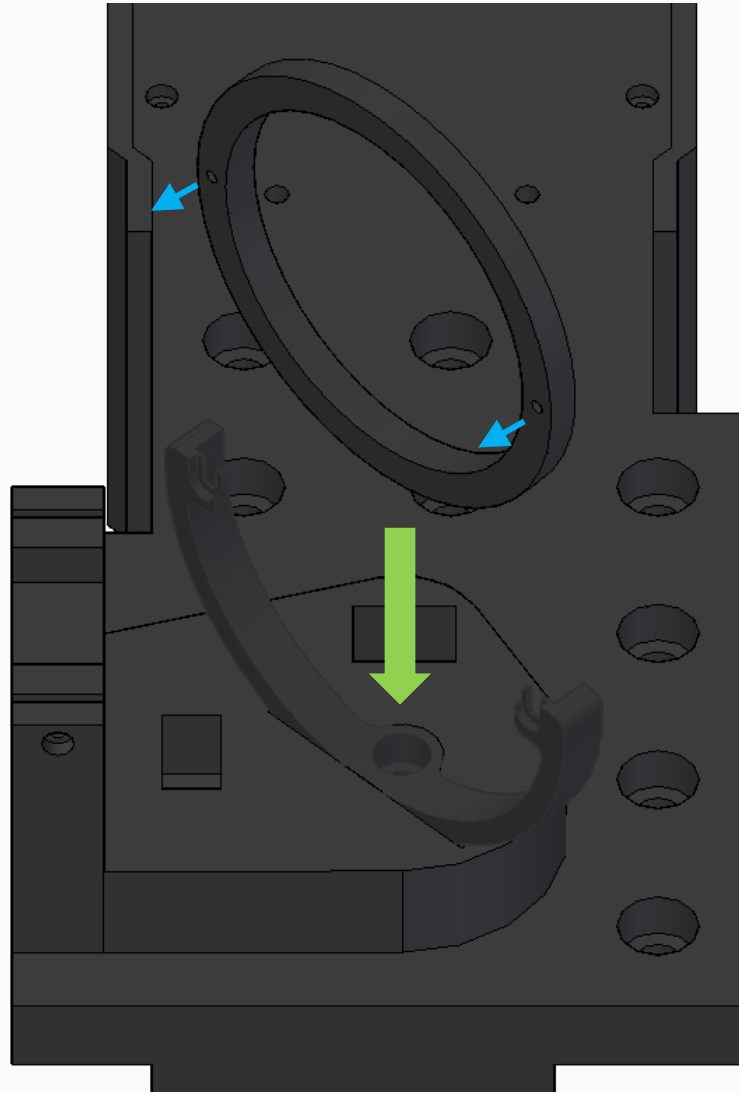


Needs to be 3D-printed:

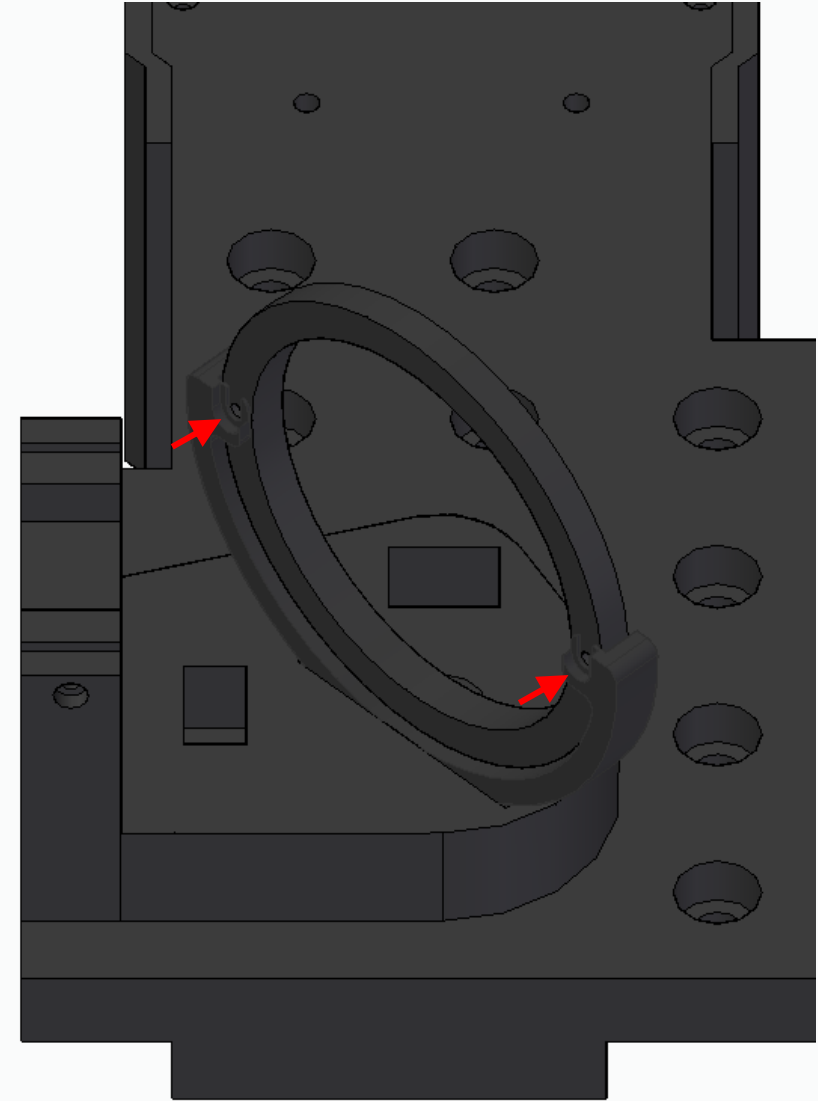
- 1 × 09_one_inch_lens_clamp_holder_stim
- 1 × 10_one_inch_lens_clamp_holder_fix
- 2 × 11_iris_mounting_rod

Main board

Installing pellicle BS



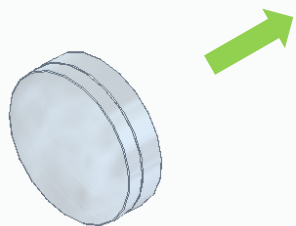
Carefully insert the pellicle into the mounting fork.
Head screws should be facing the direction indicated
by the blue arrows and fitting the clearance slots



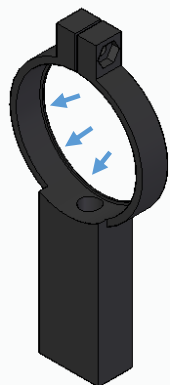
Secure with #2-56 button head screws as
indicated by the red arrows.

1.

Insert until the rim
indicated by the
blue arrows.



Convex side



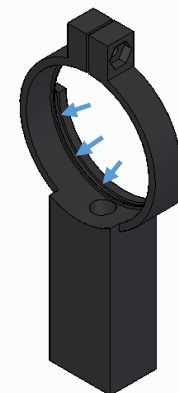
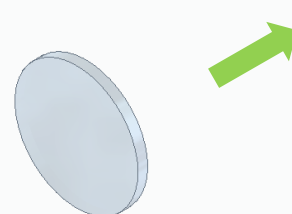
2.

Tighten with
an M2 socket
head cap
screw.



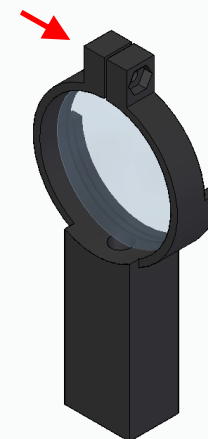
3.

Insert until the
rim indicated by
the blue arrows.



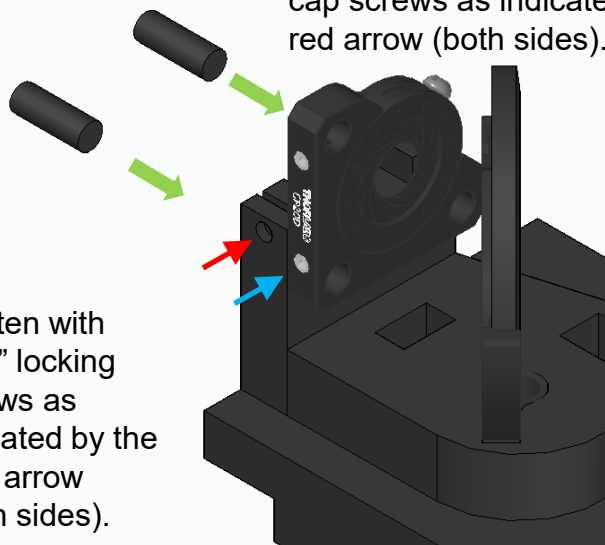
4.

Tighten with
an M2 socket
head cap
screw.



5.

Tighten with M2 socket head
cap screws as indicated by the
red arrow (both sides).



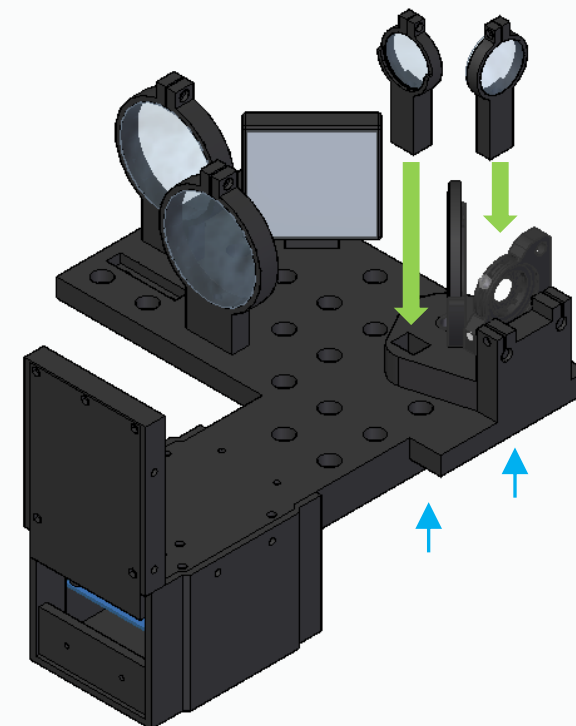
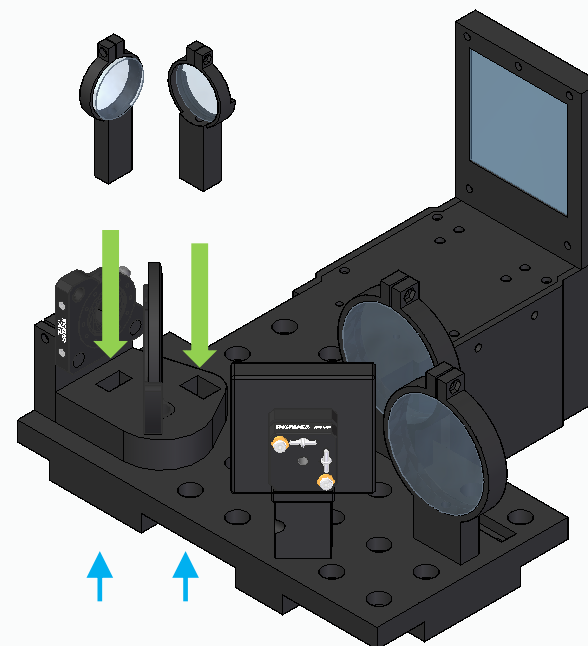
Tighten with
5/64" locking
screws as
indicated by the
blue arrow
(both sides).

Alternatively, you can use 6 mm rod off-cuts from Badal stage or 1" cage rods here (will protrude slightly as the optimal length is 17 mm).

6.

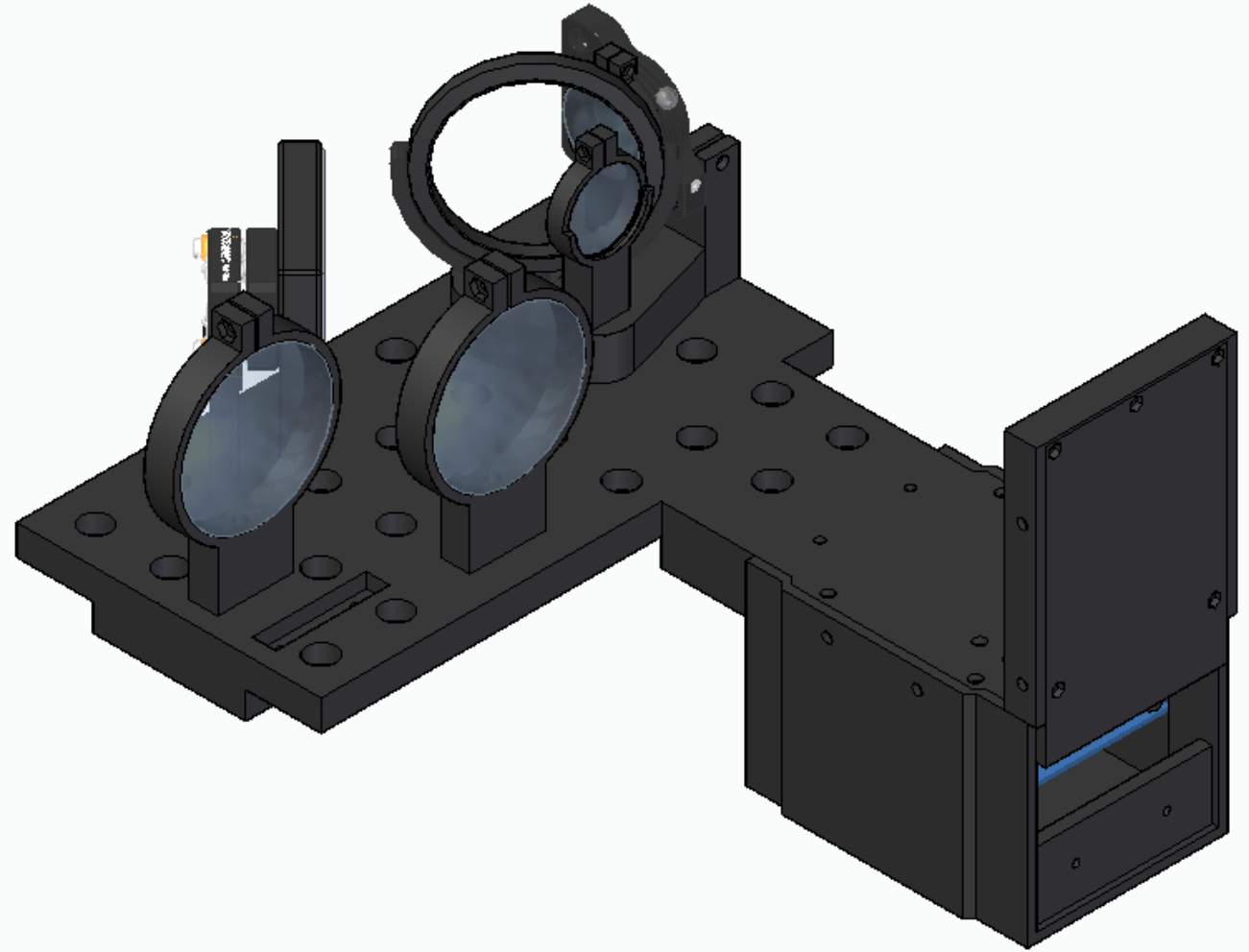
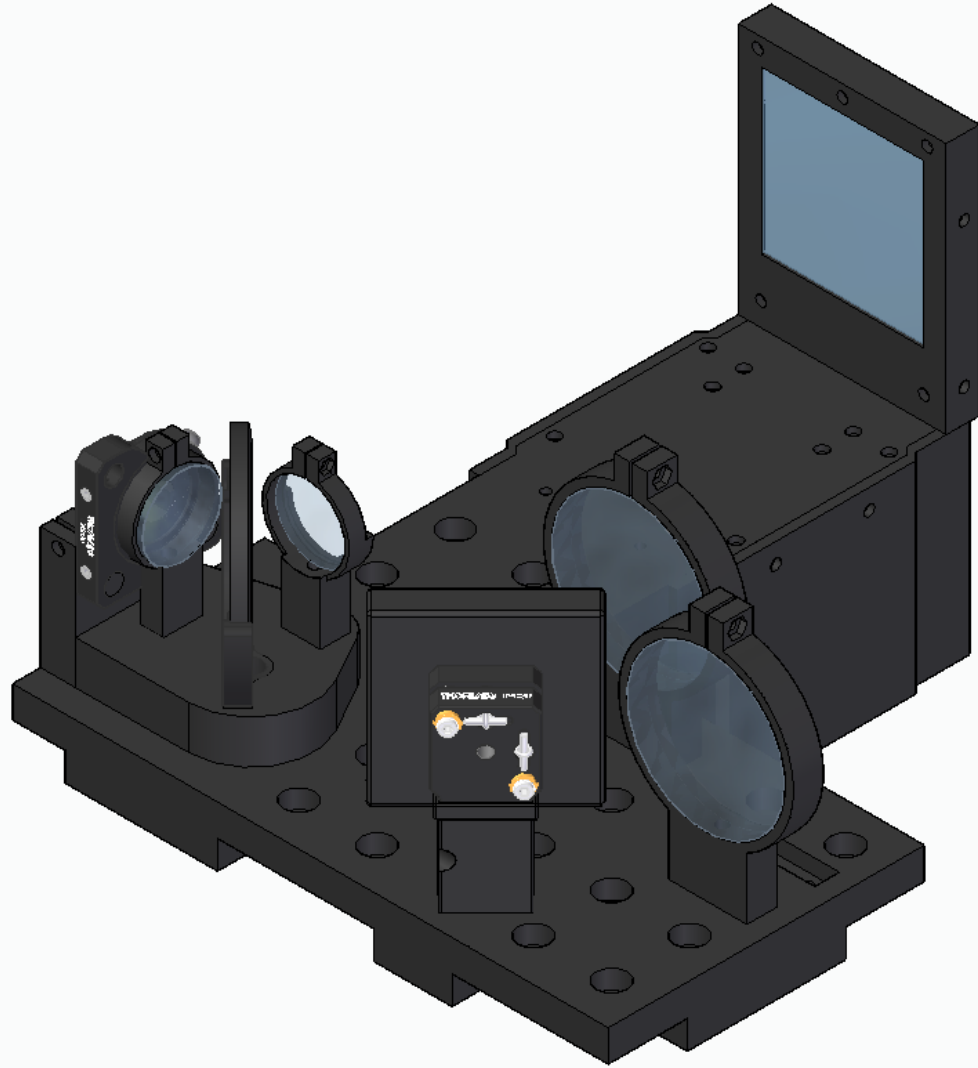
Insert both holders
tightly into
dedicated slots,
paying attention to
the orientation of
the holder.

Secure with M3
cap screws
inserted from the
bottom as
indicated by the
blue arrows.



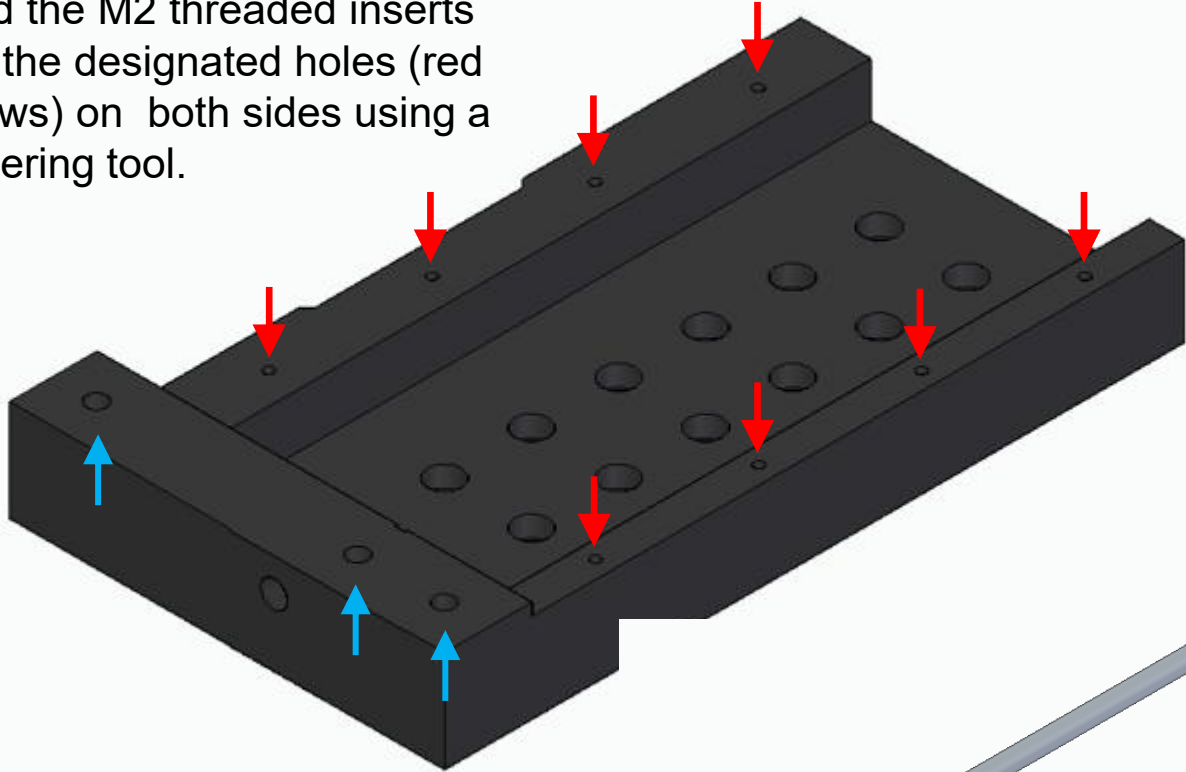
Main board

Complete!



Badal stage

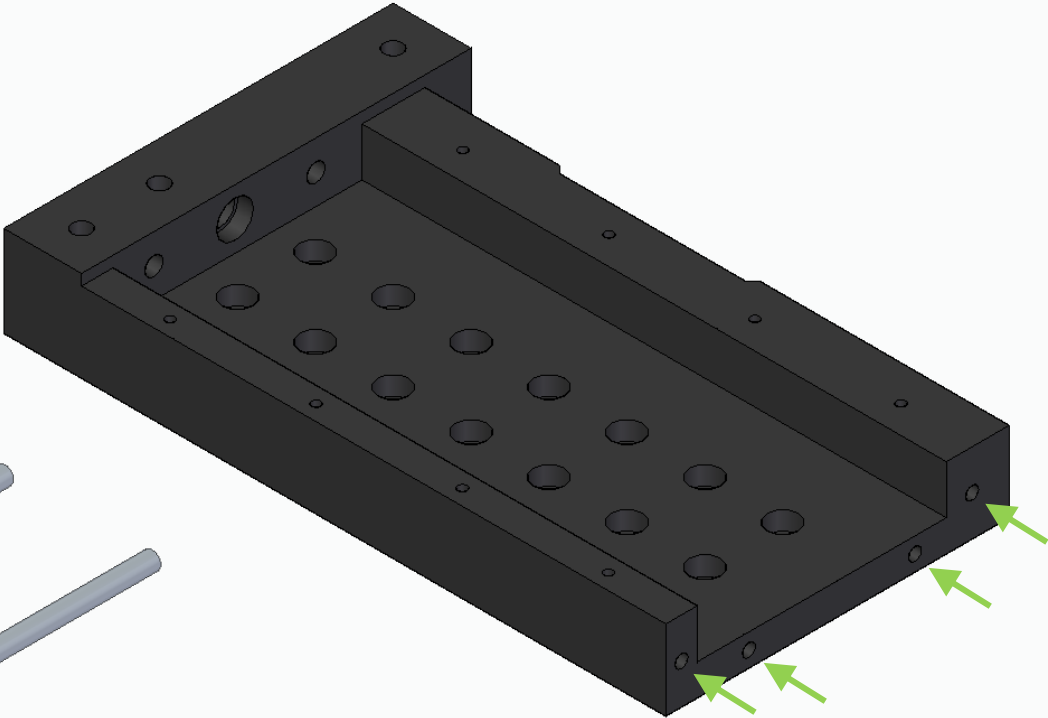
Mold the M2 threaded inserts into the designated holes (red arrows) on both sides using a soldering tool.



Mold the M6 threaded inserts into the designated holes (blue arrows) using a soldering tool.

Off-the-shelf components needed

2 × sliding rod (Ø6mm, L = 200 mm)



Mold the M3 threaded inserts into the designated holes (green arrows) using a soldering tool.

Needs to be 3D-printed:

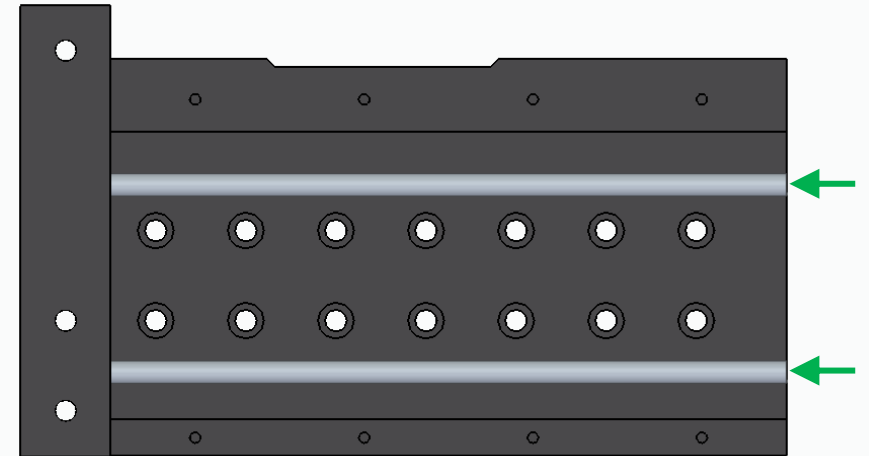
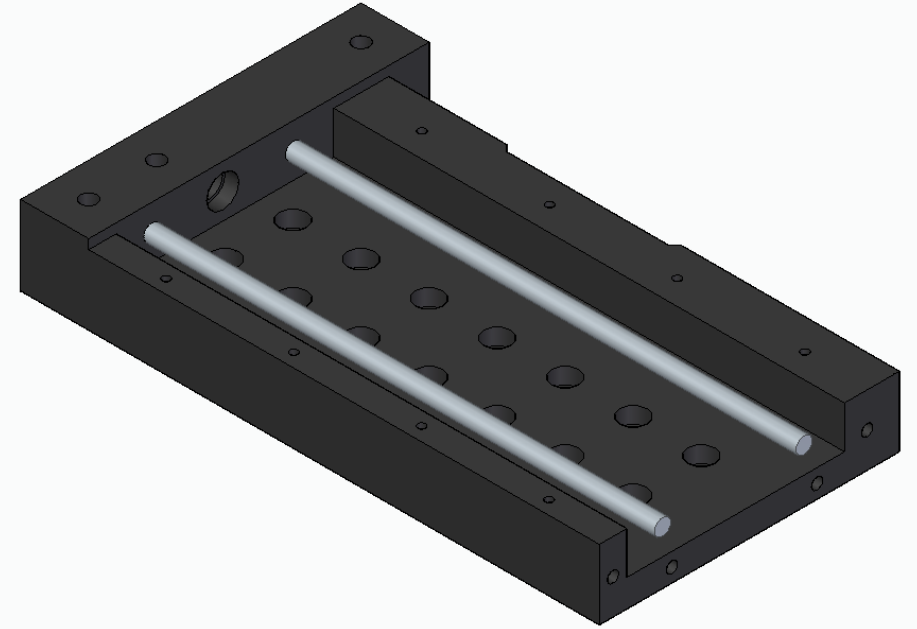
1 × 12_badal_stage_bottom



Badal stage

Translation stage base

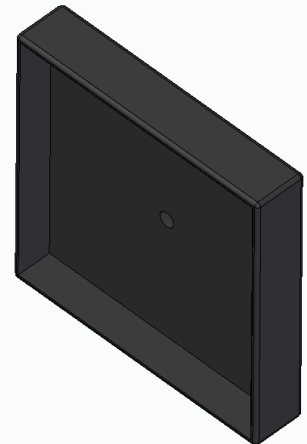
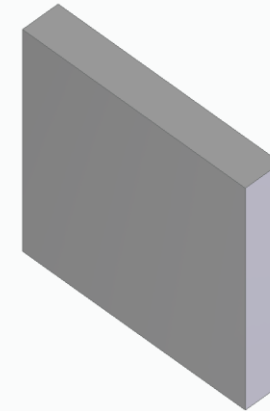
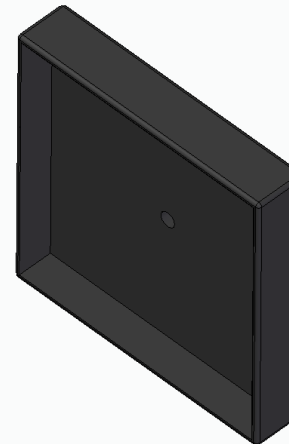
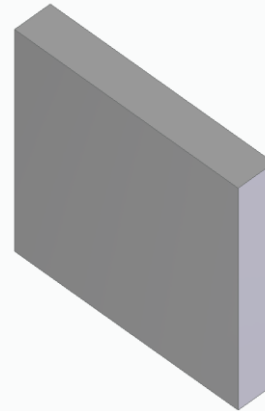
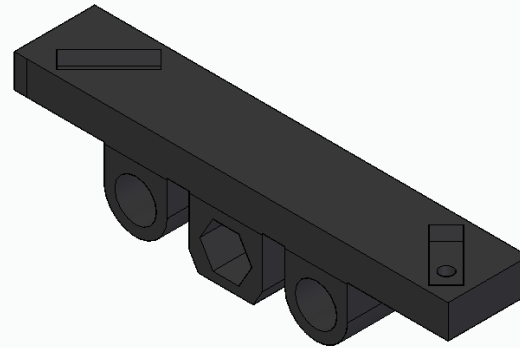
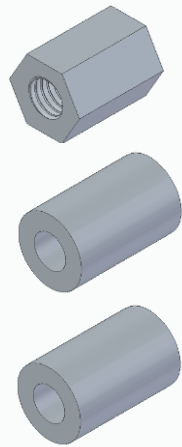
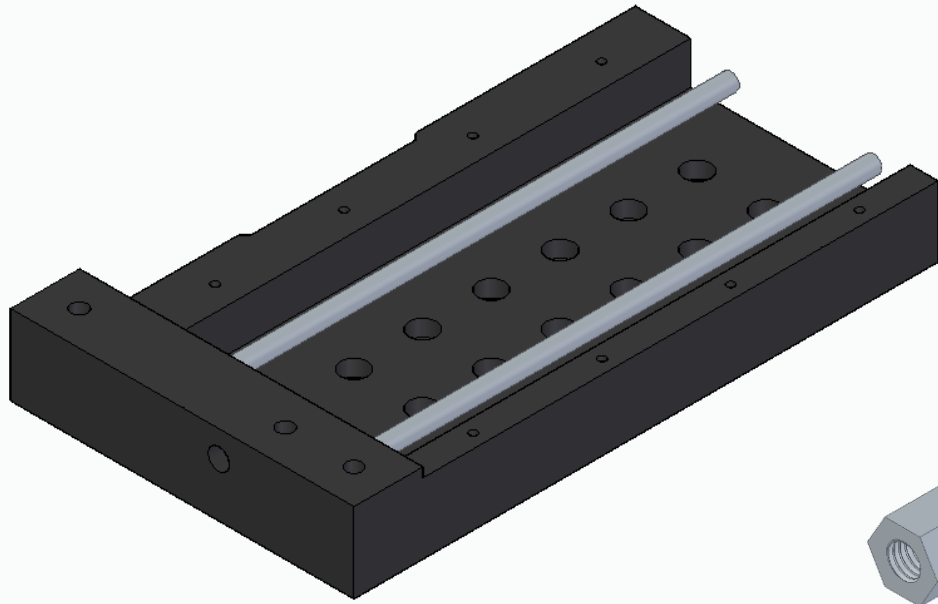
Slide the rods into the designated slots for a tight fit.



Ensure the end tips of the rods are flush with the encasing.

Badal stage

Translation stage base



Off-the-shelf components needed

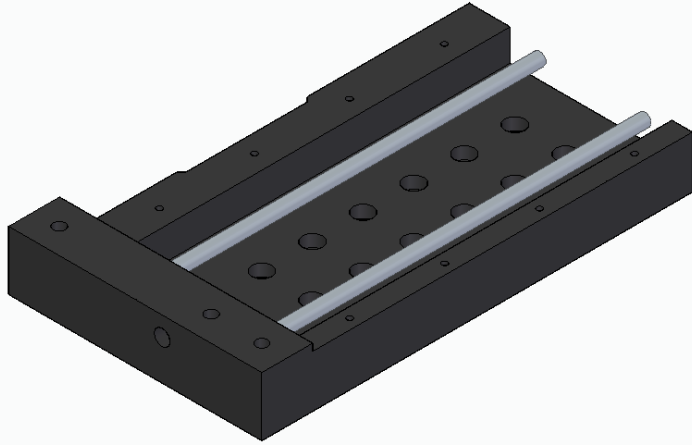
- 2** × Kinematic mirror mount ([KMSS/M](#))
- 2** × Square silver mirror ([#88-541](#))
- 1** × M8 hex coupling nut, L = 20 mm
- 2** × linear ball bearing (Ø6mm)

Needs to be 3D-printed:

- 1** × 13_badal_stage_top
- 2** × 14_square_mirror_support_badal_stage
- 1** × 15_square_mirror_holder_large
- 1** × 16_square_mirror_holder_large_cut_corner

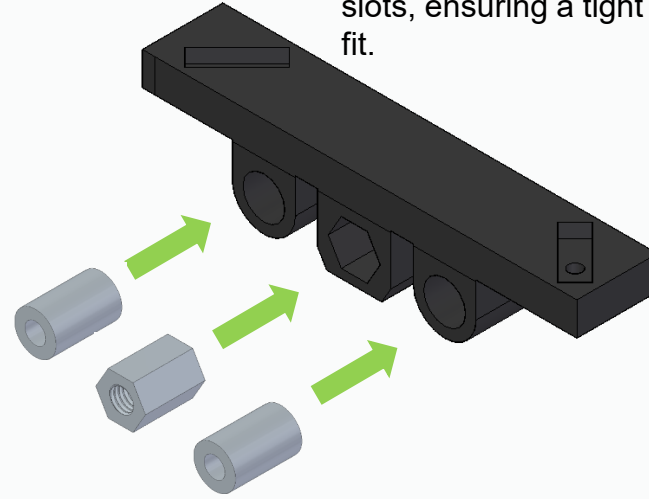
Badal stage

Mirror unit assembly



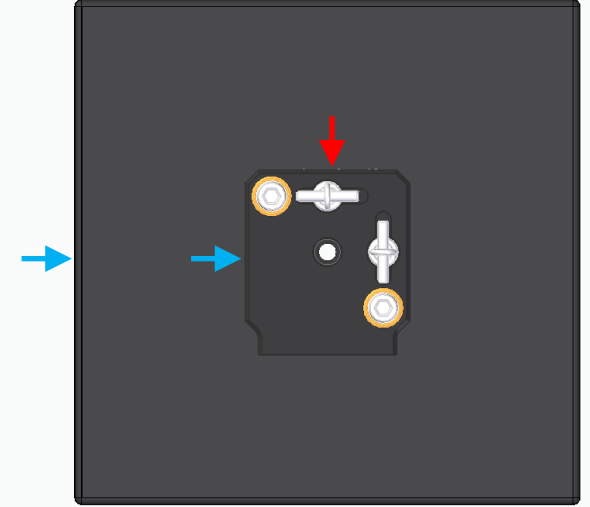
1.

Insert the ball bearings and the hex coupling nut into the designated slots, ensuring a tight fit.

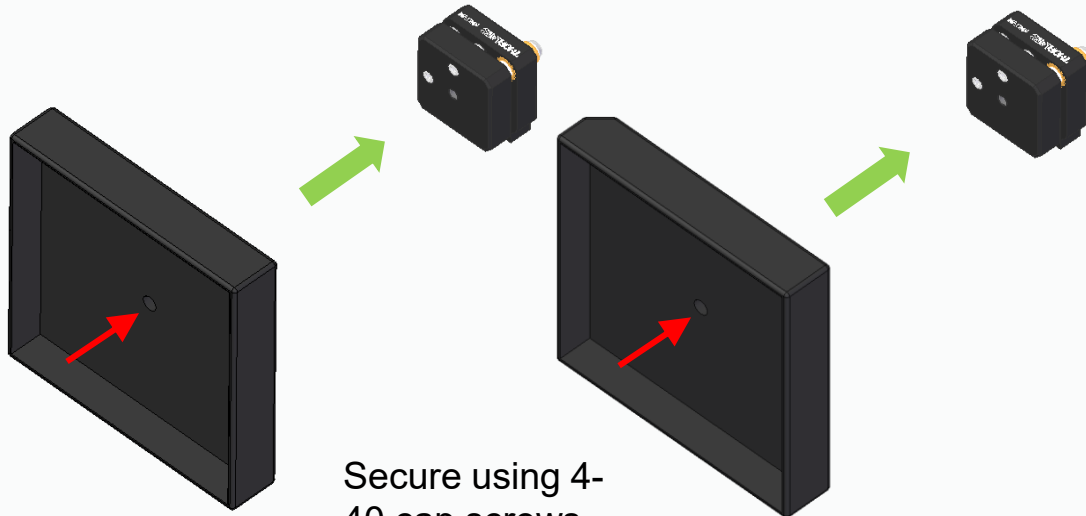


4.

Ensure the corresponding surfaces are as parallel. ↓

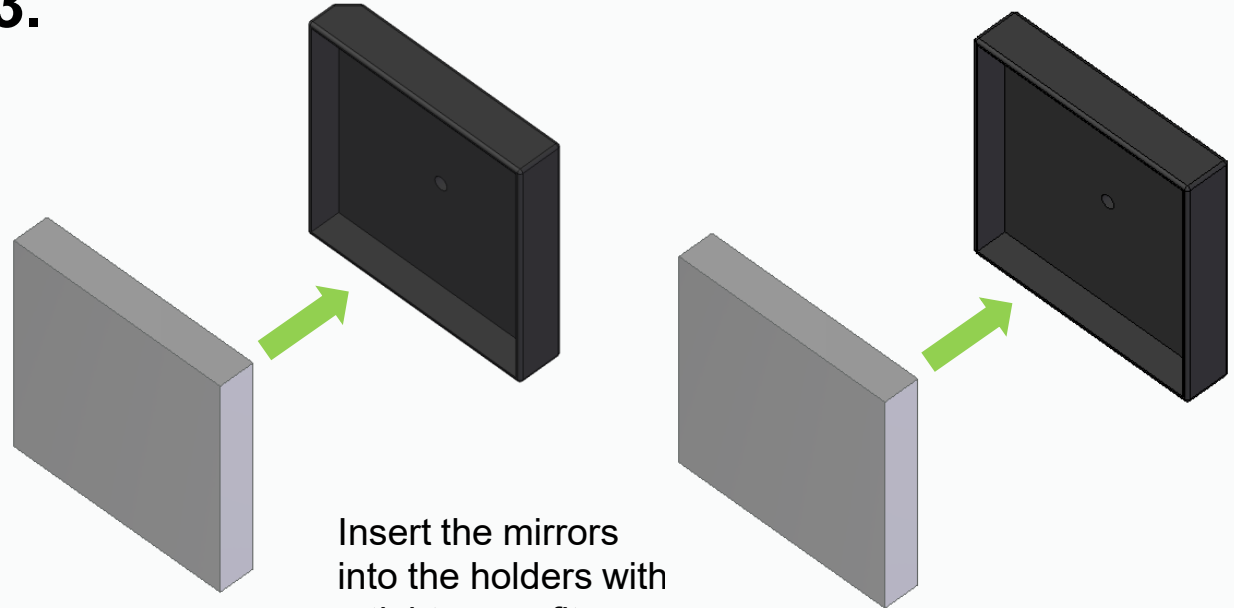


2.



Secure using 4-40 cap screws (red arrows).

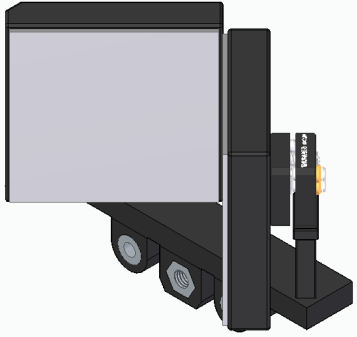
3.



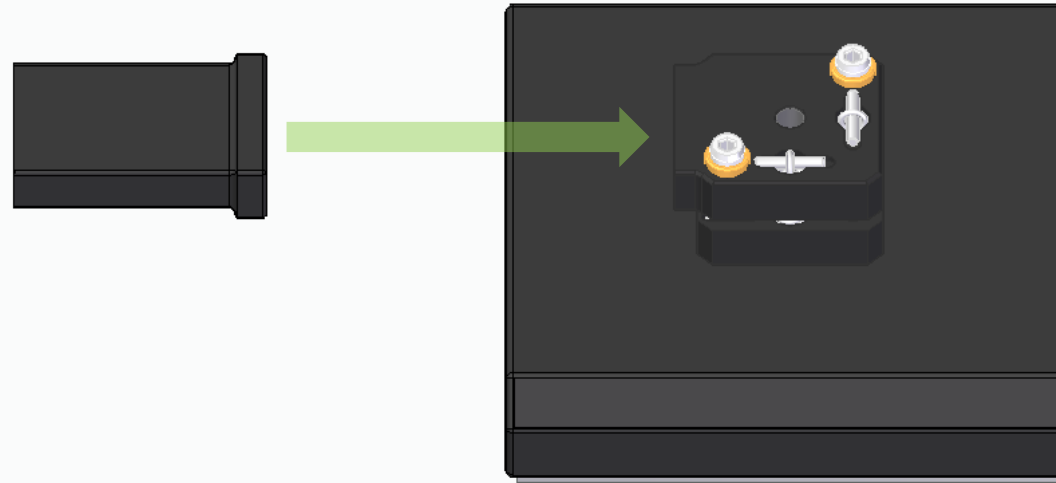
Insert the mirrors into the holders with a tight press fit.

Badal stage

Mirror unit assembly



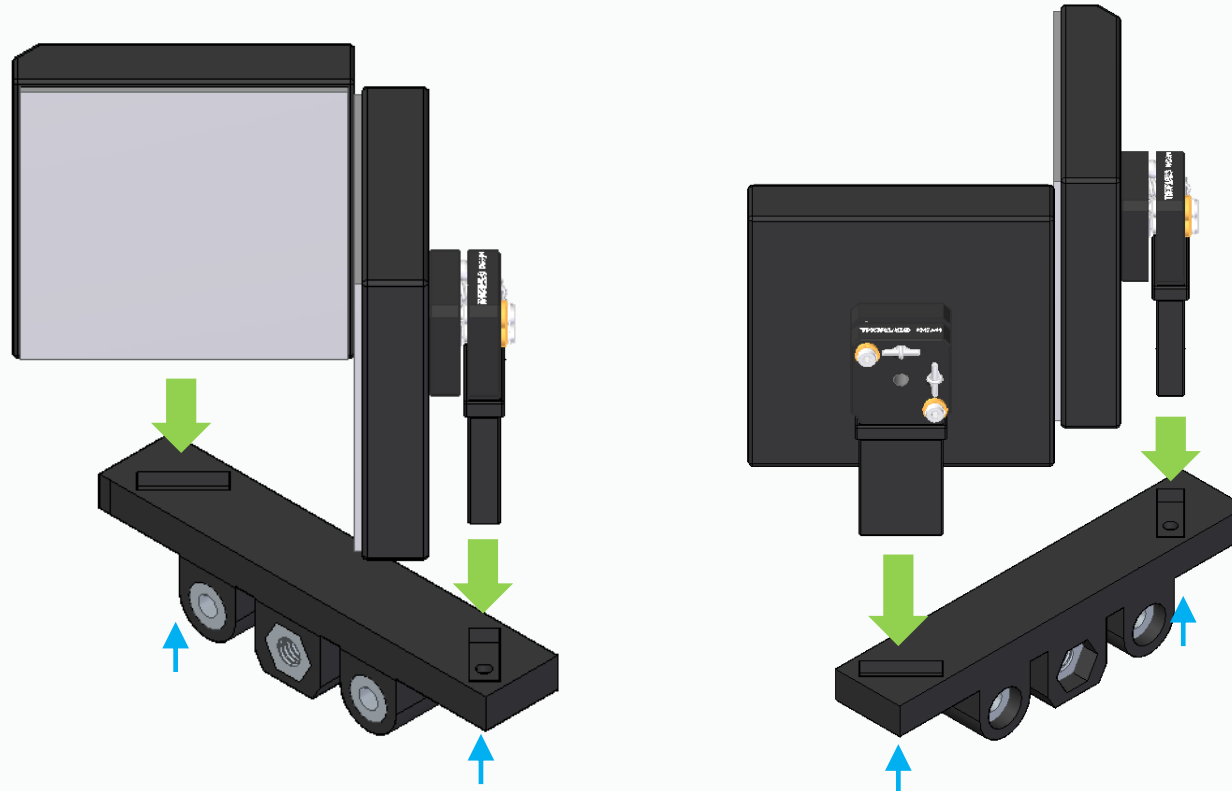
1.



Slide the kinematic mounts into the post holders as tight as possible. Be careful as it may fall off – final fixing is with the M4 cap screw in the next step.

2.

Insert both holders tightly into the designated slots, ensuring proper alignment with the slot orientation and the correct positioning of the holder with the cut corner. Secure the assembly using M4 socket head cap screws inserted from the bottom, as indicated by the blue arrows.

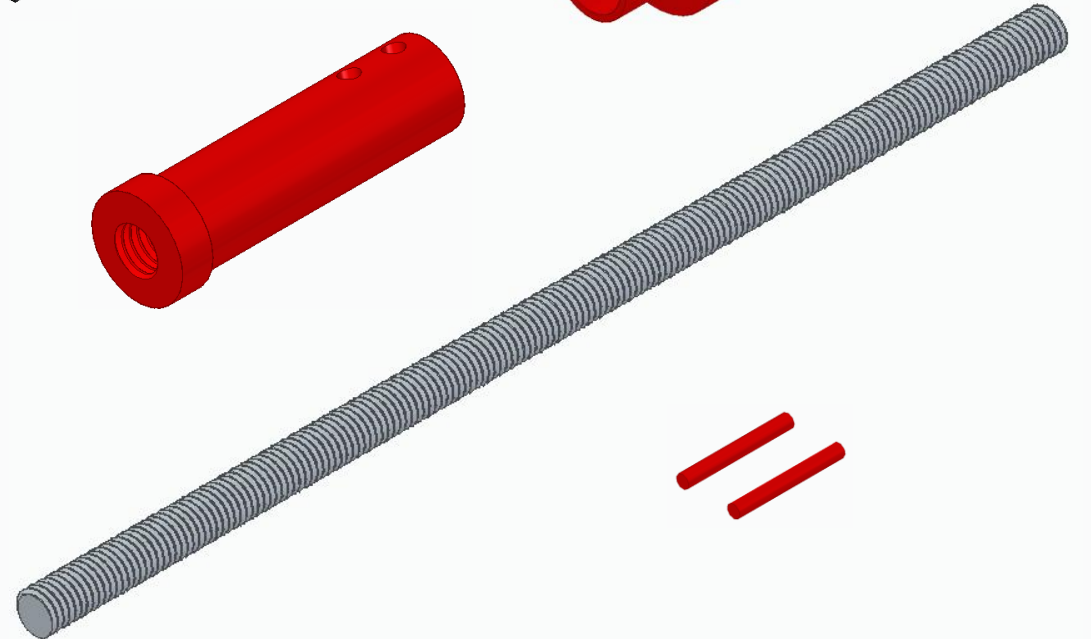
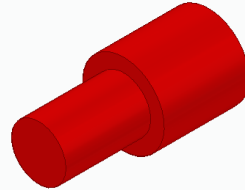
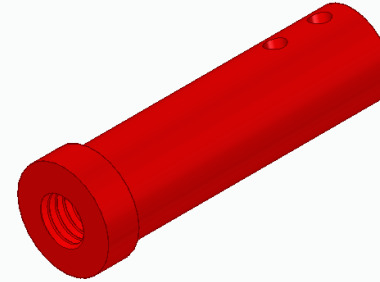
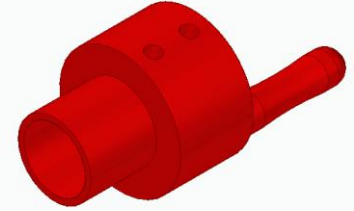
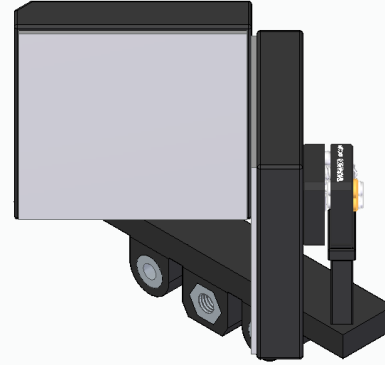
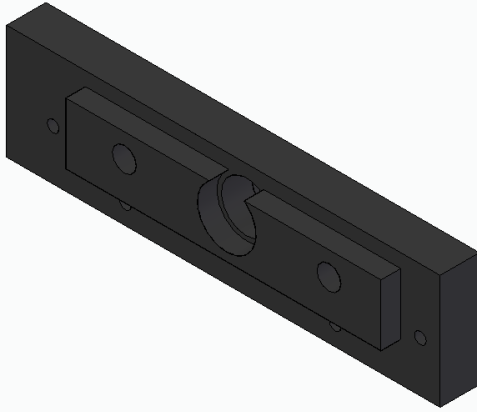
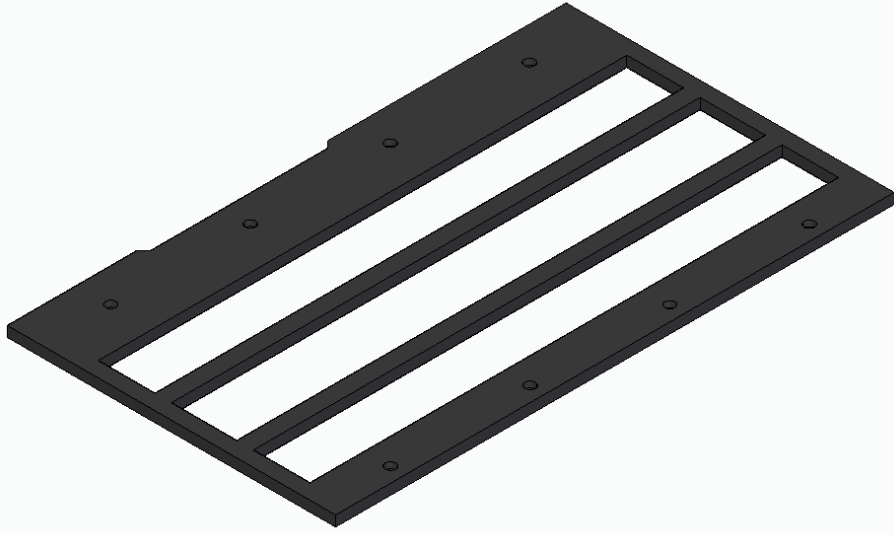


Badal stage

Movement assembly

Off-the-shelf components needed

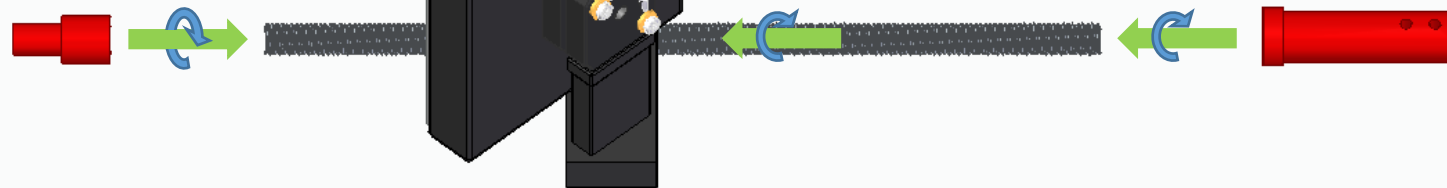
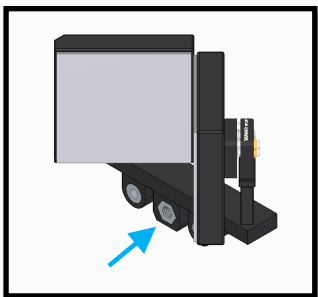
1 × M8 threaded rod, L = 215 mm



Needs to be 3D-printed:

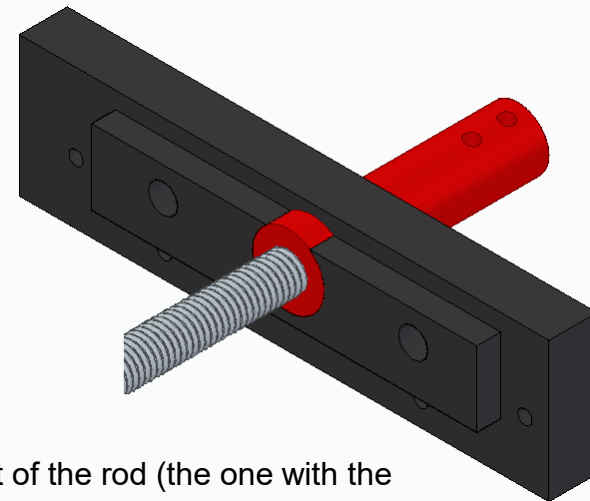
- 1 × 17_Badal_screw_8mm_end_cap
- 1 × 18_Badal_screw_8mm_front_cap
- 1 × 19_Badal_screw_8mm_rotating_handle
- 2 × 20_Badal_screw_handle_fixing_bolt
- 1 × 21_Badal_stage_cover_rear
- 1 × 22_Badal_stage_cover_top

1.



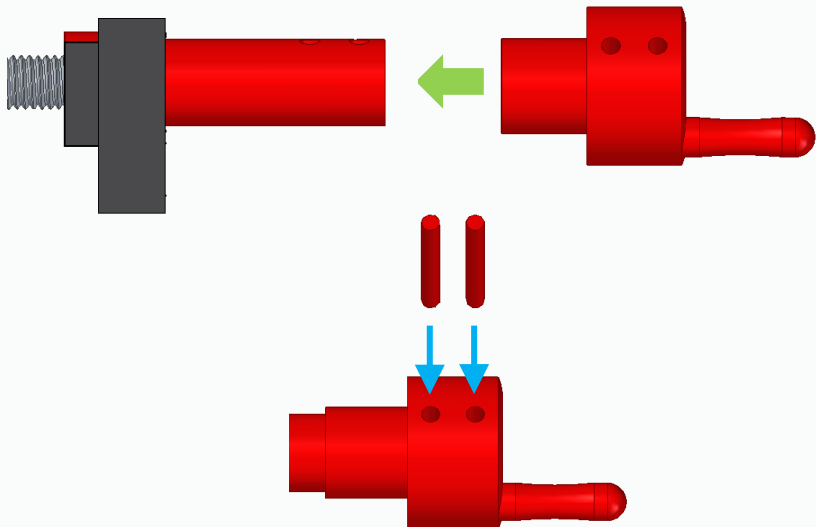
Insert the threaded rod into the central assembly through the designated slot with hex coupling (blue arrow in inset). Attach end and front caps by rotating as indicated by the blue arrows until the fit is tight.

2.



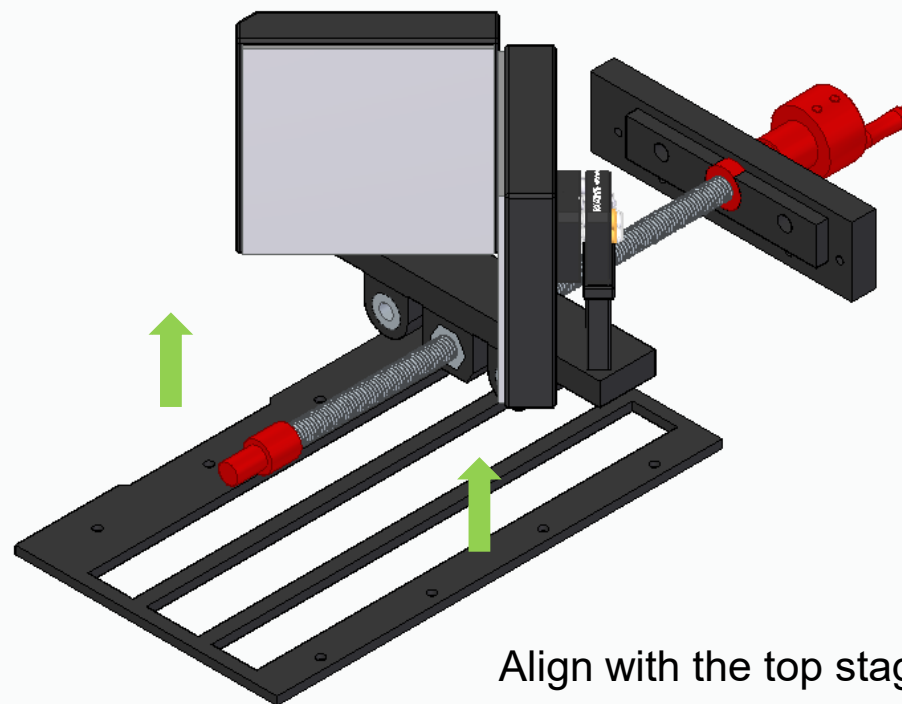
Slide the handle part of the rod (the one with the two holes) into the slot on the rear stage holder. Ensure it fits snugly and aligns with the holder.

3.



Attach the rotating handle and secure with the two fixing bolts.

4.



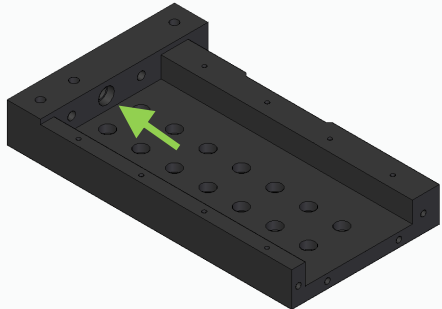
Align with the top stage cover.

Badal stage

Final assembly

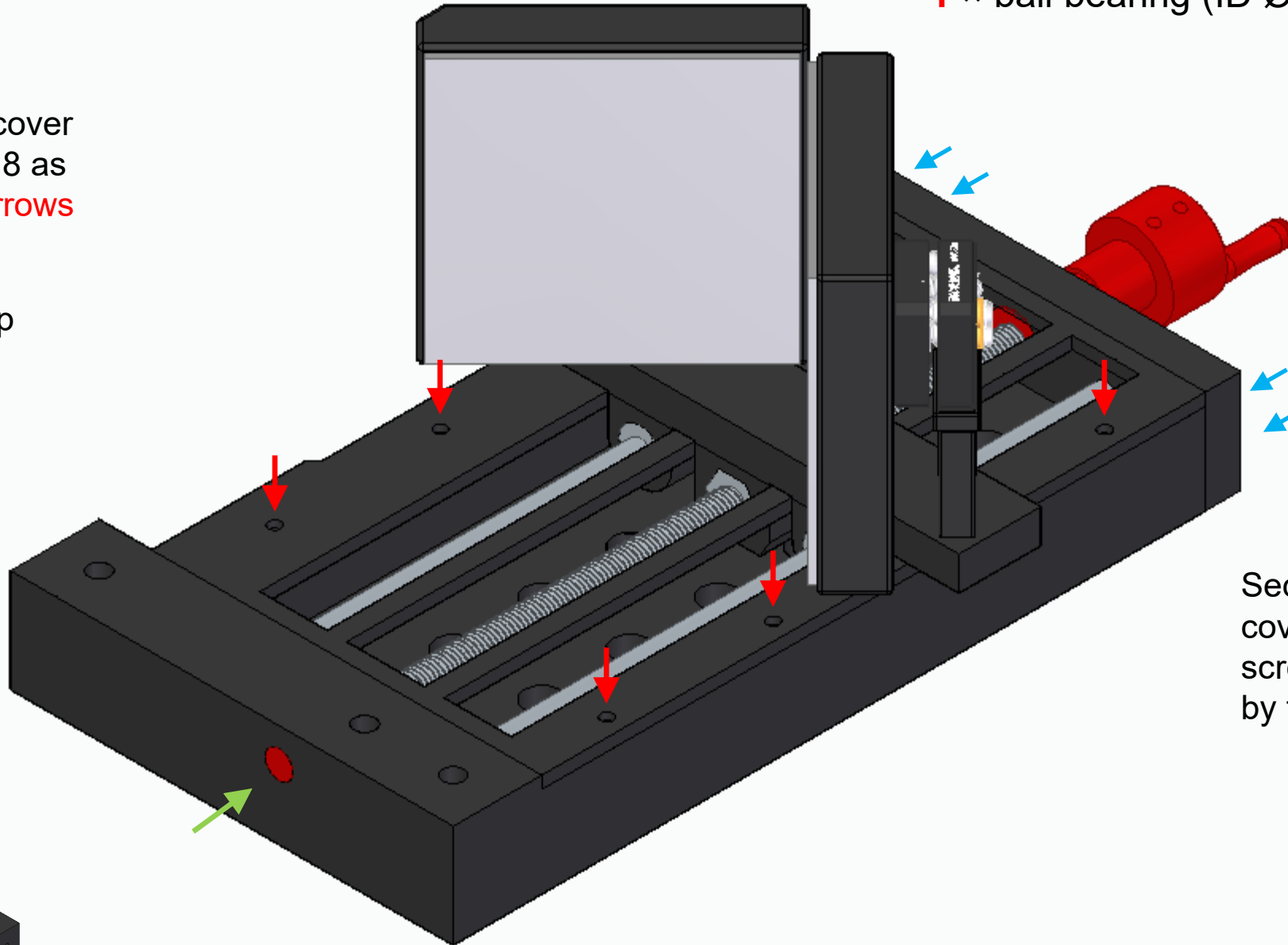
Secure the stage top cover with M2 cap screws × 8 as indicated by the **red arrows** (not all shown).

Insert and secure deep groove ball bearing in (inside) indicated by the **green arrow**.



Off-the-shelf components needed

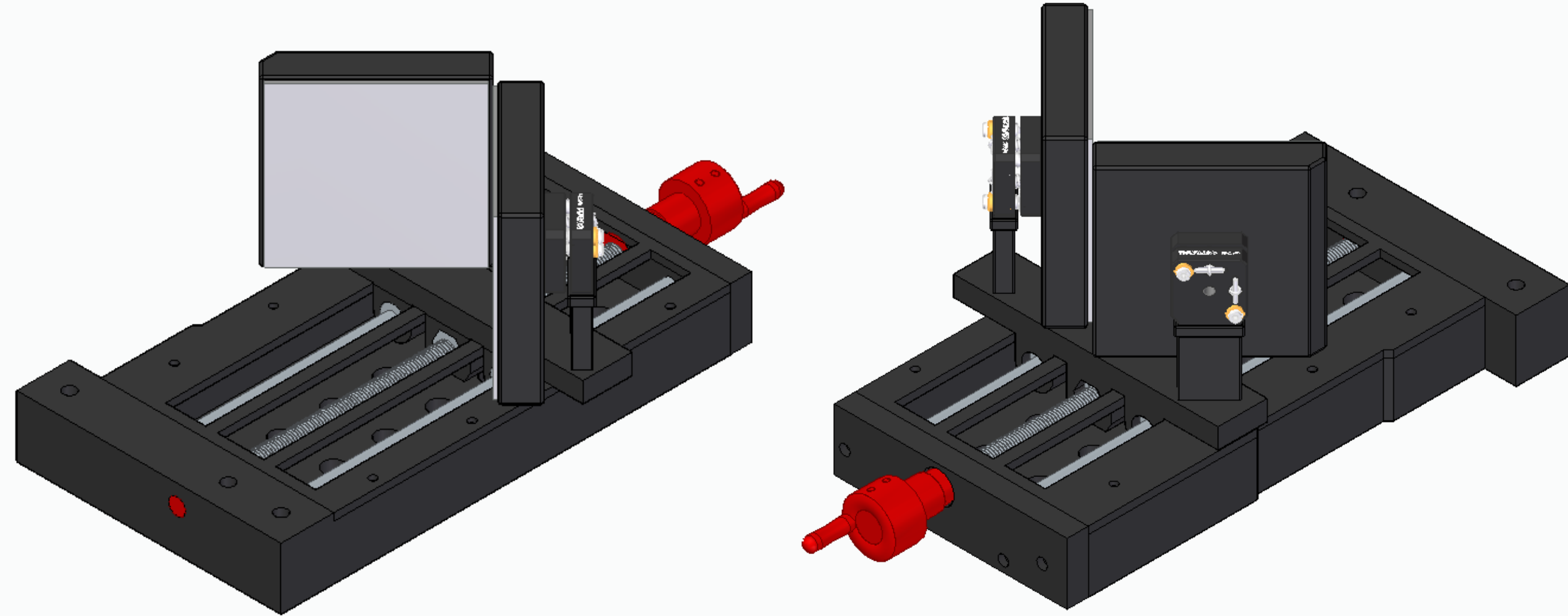
1 × ball bearing (ID Ø8mm, OD Ø12 mm)



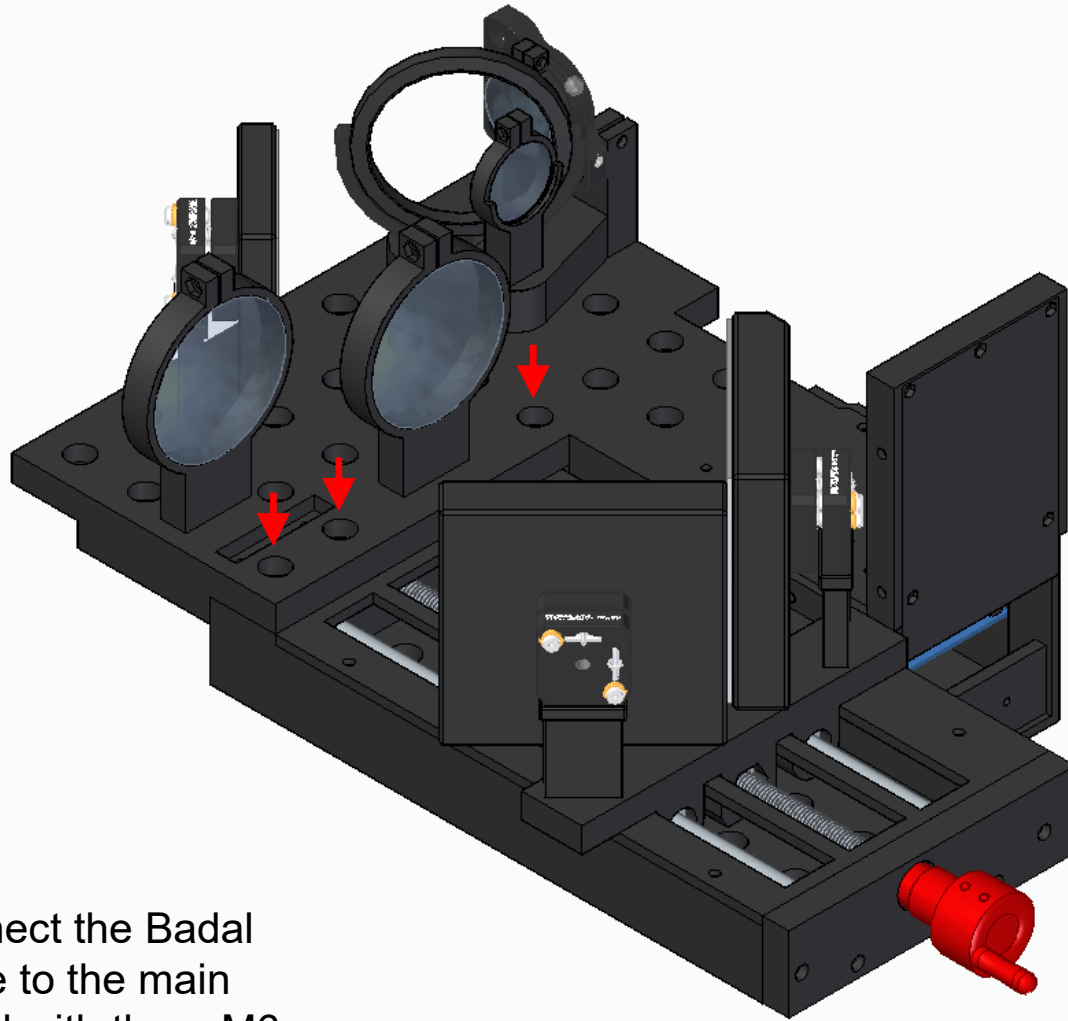
Secure the stage rear cover with M3 cap screws as indicated by the **blue arrows**.

Badal stage

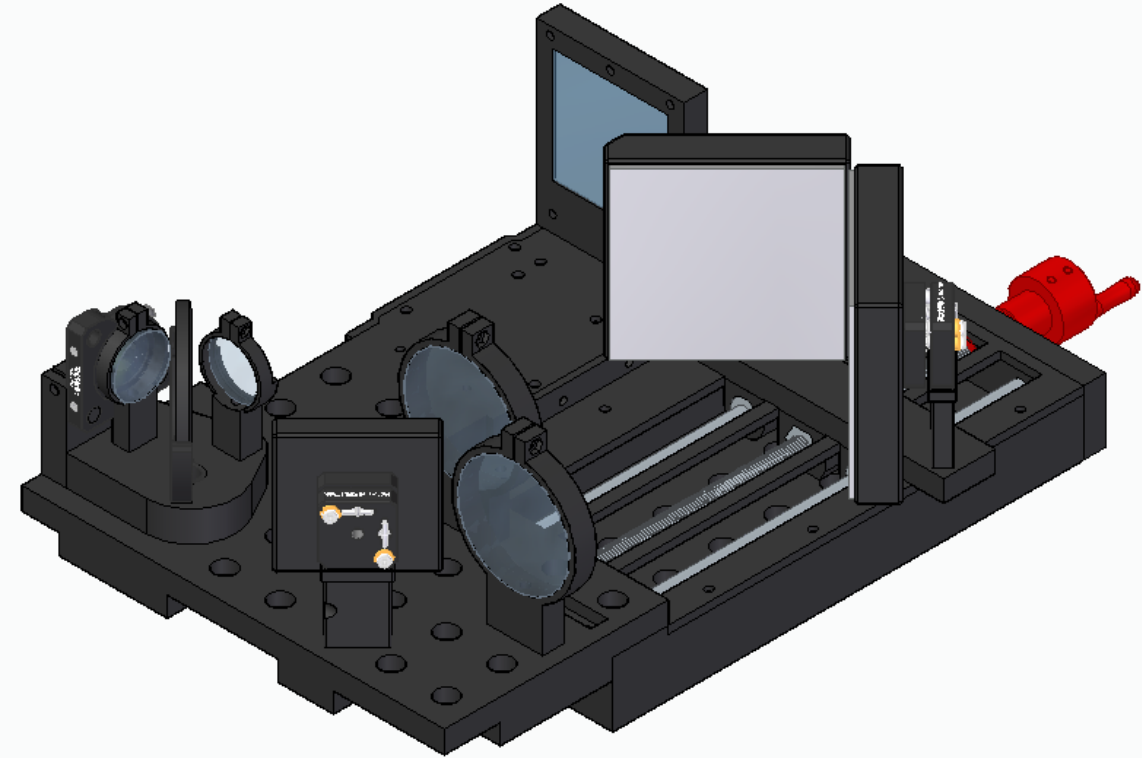
Complete!



Connecting Badal stage

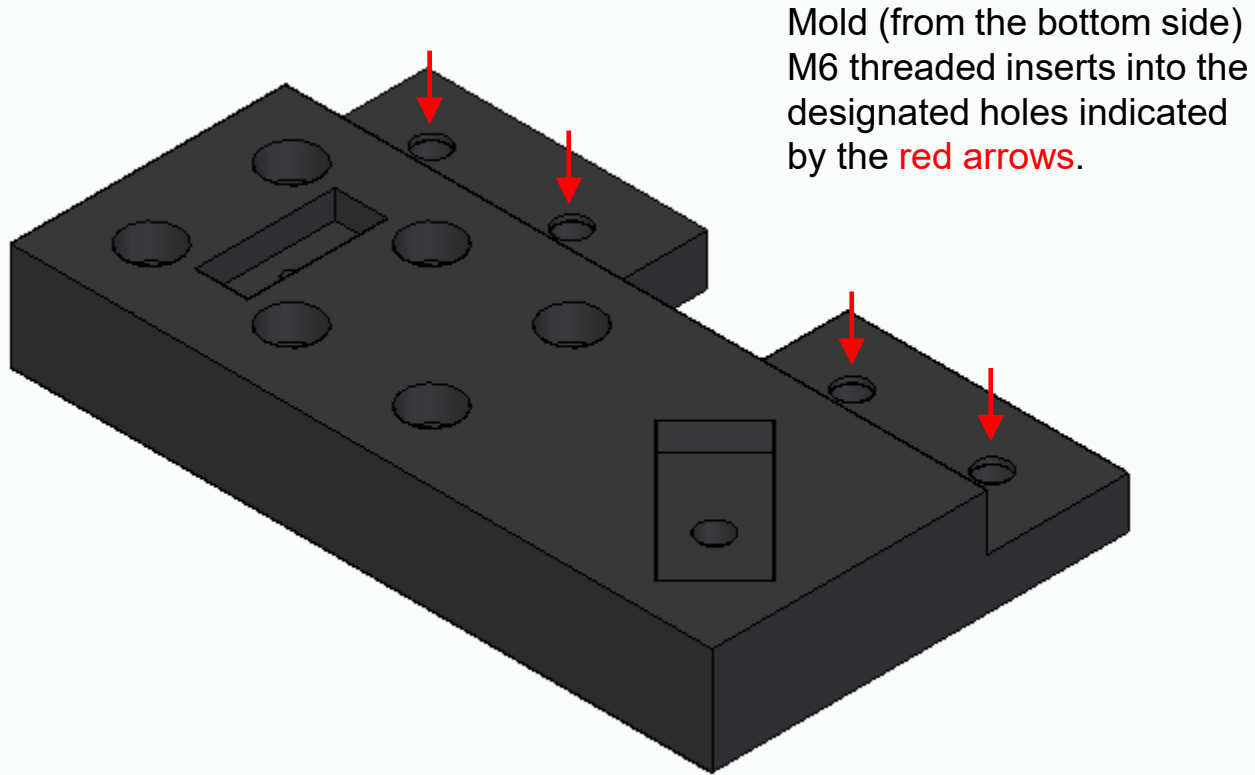


Connect the Badal stage to the main board with three M6 socket head cap screws (**red arrows**).



Imaging board

Attach element 23 to the Main board assembly first. Then proceed as the instructions indicate.

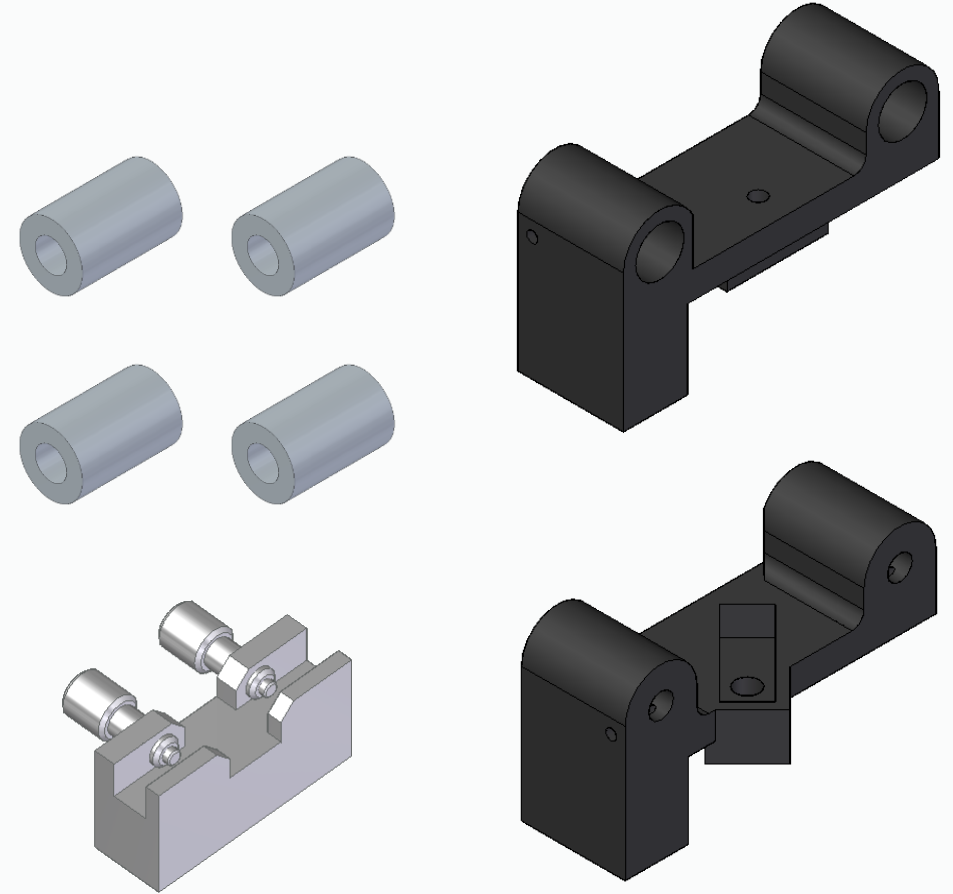


Needs to be 3D-printed:

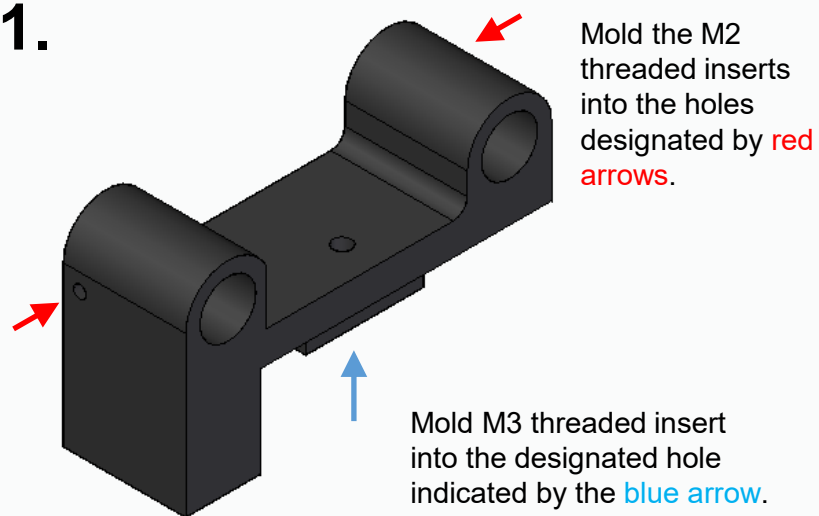
- 1** × 23_dichroic_board
- 1** × 24_dichroic_holder
- 1** × 25_rods_guiding_addon

Off-the-shelf components needed

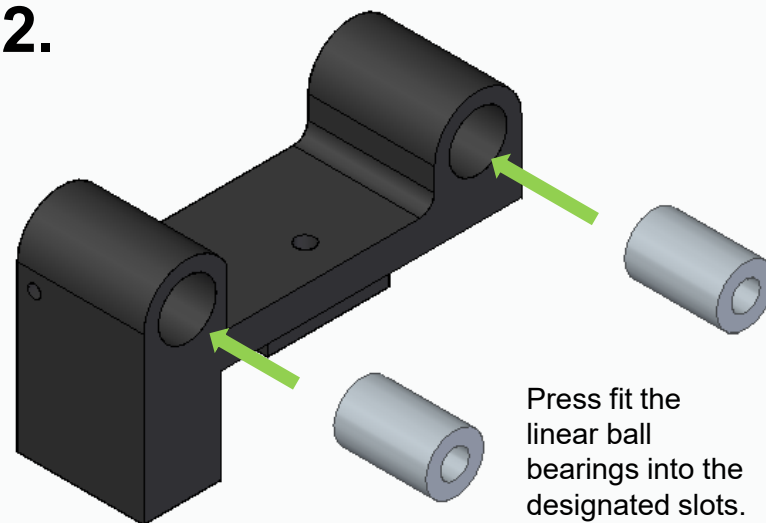
- 4** × linear ball bearing (Ø6mm)
- 1** × Filter holder ([#54-996](#))



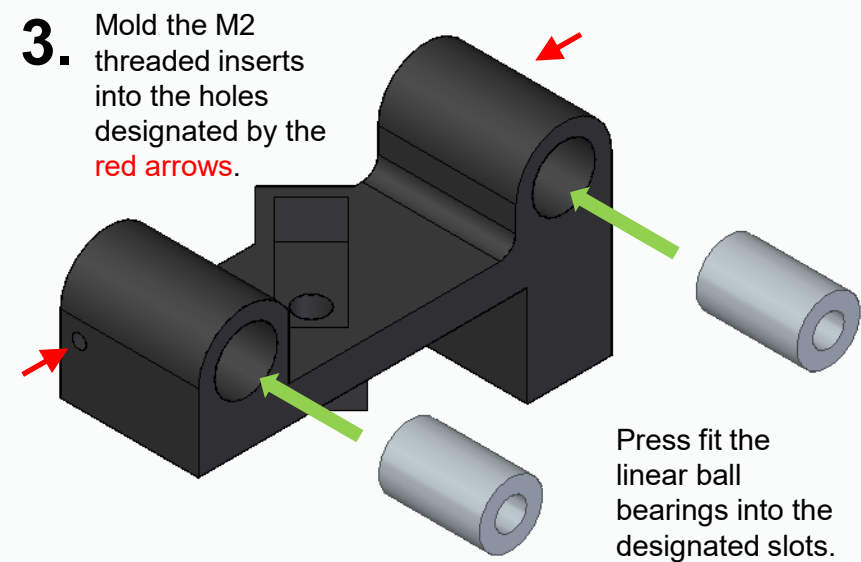
1.



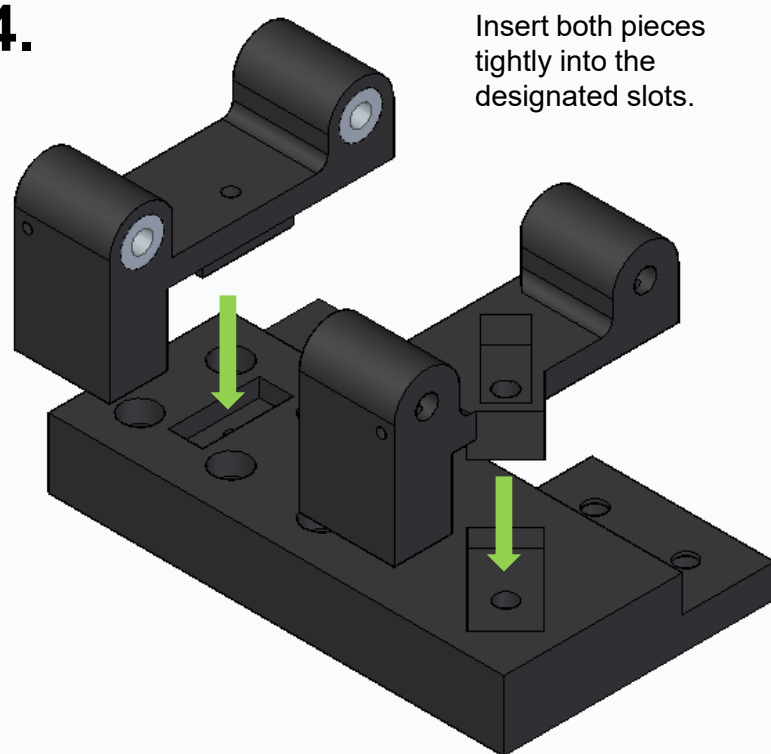
2.



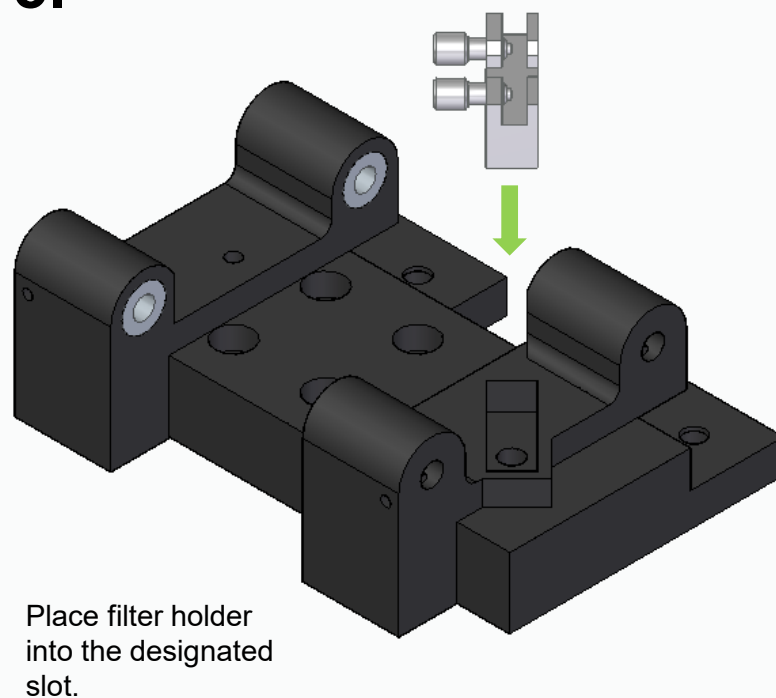
3.



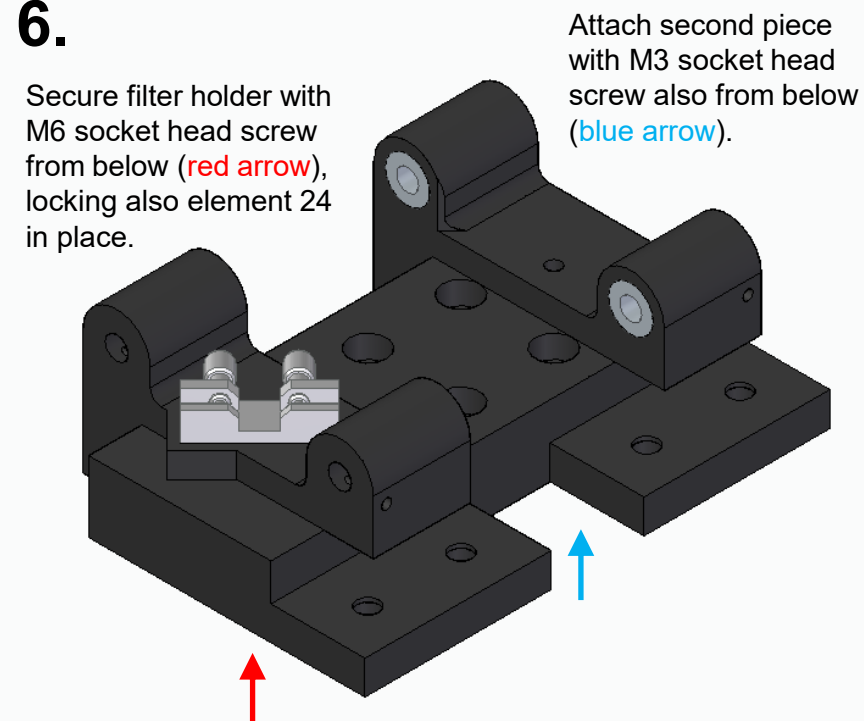
4.



5.

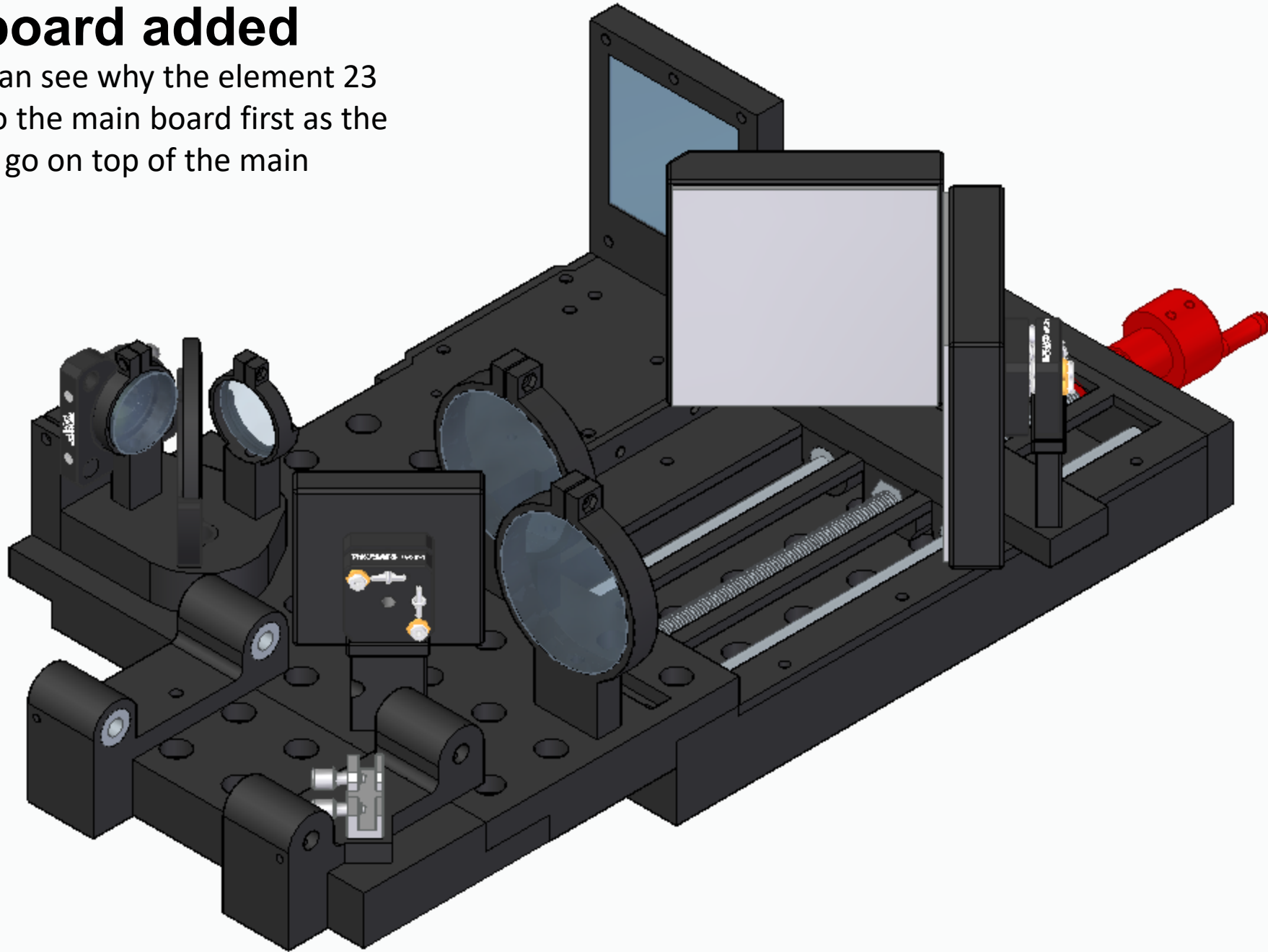


6.

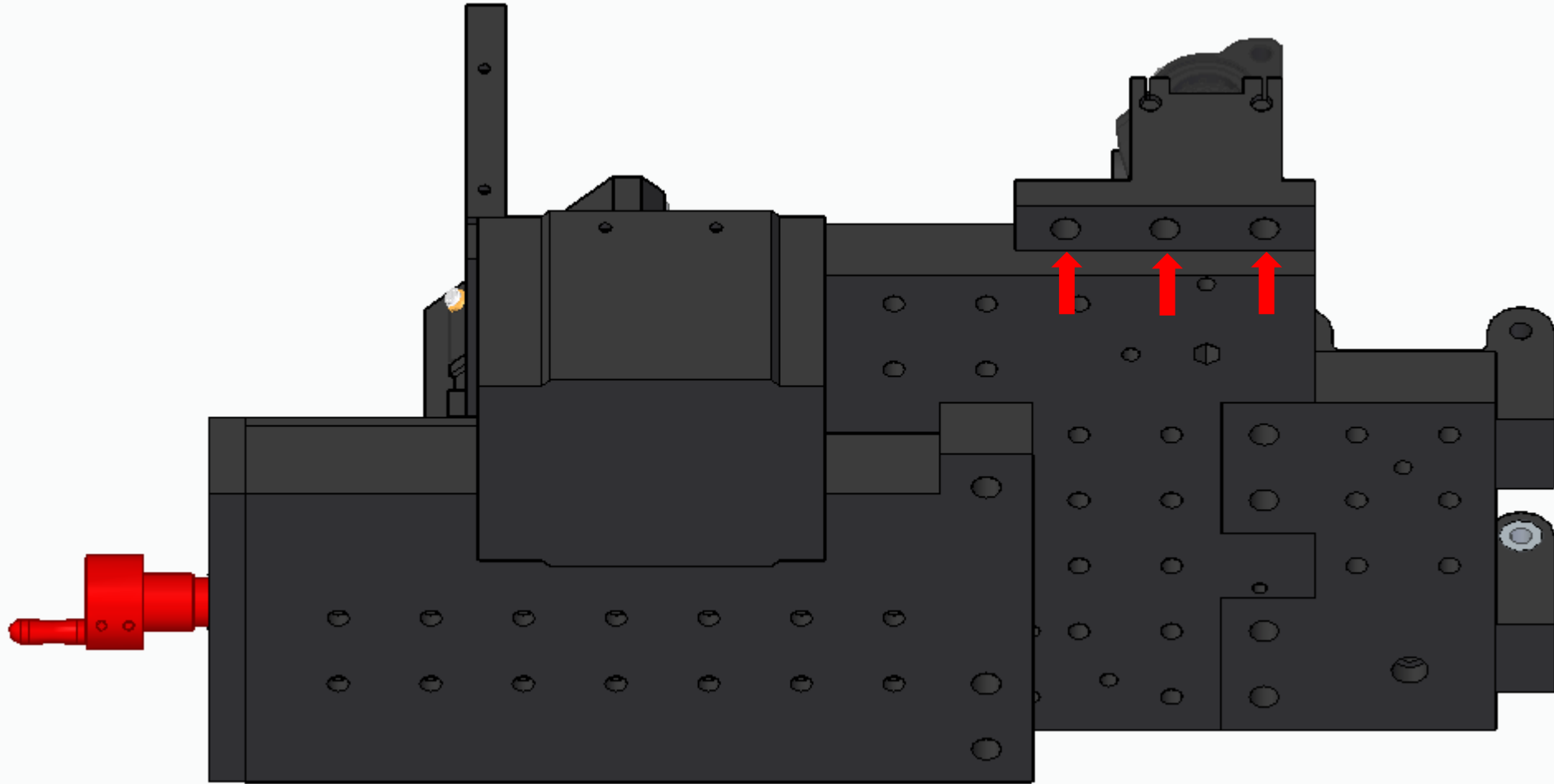


Imaging board added

Note: Here you can see why the element 23 was connected to the main board first as the two other pieces go on top of the main board.

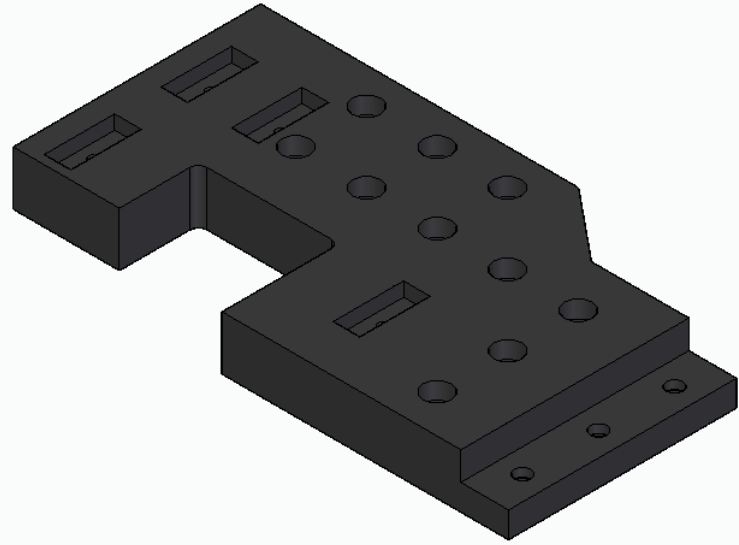


Imaging board added



Before proceeding to building the stimulus board, mold M6 threaded inserts into the holes designated by the **red arrows**.

Stimulus board

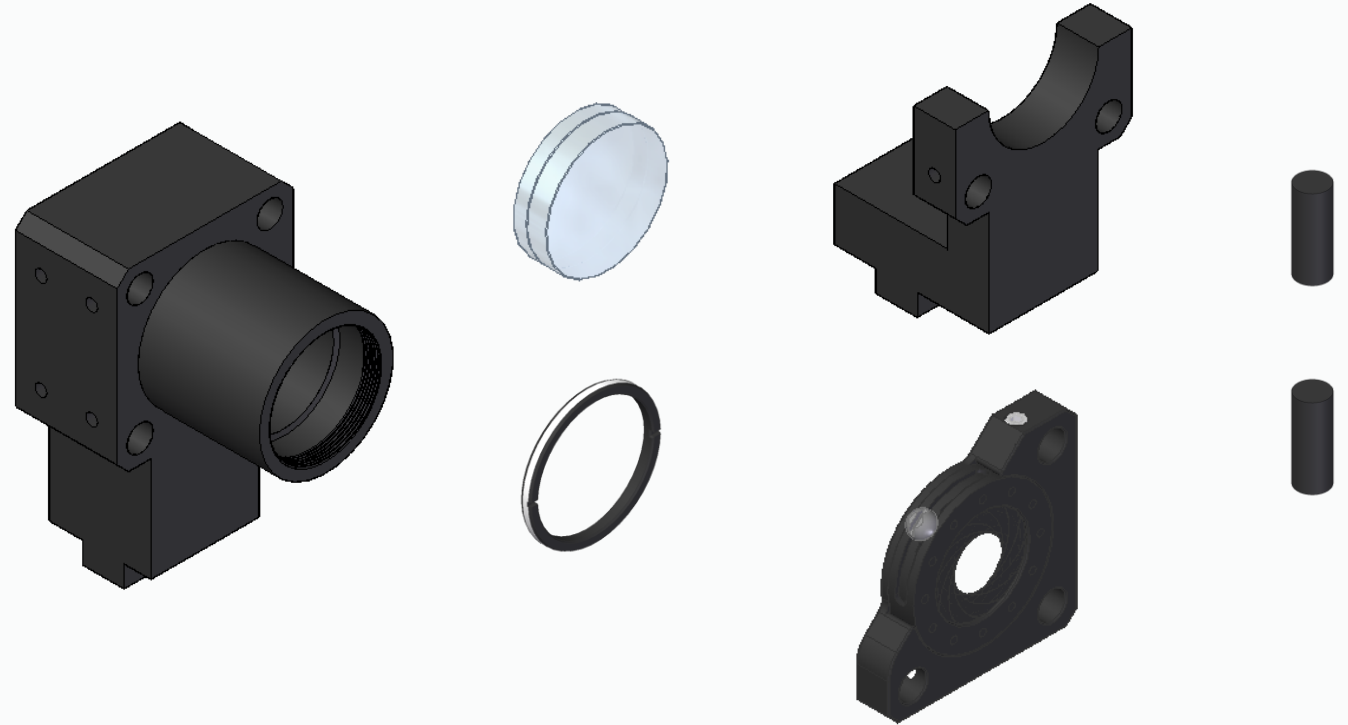


Needs to be 3D-printed:

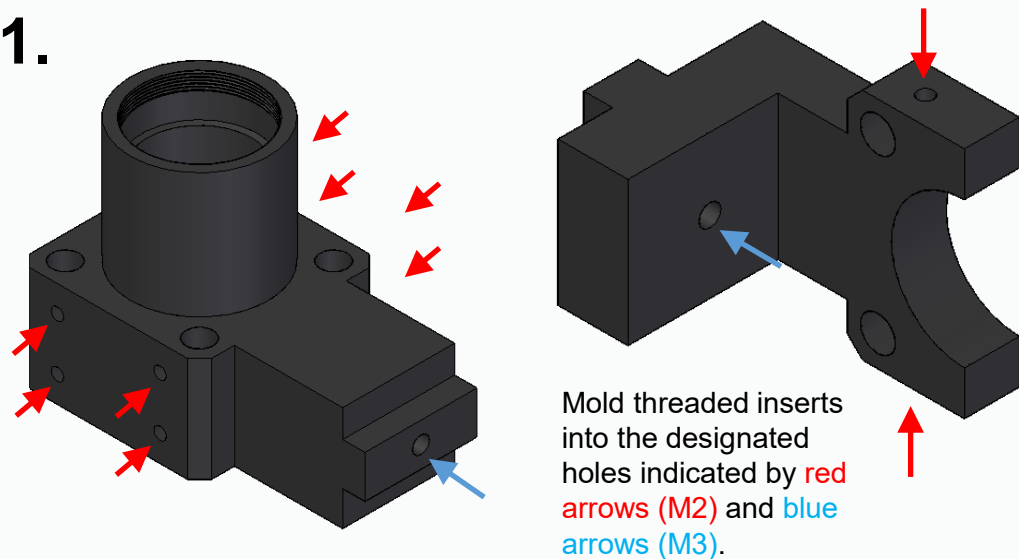
- 1 × 26_stimulus_board
- 1 × 28_stimulus_cage_system_insert
- 1 × 27_cage_system_insert_iris_holder
- 2 × 11_iris_mounting_rod

Off-the-shelf components needed

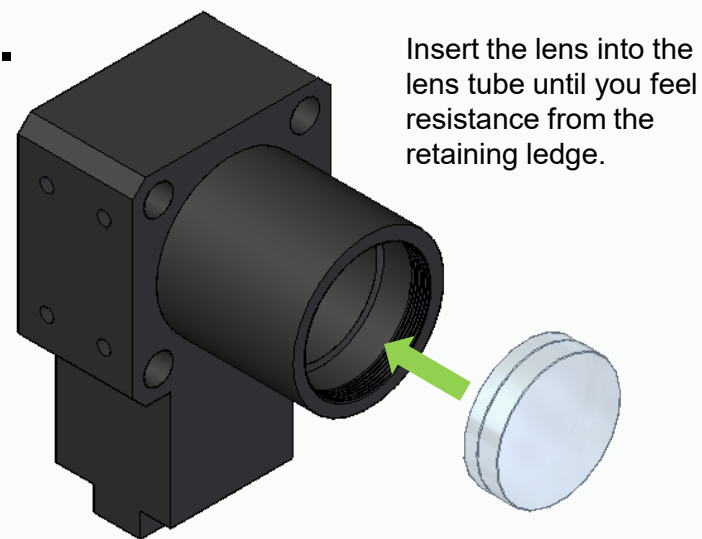
- 1 × Ø1" achromatic doublet ([AC254-075-A](#))
- 1 × retainer ring ([SM1RR](#))
- 1 × Iris Diaphragm ([CP20D](#))



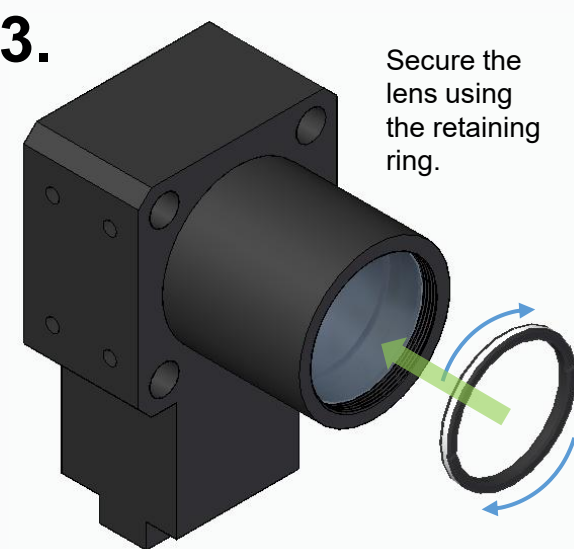
1.



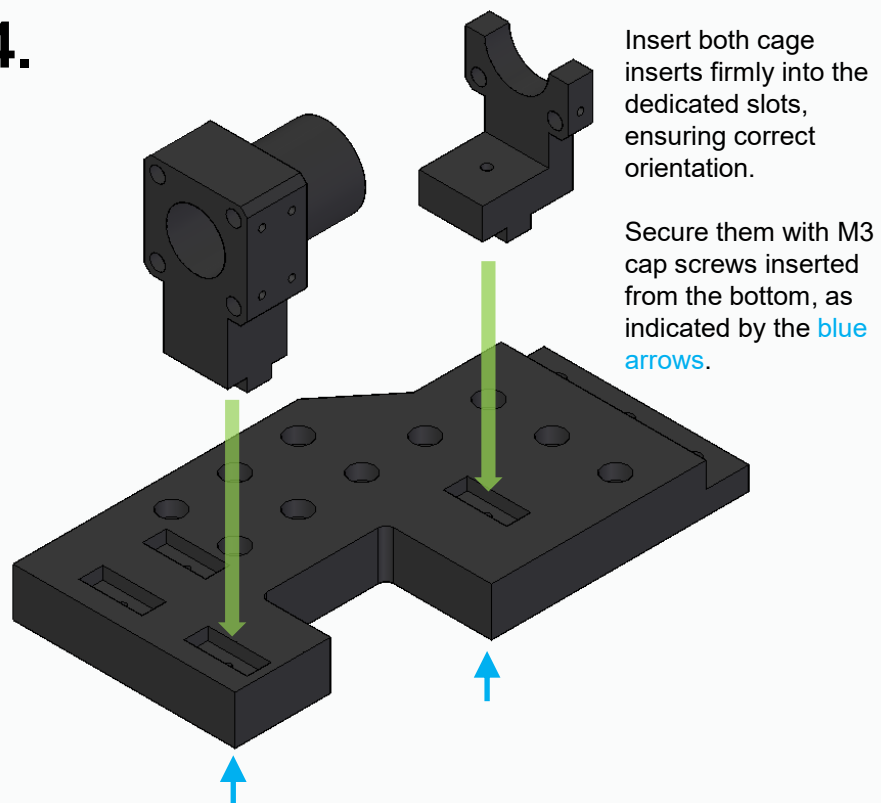
2.



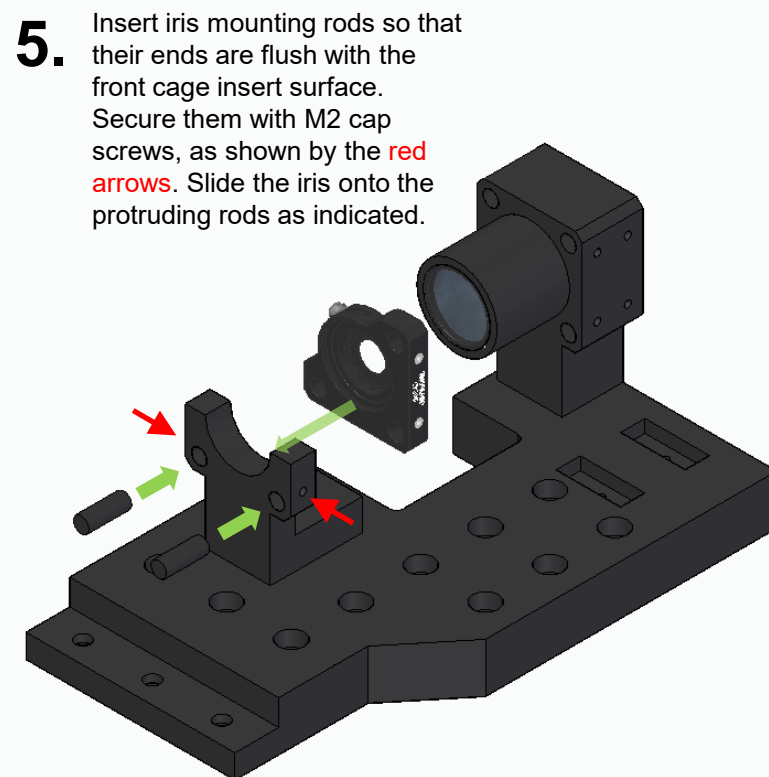
3.



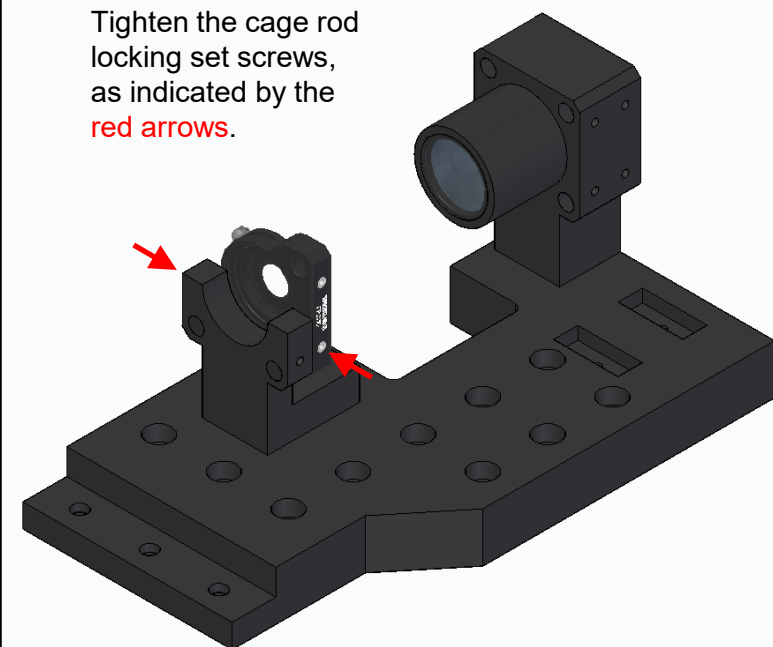
4.



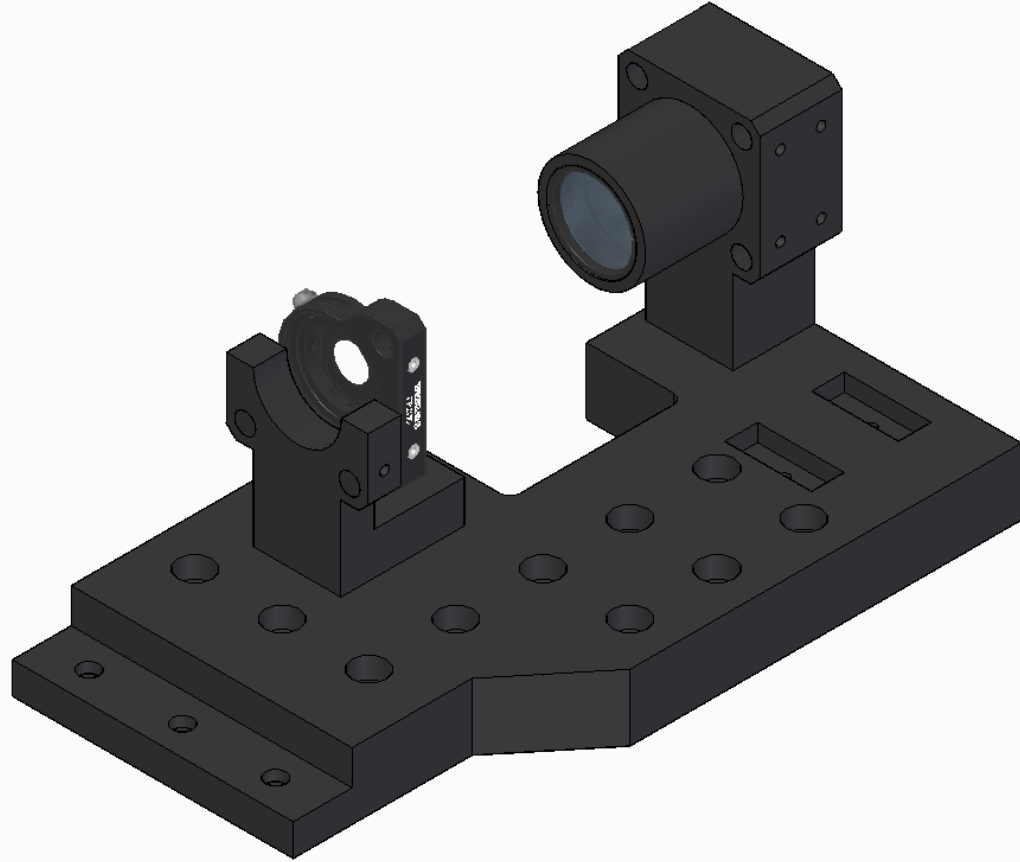
5.



6.



Stimulus board

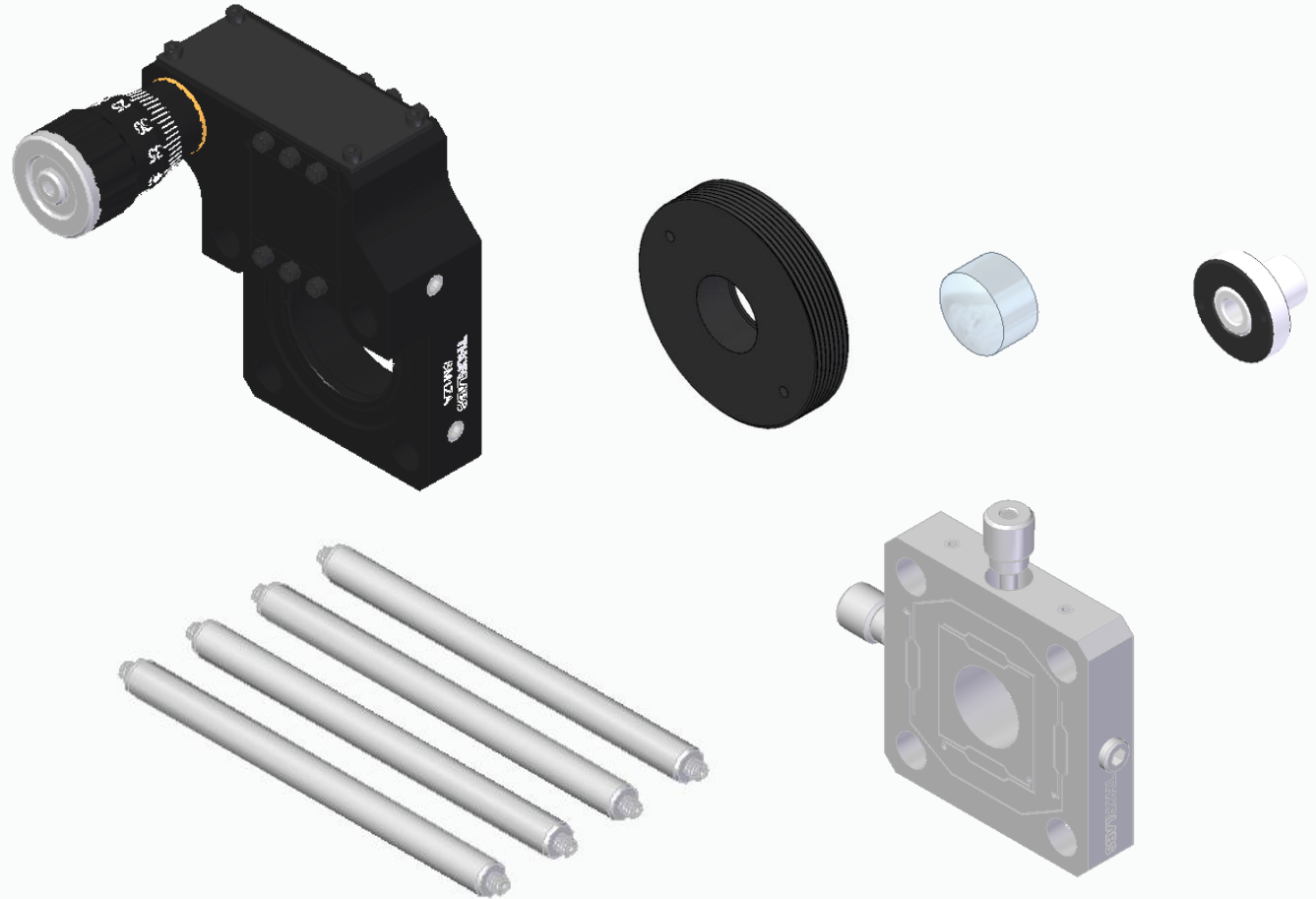


Needs to be 3D-printed:

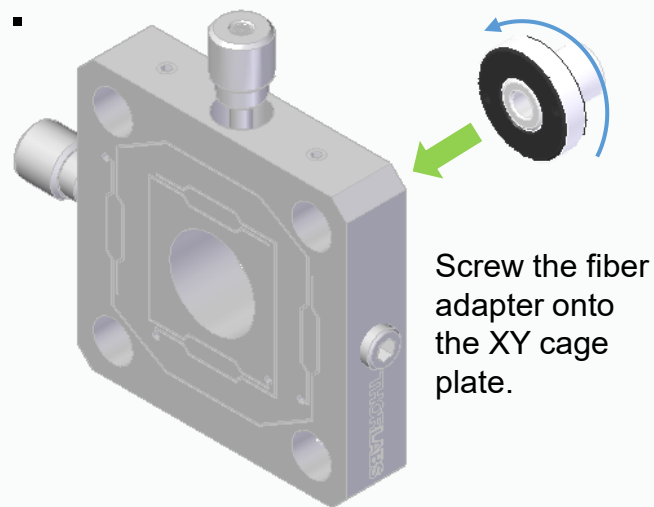
1 × 29_collimator_lens_holder_tube

Off-the-shelf components needed

- 4** × Cage rod 3 inch ([ER3](#))
- 1** × Aspherized Achromat ([#49-656](#))
- 1** × Cage XY adjustment plate ([CP1XY](#))
- 1** × SMA Fiber Adapter Plate ([SM05SMA](#))
- 1** × Z-Axis Translation Mount ([SM1ZA](#))

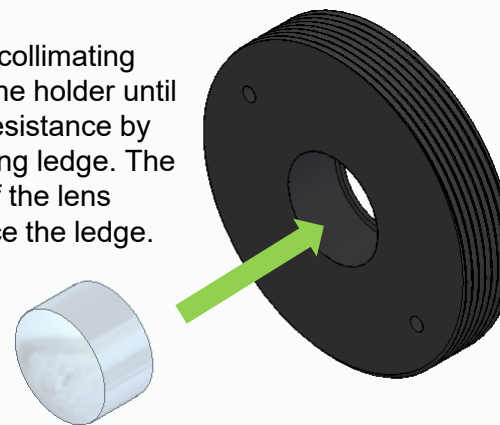


1.

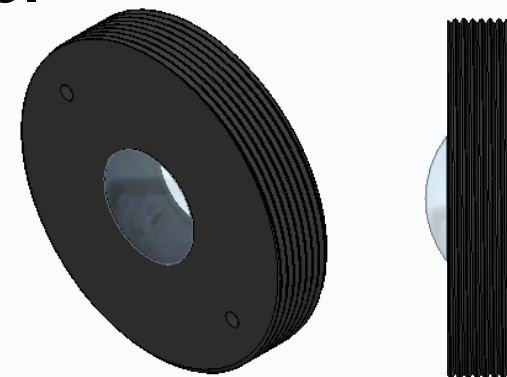


2.

Insert the collimating lens into the holder until you feel resistance by the retaining ledge. The flat side of the lens should face the ledge.

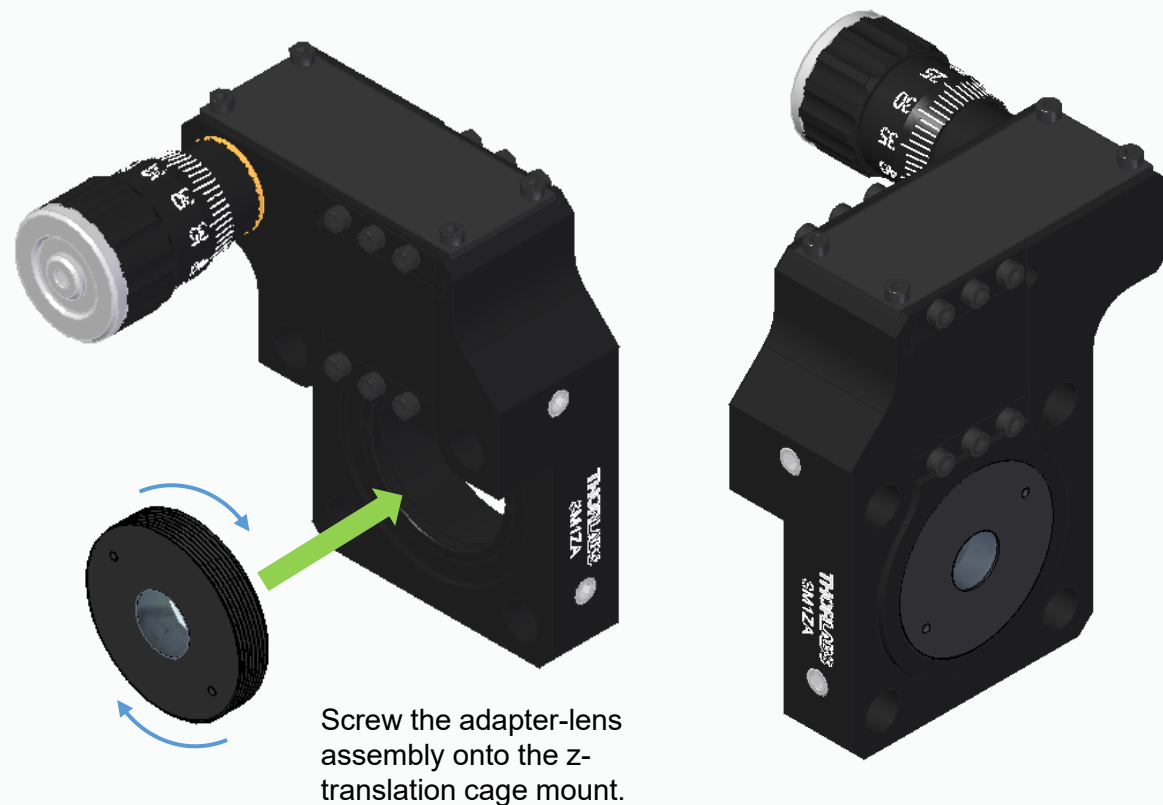


3.



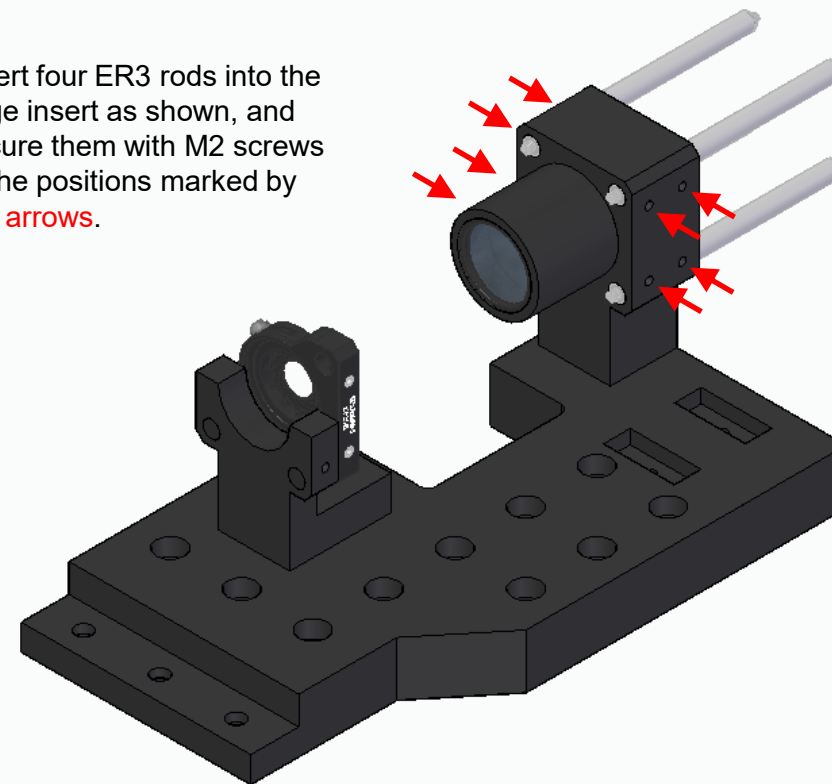
Confirm the lens is fully seated.

4.



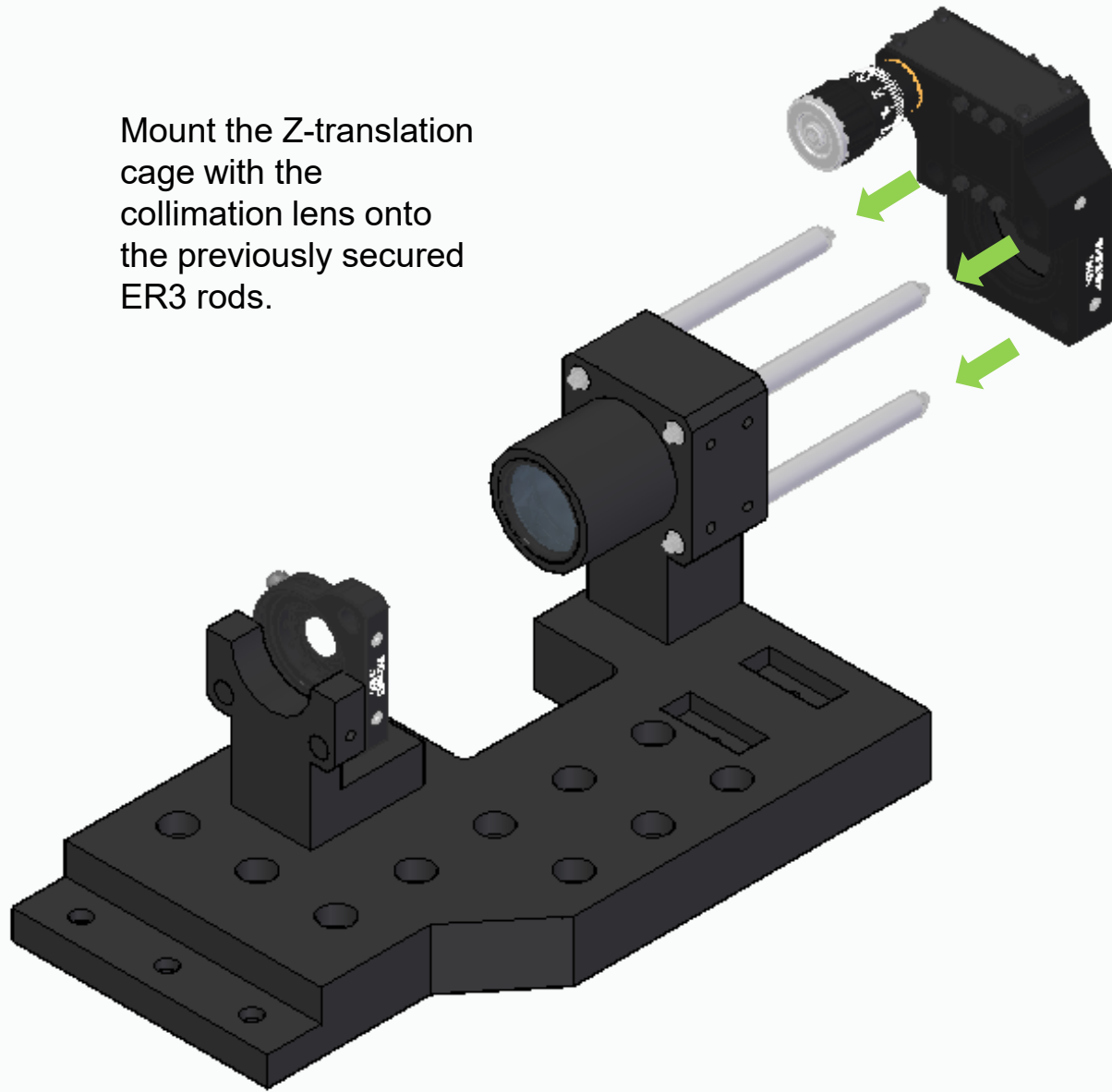
5.

Insert four ER3 rods into the cage insert as shown, and secure them with M2 screws at the positions marked by red arrows.



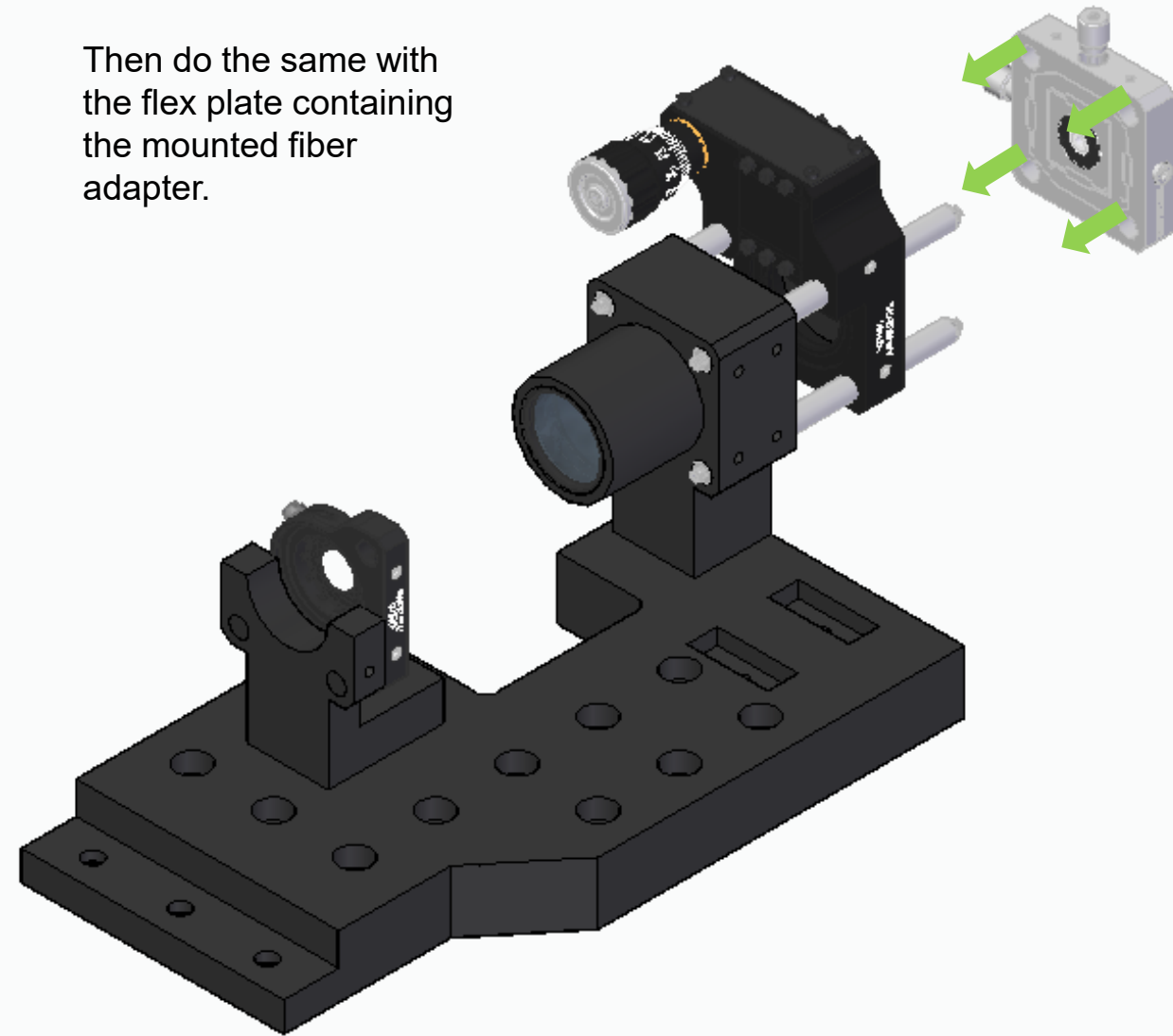
1.

Mount the Z-translation cage with the collimation lens onto the previously secured ER3 rods.

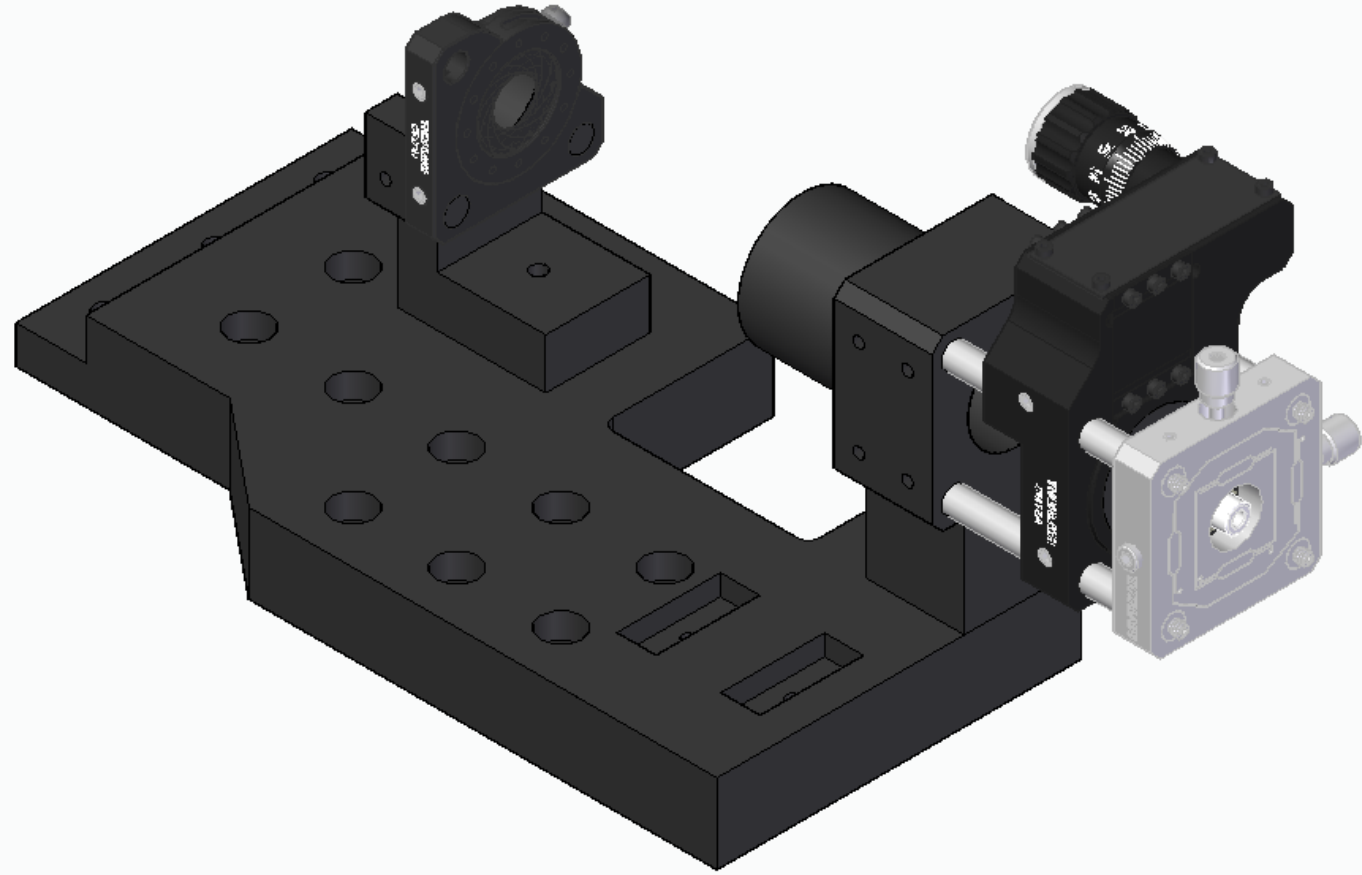
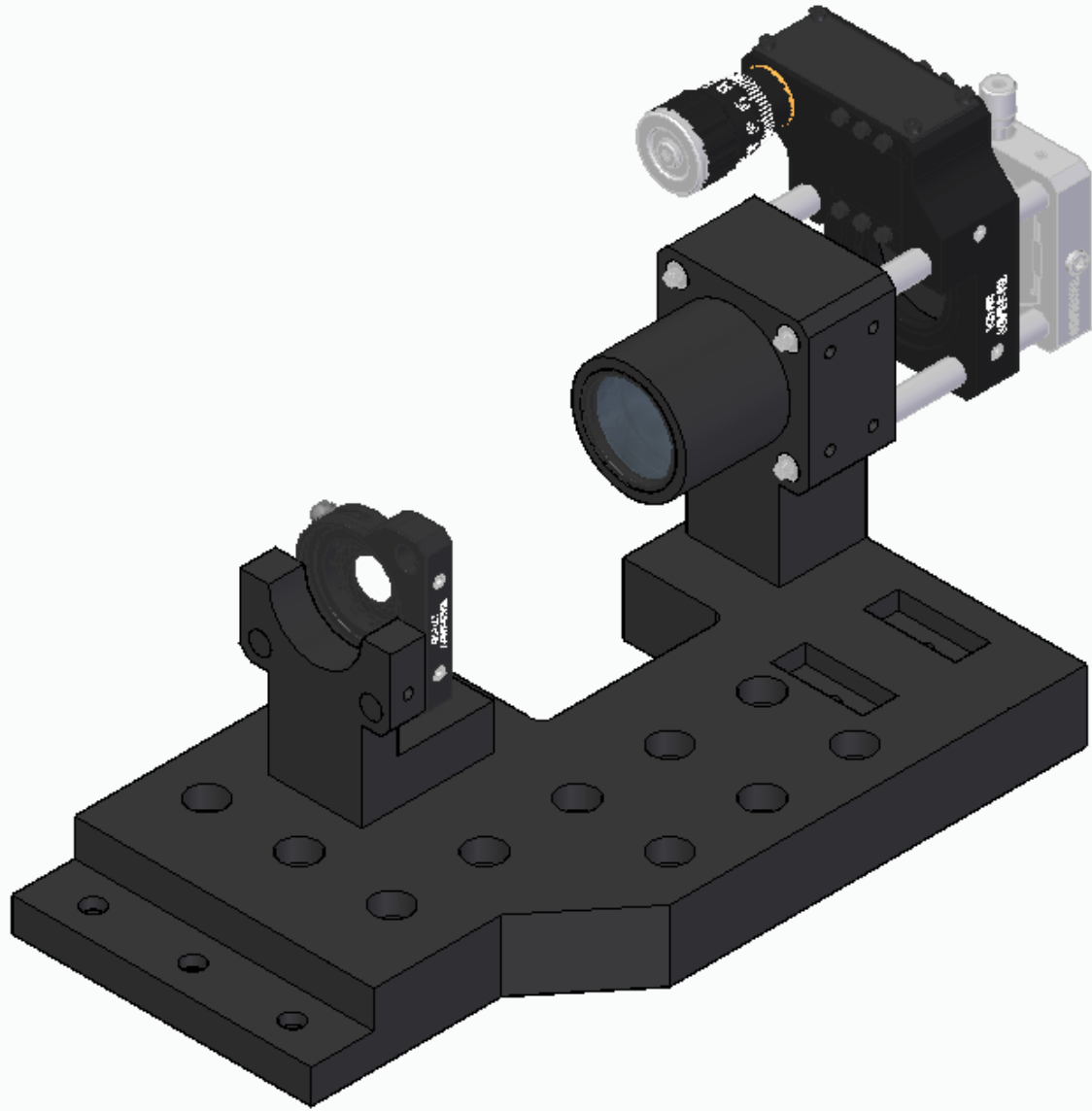


2.

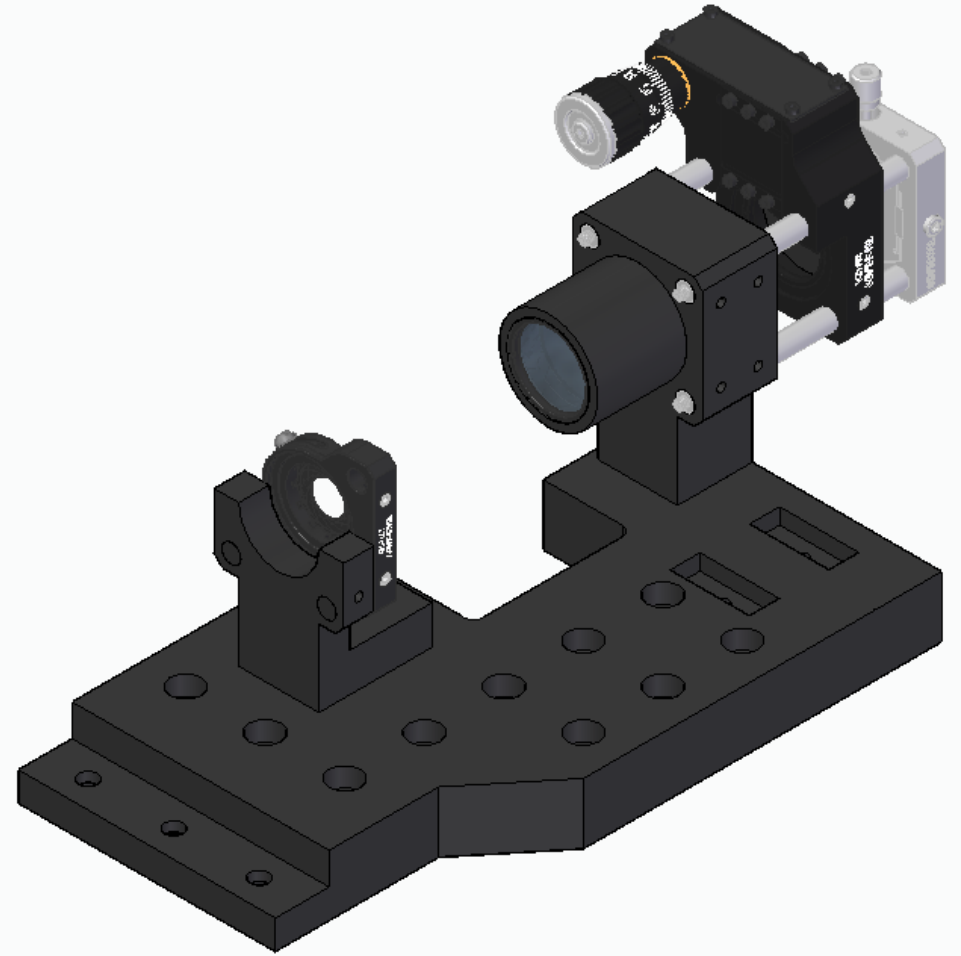
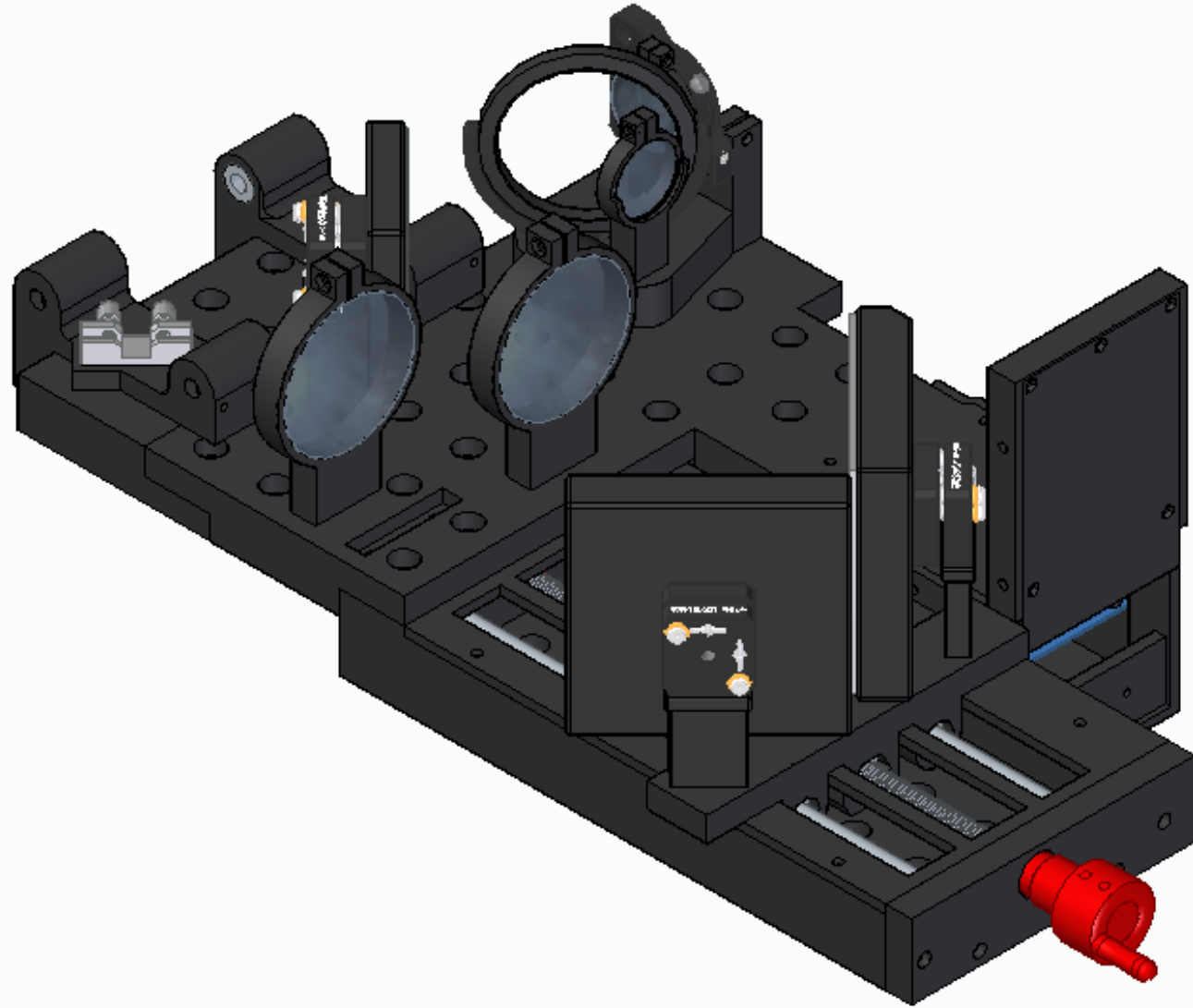
Then do the same with the flex plate containing the mounted fiber adapter.



Stimulus board

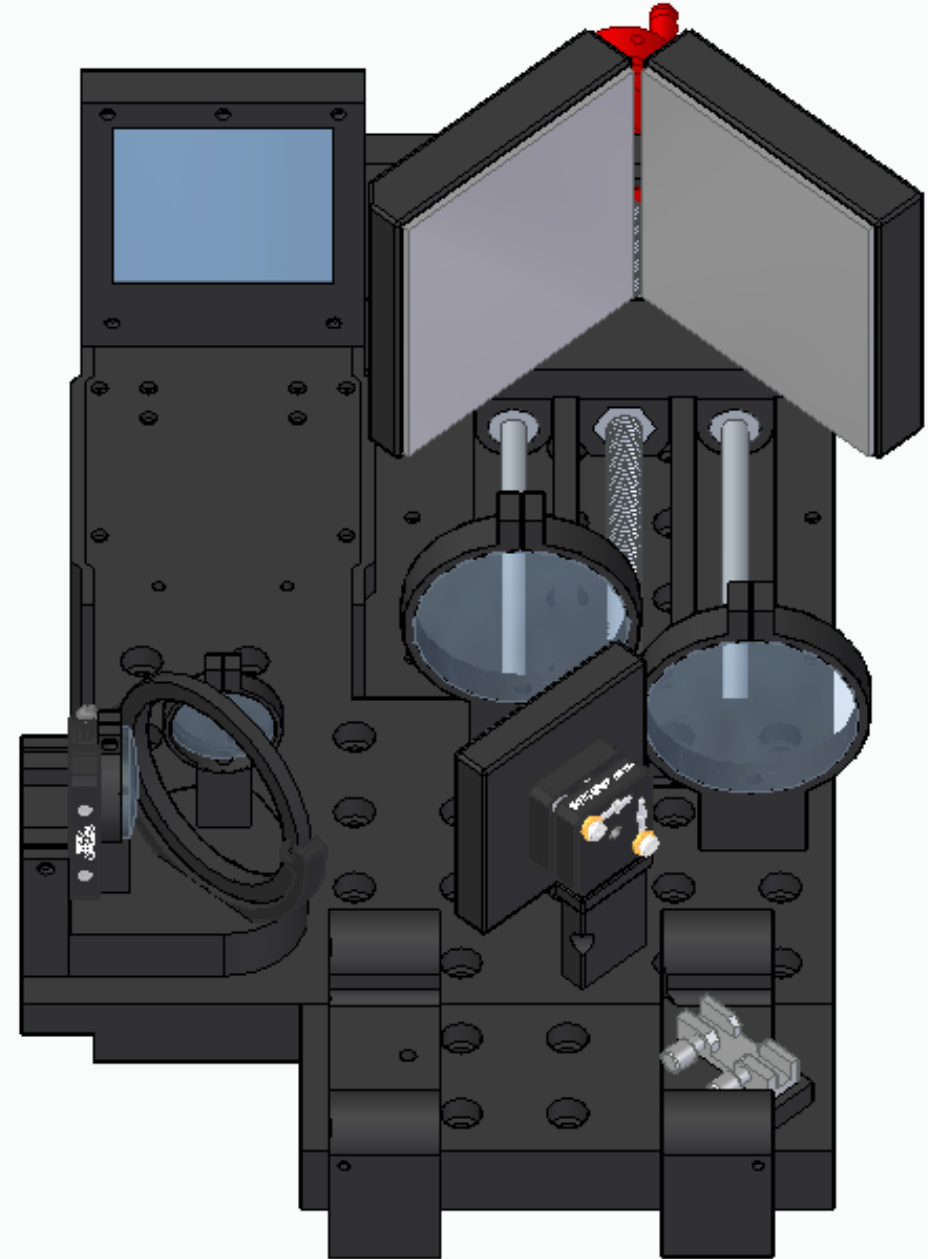
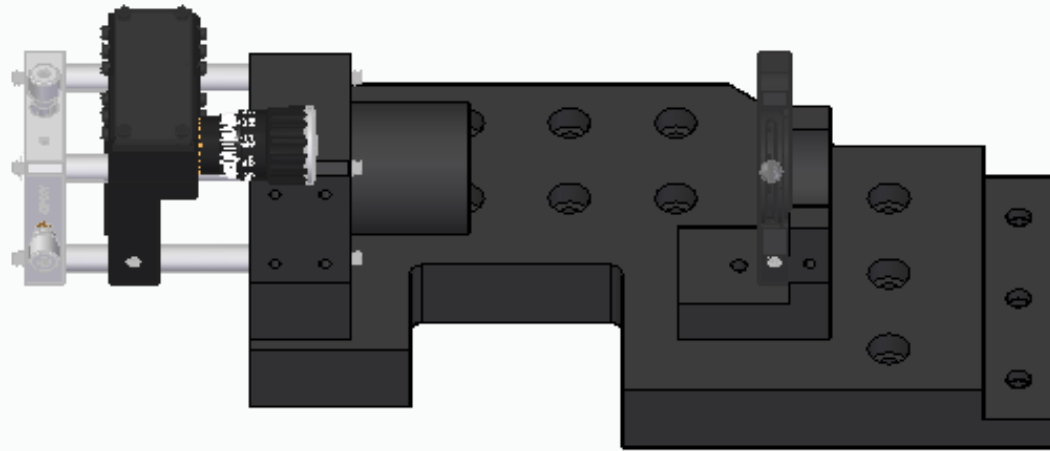


Connecting stimulus board to the main assembly

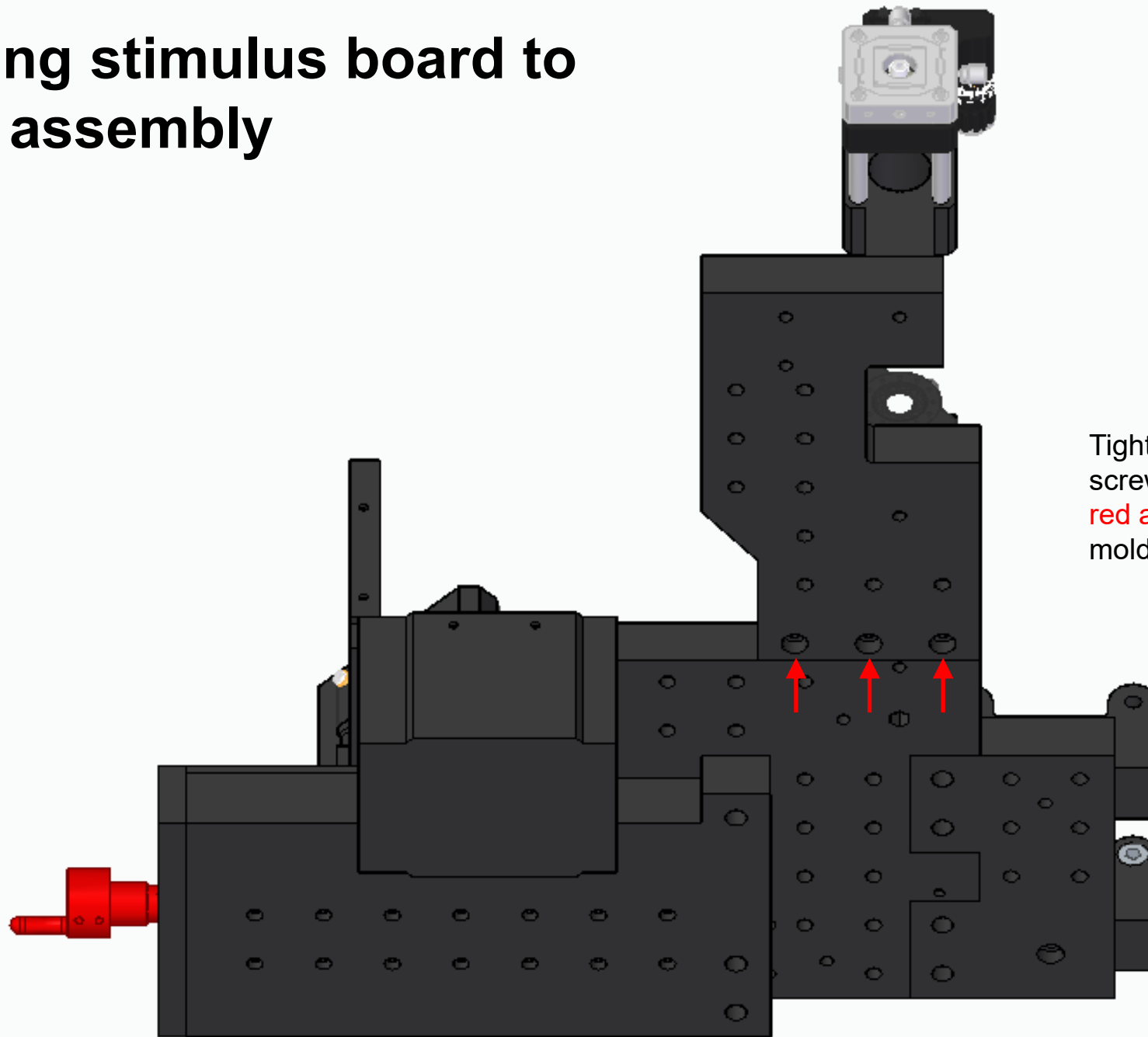


Connecting stimulus board to the main assembly

Slide the stimulus board assembly into the rest of the setup, aligning it with the matching slots as indicated.

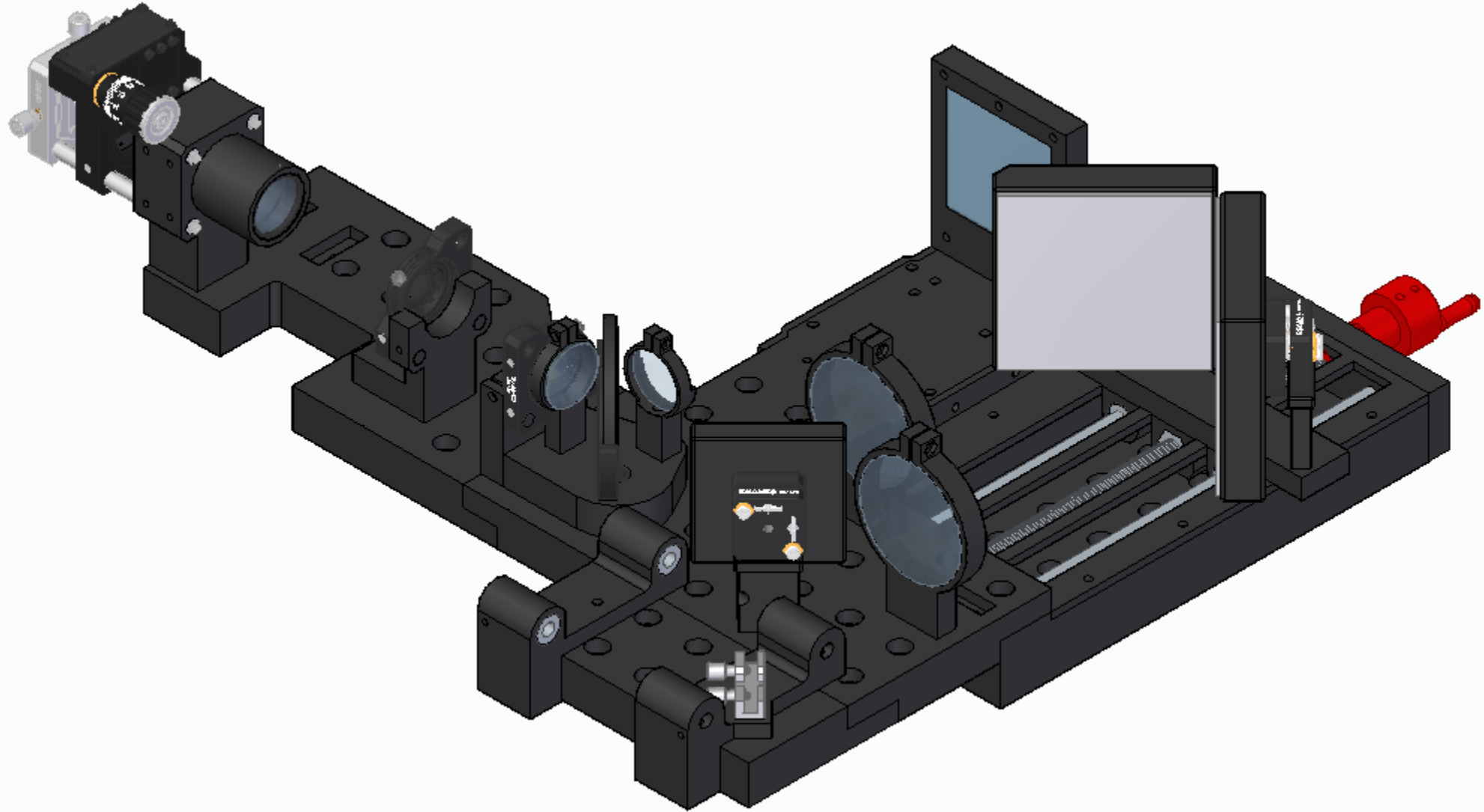


Connecting stimulus board to the main assembly

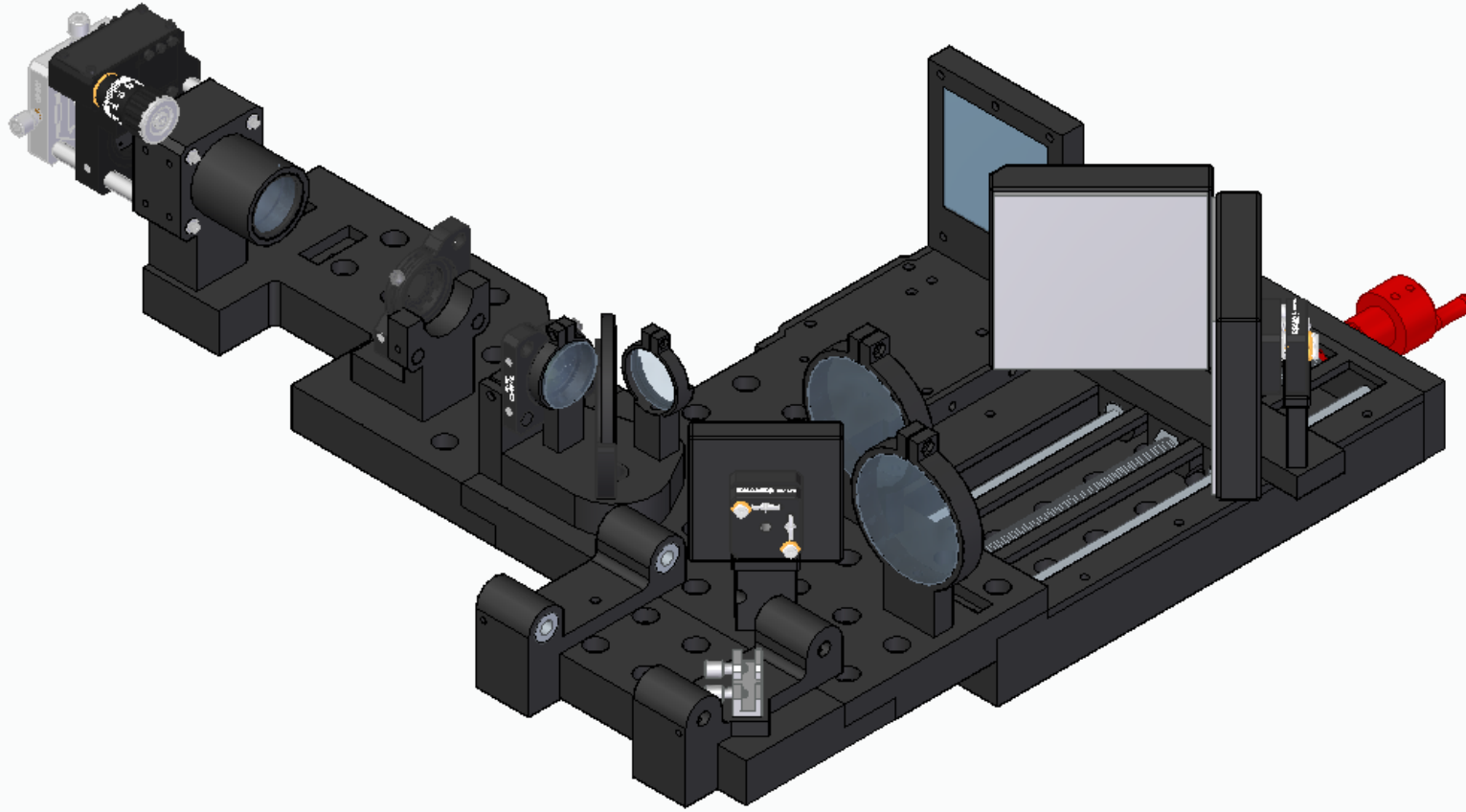


Tightly secure with M6 cap screws as indicated with the red arrows to the previously molded thread inserts.

FiSt module complete



Installing posts



Mold M6 threaded inserts into each hole at the top of each post (insert from the top side).



Note! The length of the posts can be altered to match the beam height of your optical system.

Needs to be 3D-printed:

3 × 30_mounting_post_long

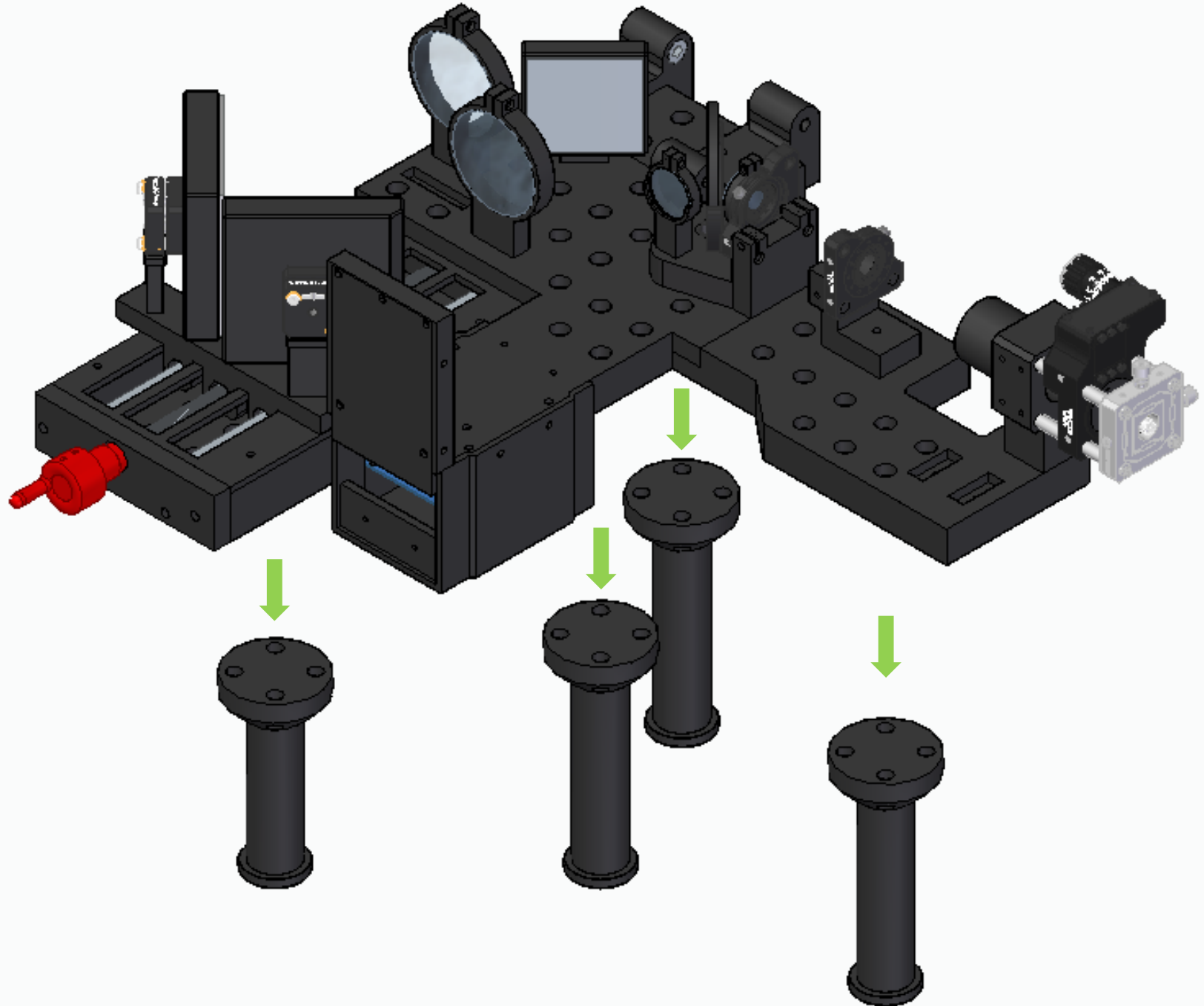
1 × 31_mounting_post_short

Installing posts

The full assembly is supported by three longer posts and one shorter post (supporting the Badal stage).

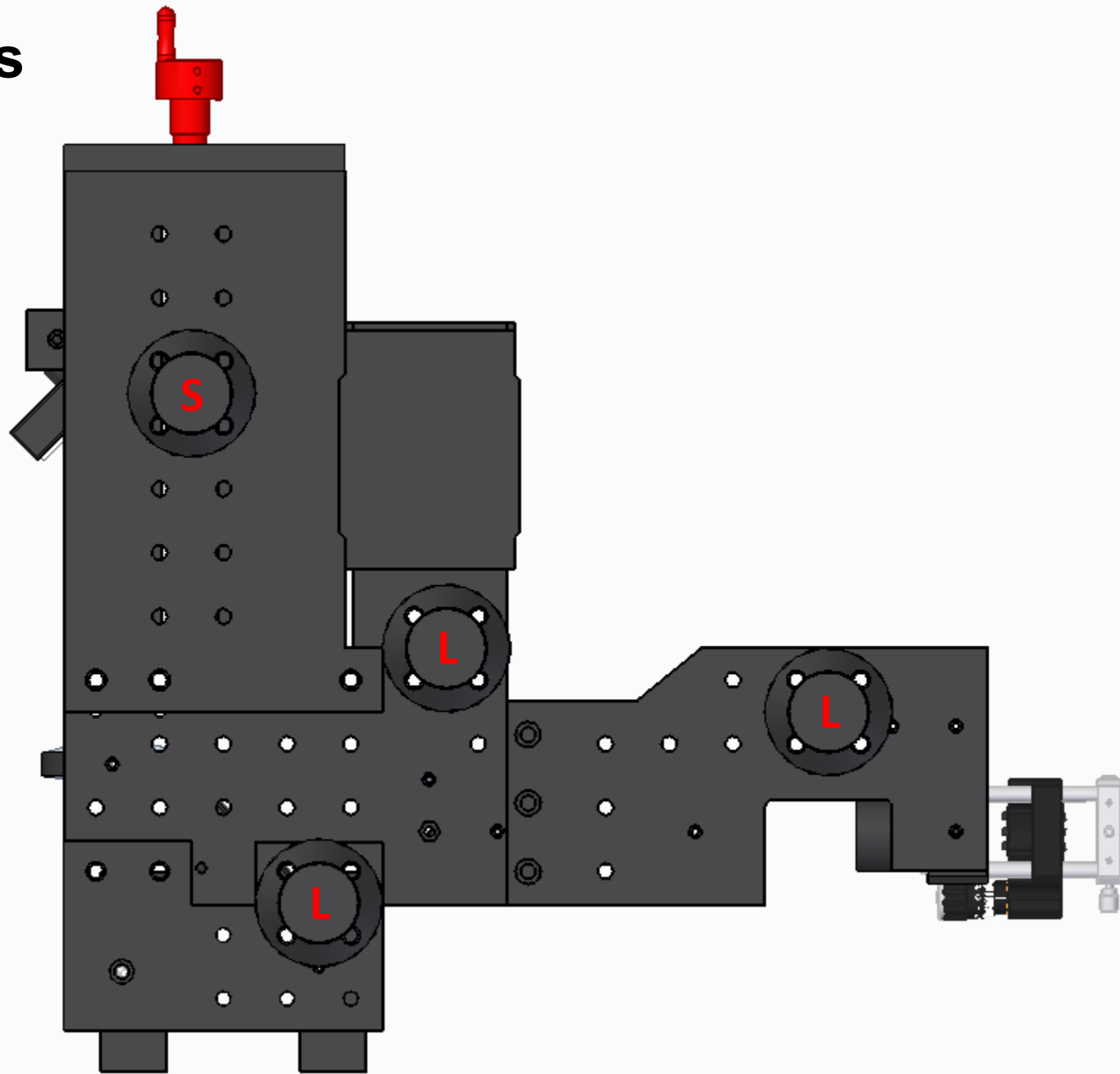
Posts can be secured to the optical table using standard optical forks. Secure the assembly to the posts using M6 cap screws inserted through the holes in the assembly, threading into the molded inserts at the top of each post.

Suggested post positions are shown on the next slide, which presents a bottom view of the assembly with precise post placement labeled (L = long, S = short).



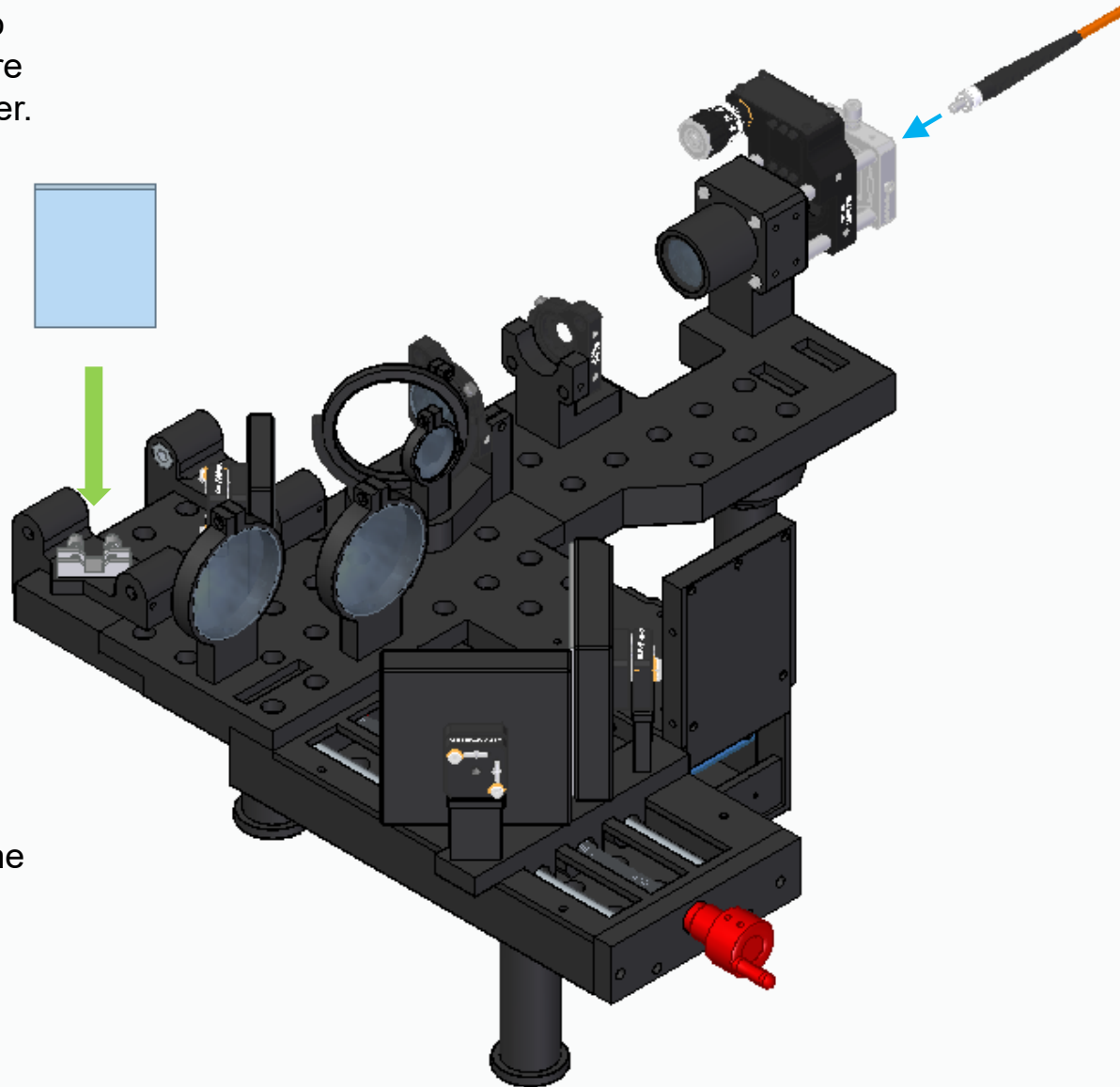
Installing posts

L = long post
S = short post



Final notes

Insert the dichroic mirror into designated holder and secure using the screws in the holder.



After inserting the fiber and the dichroic mirror you may proceed to final alignment of the optics. The stimulus-fixation channel is now ready.

Off-the-shelf components

1 × Multimode fiber ([M45L01](#))

1 × Dichroic mirror (TP715lp)

Insert the multimode fiber tip into the entry of the stimulus channel ([blue arrow](#)). Tighten the screw to the fiber adapter.

Final notes

